

Anomaly SUSY Transformation in SUSY Gauge Theory

The one-loop anomalous supersymmetry Ward identity of BPS letters in $\mathcal{N} = 4$ super-Yang–Mills: superspace, component, heat-kernel and holomorphic-twist approaches

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Abstract

We compute the one-loop anomaly of the supersymmetry transformation law — the anomalous supersymmetry Ward identity — acting on the semi-chiral (BPS-letter) composite operators of Euclidean $\mathcal{N} = 4$ super-Yang–Mills theory, quantized in $\mathcal{N} = 1$ superspace in a fixed Fermi–Feynman representative and regulated by dimensional reduction. The four letter families b , β_r , γ^r , $\partial_{\dot{\alpha}} c$ ($r = 1, 2, 3$) give nine component letters and eighty-one ordered composite insertions. The entire quantum correction is the failure of the regulated cutting rule, a single universal evanescent insertion $\mu_{\ell}^2/(D_0 D_1 D_2)$, and is therefore local. Twenty-nine ordered channels receive an exact nonzero local anomaly and fifty-two vanish exactly; every nonzero channel carries the universal coefficient $\hbar g^2/(16\pi^2)$ times the structure-constant tensor $f^{ACD} f^{BCE}$, and the full tower of holomorphic derivatives is generated in closed form by the kernel $K_{m,n;k,\ell} = 2 \binom{m}{k} \binom{n}{\ell} / [(m+n+2)(k+\ell+1)]$. We develop the result in four parallel frameworks. The superspace Feynman-diagram derivation is complete and constitutes the main result. The component Feynman-diagram approach yields one exact channel by bottom projection, $Q_{-}(\psi_{r+}^A \tilde{\phi}_s^B) = -(\sqrt{2} \hbar g^2/32\pi^2) \delta_{rs} f^{ACD} f^{BCE} \tilde{\chi}_{\dot{\alpha}}^D \tilde{\chi}^{E\dot{\alpha}}$, with its remaining sectors open. The heat-kernel approach supplies exact semigroup lemmas but is presently blocked: no single elliptic generator with the correct blockwise signs and normalizations has yet been constructed. Finally, the result agrees exactly — all eighty-one ordered output words and the derivative tower to all orders — with the holomorphic-twist prediction of Budzik–Kulp after correcting an internal factor-two inconsistency of that reference, the missing $\Gamma(3) = 2$ of the triangle parametrization; we discuss precisely what this agreement does and does not prove. All conventions — Lorentzian and Euclidean spinors, Weyl and Majorana fermions, covariant derivatives, θ -expansions, chiral and vector representations, and the BV–BRST quantization — are developed self-contained from the ground up.

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1 Introduction

1.1 The anomalous supersymmetry transformation law

Global supersymmetry Ward identities are among the most rigid constraints of quantum field theory: they organize composite operators into supermultiplets and underlie the non-renormalization properties of supersymmetric gauge theories [2]. A sharper question than the validity of such identities is whether the *transformation law itself* — the statement that a supercharge acts on a composite operator through the classical variations of its factors — survives quantization, or whether it is deformed at loop level. When such a deformation is ultraviolet-finite, local in the external momenta, and scheme-independent, it is an anomaly of the supersymmetry transformation law, in the same lineage as the Konishi anomaly, rather than a renormalization effect.

This paper determines the complete one-loop answer for the maximally supersymmetric case: Euclidean $\mathcal{N} = 4$ super-Yang–Mills theory with a compact gauge algebra, quantized in $\mathcal{N} = 1$ superspace [3]. The sharp probe is supplied by the BPS letters and their semi-chiral composites. In the Euclidean plus/minus spin frame, the gauge-covariant superspace spinor derivative splits into two components $\nabla_{\pm}$, and the elementary fields assemble into nine BPS letters: one built from the chiral field-strength superfield, three from the chiral matter multiplets, three from the independent Euclidean antichiral multiplets, and two from the antichiral field strength (Secs. 2 and 4). At tree level the semi-chiral derivative ∇_- of every letter is either zero or a sum of equation-of-motion (Euler) operators and explicit composites, so the ordered products $L_i L_j$ — the semi-chiral composites — obey a closed Ward identity modulo equations of motion. Precisely because the tree reduction is exact and Euler insertions are eliminated by the Schwinger–Dyson identity, the one-loop correction to this transformation law is finite and local: any nonzero answer is an anomaly, not an artifact of subtraction. The same letters are the avatars of the 1/16-BPS sector under the holomorphic twist of the theory [7], where the one-loop deformation of the supercharge action on letter pairs has been computed from the holomorphic anomaly by Budzik and Kulp [1].

Microscopically, the entire anomaly originates in a single mechanism: dimensional reduction’s evanescent failure of the cutting rule [5]. In the superspace Schwinger–Dyson derivation, an Euler operator attached to a propagator collapses it, and every one-loop graph reduces to a cut line times a tree subgraph. In dimensional reduction (DRED) with $d = 4 - 2\epsilon$, the superspace spin algebra is kept four-dimensional while the propagator denominators are d -dimensional; the two structures are kept strictly separate and are never fused into one another before the Schwinger sum is performed. For every marked internal line of every one-loop graph, with four-dimensional numerator square $\bar{r}_e^2$ and d -dimensional denominators D_i , the cutting rule therefore fails by one universal evanescent insertion,

$$\frac{\bar{r}_e^2}{D_0 D_1 D_2} - \frac{1}{D_1 D_2} = \frac{\mu_\ell^2}{D_0 D_1 D_2}, \quad \lim_{\epsilon \rightarrow 0} \mu^{2\epsilon} \int \frac{d^{4-2\epsilon} \ell}{(2\pi)^{4-2\epsilon}} \frac{\mu_\ell^2}{(\ell^2 + \Delta)^3} = \frac{1}{32\pi^2}, \quad (1.1)$$

with no graph-specific factor of $4-d$. The evanescent master integral is finite because $\epsilon \Gamma(\epsilon) \rightarrow 1$, and its would-be nonlocal Δ -dependence enters only at order ϵ : the anomaly is a polynomial in the external momenta, i.e. the insertion of a local composite operator. Replacing the numerator by its d -dimensional form before the subtraction would annihilate (1.1) identically and erase the anomaly.

1.2 Main result

Consider Euclidean $\mathcal{N} = 4$ super-Yang–Mills theory, with gauge generators $[T_A, T_B] = i f_{AB}^C T_C$ and Killing form normalized to $\kappa_{AB} = \delta_{AB}$, quantized in $\mathcal{N} = 1$ superspace. The main result of

this paper is the complete one-loop anomalous supersymmetry Ward identity of the BPS-letter composites: for each of the $9 \times 9 = 81$ ordered pairs (L_i, L_j) ,

$$\nabla_-(L_i^A L_j^B) \Big|_{1\text{-loop}} = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \Delta(L_i, L_j)^{DE}, \quad (1.2)$$

with the sum over C understood. Three features are universal. First, the coefficient $\hbar g^2/(16\pi^2)$ and the ordered color tensor $f^{ACD} f^{BCE}$ are the same in every channel. Second, the letter bilinear $\Delta(L_i, L_j)^{DE}$ is fixed explicitly (Sec. 4): exactly 29 of the 81 ordered channels are nonzero and the remaining 52 vanish exactly, for four independent structural reasons; reversed orderings (L_j, L_i) are independent rows, not quotients of (L_i, L_j) . Third, the anomaly is local, finite, and — up to a finite normal-product term fixed by the residual supersymmetries of the vacuum, under which the 29 nonzero channels close — scheme-independent.

Resumming the holomorphic-derivative tower, the anomaly of an arbitrary derivative composite is governed by the closed-form kernel

$$K_{m,n;k,\ell} = \frac{2 \binom{m}{k} \binom{n}{\ell}}{(m+n+2)(k+\ell+1)}, \quad 0 \leq k \leq m, \quad 0 \leq \ell \leq n, \quad (1.3)$$

where the overall factor 2 is $\Gamma(3)$ of the triangle Feynman parametrization, not an orientation multiplicity. The same 81 channels and the same tower were predicted, up to normalization, by the holomorphic-twist computation of [1]. After a factor-two correction of the source — its printed master-triangle and Appendix-B tower coefficients miss exactly $\Gamma(3) = 2$ relative to its own displayed component results — the agreement is exact on all 81 ordered output words and on the full derivative tower (Sec. 7). The comparison simultaneously adjudicates a second internal inconsistency of the source, whose compact one-loop coefficient is printed both as $-1/4$ and as $-\kappa^2$ with $\kappa^2 = 1/4$, selecting the main-text branch.

1.3 Four approaches, four statuses

The result (1.2) was attacked along four parallel routes, which have reached sharply different levels of completion; we state each status honestly, and Sec. 8 collects the open problems.

The first route is the direct *superspace Feynman-diagram* computation (Sec. 4): the Schwinger–Dyson reduction of every ordered letter pair, the occurrence-resolved cutting failure (1.1), and exact-arithmetic evaluation of all 81 channels.

Remark 1.1. The superspace route is complete and is the accepted derivation of (1.2): 81 channels computed exactly, 29 nonzero and 52 exactly zero, with no unresolved coefficient row. An adversarial census of potentially omitted graph sectors remains open as a completeness question; it does not touch any accepted coefficient (Sec. 7 and App. A).

The second route is the *component Feynman-diagram* formulation (Sec. 5), which has locked the exact Euclidean $\mathcal{N} = 4$ component actions, the BV–BRST data [4], and one exact physical channel: the bottom projection of the matter-fermion/antichiral-scalar composite,

$$Q_-(\psi_{r+}^A \tilde{\phi}_s^B) \Big|_{1\text{-loop}} = -\frac{\sqrt{2} \hbar g^2}{32\pi^2} \delta_{rs} f^{ACD} f^{BCE} \tilde{\chi}_\alpha^D \tilde{\chi}^{E\dot{\alpha}}, \quad (1.4)$$

whose dotted-gaugino bilinear is color-symmetric, so every antisymmetric color word annihilates it.

Remark 1.2. The component route is partial. Equation (1.4) is an exact projection of the superfield result, not an independent component derivation: the would-be ordinary two-Yukawa parent graph does not exist because the mixed propagators vanish identically. The complete component Euler operators and the announced termwise off-shell Noether checks, the complete component Feynman rules, an independent all-pairs component sweep, and the Lorentzian vector integration cycle all remain open (Sec. 8).

The third route is a non-perturbative *heat-kernel* derivation (Sec. 6) in the spirit of [6]: regulate the one-loop fluctuation determinant by the semigroup of a single elliptic operator and read the anomaly off a short-time coefficient.

Remark 1.3. The heat-kernel route has produced exact, machine-verified auxiliary lemmas — the ordered-simplex moment $1/((p+1)(p+q+2))$, the second-order semigroup/Duhamel coefficient, the vanishing short-time limit at fixed nonzero shift, and consistency with the master integral of (1.1) — but the full derivation is blocked: the proposed regulator’s matter blocks give $16\Box\mathcal{P}_+$ and its vector block gives $+\Box$, against the declared $-\Box$ generator, and applying different powers of the operator on different blocks is not one functional calculus. The accepted coefficient does not come from this route.

The fourth route is the *holomorphic-twist* comparison (Sec. 7): translate the external one-loop prediction of [1] into the present alphabet and compare, word by word and jet by jet.

Remark 1.4. The holomorphic-twist agreement is settled as an external comparison only: the independently sealed coefficient table matches the factor-two-corrected prediction of [1] on all 81 ordered output words and all checked tower coefficients. Equality of totals is a cross-check, not an internal completeness proof — omitted contributions could cancel in the sum — and the 17-candidate adversarial census remains open. The twisted source is also not one-loop exact: its own algebra requires higher-loop corrections.

1.4 Organization

Section 2 fixes notation and conventions. Section 3 presents the gauge fixing and BV–BRST quantization. Section 4 contains the superspace derivation of (1.2), the 81-channel census, and the kernel (1.3). Section 5 treats the component formulation and proves (1.4). Section 6 presents the heat-kernel auxiliary lemmas and the blocker. Section 7 gives the holomorphic-twist comparison and its admissibility verdict. Section 8 summarizes and lists the open problems. Appendix A records the verification status: the machine-locked exact-arithmetic checks and the independent review trail.

2 Conventions: spinors, superspace and superfields

We use two-component Weyl spinors throughout, in both Lorentzian and Euclidean signature; textbook references for the superspace conventions are [3, 2]. Lorentzian vector indices are $\mu, \nu, \rho = 0, 1, 2, 3$, Euclidean vector indices are $m, n, r = 1, 2, 3, 4$, undotted spinor indices are $\alpha, \beta, \gamma = 1, 2$, and dotted spinor indices are $\dot{\alpha}, \dot{\beta}, \dot{\gamma} = \dot{1}, \dot{2}$. The index alphabet already distinguishes the two signatures; where confusion is possible we add the label L (Lorentzian) or E (Euclidean) explicitly. All commutators are graded, $[A, B] := AB - (-1)^{|A||B|}BA$, with $|X| \in \{0, 1\}$ the Grassmann parity.

2.1 Lorentzian spinors

We use the mostly-plus metric and the orientation

$$\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1), \quad v_\mu = \eta_{\mu\nu}v^\nu, \quad \epsilon^{0123} = +1. \quad (2.1)$$

The two-component epsilon tensors and index raising and lowering are

$$\epsilon^{12} = \epsilon^{\dot{1}\dot{2}} = +1, \quad \epsilon_{12} = \epsilon_{\dot{1}\dot{2}} = -1, \quad \psi^\alpha = \epsilon^{\alpha\beta}\psi_\beta, \quad \bar{\psi}^{\dot{\alpha}} = \epsilon^{\dot{\alpha}\dot{\beta}}\bar{\psi}_{\dot{\beta}}, \quad (2.2)$$

with $\epsilon^{\alpha\beta}\epsilon_{\beta\gamma} = \delta^\alpha_\gamma$, $\epsilon_{\alpha\beta}\epsilon^{\beta\gamma} = \delta_\alpha^\gamma$, and the same relations with dotted indices. Spinor contractions follow the *asymmetric* summation convention: undotted indices are contracted from

upper left to lower right, dotted indices from lower left to upper right,

$$\xi\chi := \xi^\alpha\chi_\alpha, \quad \bar{\xi}\bar{\chi} := \bar{\xi}_{\dot{\alpha}}\bar{\chi}^{\dot{\alpha}}, \quad \xi^\alpha\chi_\alpha = -\xi_\alpha\chi^\alpha, \quad \bar{\xi}^{\dot{\alpha}}\bar{\chi}_{\dot{\alpha}} = -\bar{\xi}_{\dot{\alpha}}\bar{\chi}^{\dot{\alpha}}. \quad (2.3)$$

For Grassmann-odd spinors this gives $\xi\chi = \chi\xi$ and $\bar{\xi}\bar{\chi} = \bar{\chi}\bar{\xi}$. With $\sigma^1, \sigma^2, \sigma^3$ the standard Pauli matrices, the sigma matrices are

$$(\sigma^\mu)_{\alpha\dot{\alpha}} = (\mathbf{1}_2, \sigma^1, \sigma^2, \sigma^3)_{\alpha\dot{\alpha}}, \quad (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} := \epsilon^{\alpha\beta}\epsilon^{\dot{\alpha}\dot{\beta}}(\sigma^\mu)_{\beta\dot{\beta}} = (\mathbf{1}_2, -\sigma^1, -\sigma^2, -\sigma^3)^{\dot{\alpha}\alpha}, \quad (2.4)$$

and obey the Clifford relations

$$(\sigma^\mu)_{\alpha\dot{\alpha}}(\bar{\sigma}^\nu)^{\dot{\alpha}\beta} + (\sigma^\nu)_{\alpha\dot{\alpha}}(\bar{\sigma}^\mu)^{\dot{\alpha}\beta} = -2\eta^{\mu\nu}\delta_{\alpha}^{\beta}, \quad (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha}(\sigma^\nu)_{\alpha\dot{\beta}} + (\bar{\sigma}^\nu)^{\dot{\alpha}\alpha}(\sigma^\mu)_{\alpha\dot{\beta}} = -2\eta^{\mu\nu}\delta_{\dot{\beta}}^{\dot{\alpha}}. \quad (2.5)$$

The (anti-)self-dual Lorentz generators are

$$(\sigma^{\mu\nu})_{\alpha}^{\beta} := \frac{1}{4}[(\sigma^\mu)_{\alpha\dot{\gamma}}(\bar{\sigma}^\nu)^{\dot{\gamma}\beta} - (\sigma^\nu)_{\alpha\dot{\gamma}}(\bar{\sigma}^\mu)^{\dot{\gamma}\beta}], \quad (\bar{\sigma}^{\mu\nu})^{\dot{\alpha}}_{\dot{\beta}} := \frac{1}{4}[(\bar{\sigma}^\mu)^{\dot{\alpha}\gamma}(\sigma^\nu)_{\gamma\dot{\beta}} - (\bar{\sigma}^\nu)^{\dot{\alpha}\gamma}(\sigma^\mu)_{\gamma\dot{\beta}}], \quad (2.6)$$

with components $\sigma^{0i} = -\frac{1}{2}\sigma^i$, $\bar{\sigma}^{0i} = +\frac{1}{2}\sigma^i$ and $\sigma^{ij} = \bar{\sigma}^{ij} = -\frac{i}{2}\epsilon^{ijk}\sigma^k$.

Weyl versus Majorana spinors. The formalism is strictly two-component. In Lorentzian signature the barred spinor is the Hermitian conjugate of the unbarred one,

$$\bar{\chi}^{\dot{\alpha}} := (\chi^\alpha)^\dagger; \quad (2.7)$$

this is the Majorana reality condition: a Majorana four-spinor is the pair $(\chi_\alpha, \bar{\chi}^{\dot{\alpha}})$, while a Weyl spinor is half of this data and is the fundamental object here. In particular the Lorentzian supercharges obey $(Q_\alpha)^\dagger = \bar{Q}_{\dot{\alpha}}$.

2.2 Euclidean spinors

The Wick rotation keeps the spatial coordinates and the Grassmann coordinates fixed and rotates the time coordinate; the generators acquire phases,

$$x_E^i = x_L^i, \quad x_E^4 = ix_L^0, \quad P_i^E = -iP_i^L, \quad P_4^E = P_L^0 = H, \quad Q_\alpha^E = -iQ_\alpha^L, \quad \bar{Q}_{\dot{\alpha}}^E = -i\bar{Q}_{\dot{\alpha}}^L. \quad (2.8)$$

The Euclidean sigma matrices and Clifford relations are

$$(\sigma^m)_{\alpha\dot{\alpha}} = (-i\sigma^1, -i\sigma^2, -i\sigma^3, \mathbf{1}_2)_{\alpha\dot{\alpha}}, \quad (\bar{\sigma}^m)^{\dot{\alpha}\alpha} := \epsilon^{\alpha\beta}\epsilon^{\dot{\alpha}\dot{\beta}}(\sigma^m)_{\beta\dot{\beta}} = (+i\sigma^1, +i\sigma^2, +i\sigma^3, \mathbf{1}_2)^{\dot{\alpha}\alpha}, \quad (2.9)$$

$$(\sigma^m)_{\alpha\dot{\alpha}}(\bar{\sigma}^n)^{\dot{\alpha}\beta} + (\sigma^n)_{\alpha\dot{\alpha}}(\bar{\sigma}^m)^{\dot{\alpha}\beta} = 2\delta^{mn}\delta_{\alpha}^{\beta}, \quad (2.10)$$

so that $(\bar{\sigma}^m)^{\dot{\alpha}\alpha} = ((\sigma^m)_{\alpha\dot{\alpha}})^\dagger$ as matrices. The (anti-)self-dual rotation generators $\sigma^{mn}, \bar{\sigma}^{mn}$ are defined as in (2.6) with $\mu \rightarrow m$, and have components

$$\sigma^{ij} = \bar{\sigma}^{ij} = \frac{i}{2}\epsilon^{ijk}\sigma^k, \quad \sigma^{4i} = +\frac{i}{2}\sigma^i, \quad \bar{\sigma}^{4i} = -\frac{i}{2}\sigma^i. \quad (2.11)$$

The $SO(4) \simeq SU(2) \times SU(2)$ structure. In Euclidean signature the undotted index α and the dotted index $\dot{\alpha}$ are fundamental indices of two *independent* $SU(2)$ factors: $(\sigma^{mn})_{\alpha}^{\beta}$ acts only on undotted spinors and $(\bar{\sigma}^{mn})^{\dot{\alpha}}_{\dot{\beta}}$ only on dotted ones. From (2.11), σ^{mn} and $\bar{\sigma}^{mn}$ are the (anti-)self-dual generators of $\mathfrak{so}(4) = \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-$, interchanged by $\sigma \leftrightarrow \bar{\sigma}$; the vector $(\sigma^m)_{\alpha\dot{\alpha}}$ is the **(2, 2)**.

No Euclidean conjugation. In Euclidean signature no dagger relation is imposed between Q_α and $\bar{Q}_{\dot{\alpha}}$: the two Weyl sectors are independent. Consequently the chiral and antichiral fields are independent Euclidean fields, not conjugates. This holds for the matter multiplets $(\Phi, \tilde{\Phi})$ and for the two gaugini $(\lambda_\alpha, \tilde{\lambda}_{\dot{\alpha}})$ alike, and it is used throughout this paper: the superspace calculation of Section 4 and the component reduction of Section 5 treat untilded and tilded fields as independent integration variables.

Remark 2.1. Standing convention: *no Euclidean conjugation relates Φ to $\tilde{\Phi}$ (nor λ to $\tilde{\lambda}$)*. The Lorentzian identifications $\tilde{\Phi} = \bar{\Phi}$, $\tilde{\lambda} = \bar{\lambda}$ are Wick-transported relations only; no intrinsic Euclidean reality condition is imposed anywhere in this paper.

The plus/minus frame. In the Euclidean computation of Sections 4–5 the undotted $SU(2)$ index is resolved in the frame $\alpha = (+, -)$: an undotted spinor X_α has components $X_\pm$, and correspondingly the gauge-covariant superspace derivatives ∇_α (undotted) and $\tilde{\nabla}_{\dot{\alpha}}$ (dotted) have frame components

$$\nabla_\pm := \nabla_{\alpha=\pm}, \quad \tilde{\nabla}_\pm := \tilde{\nabla}_{\dot{\alpha}=\pm}. \quad (2.12)$$

Here ∇_α carries the chiral gauge connection and $\tilde{\nabla}_{\dot{\alpha}}$ the antichiral one, adapted to the independent Euclidean antichiral fields. In the one-loop calculation every remaining spinor index is written dotted, the undotted $SU(2)$ index being carried by the symbol $+$: the vector appears through the bispinor map $(\sigma^m)_{+\dot{\alpha}}$ and $(\sigma^{mn})_{++}$ generates the undotted (plus) $SU(2)$.

Remark 2.2. Only the frame and the names $\nabla_\pm, \tilde{\nabla}_\pm$ are fixed here. The connection-level definitions of the gauge-covariant derivatives are deferred to Section 3, their use in the letter algebra to Section 4.

2.3 Flat $N = 1$ superspace

Flat $N = 1$ superspace has coordinates $z = (x^m, \theta^\alpha, \bar{\theta}_{\dot{\alpha}})$ (or $(x^\mu, \theta, \bar{\theta})$ in Lorentzian signature); the odd coordinates are distinct from the supersymmetry transformation parameters $\varepsilon, \bar{\varepsilon}$. We write

$$\theta \sigma^m \bar{\theta} := \theta^\alpha (\sigma^m)_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}}, \quad \theta \sigma^\mu \bar{\theta} := \theta^\alpha (\sigma^\mu)_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}}. \quad (2.13)$$

Odd derivatives are *left* derivatives, $\partial_\alpha := \vec{\partial}/\partial\theta^\alpha$, $\bar{\partial}^{\dot{\alpha}} := \vec{\partial}/\partial\bar{\theta}_{\dot{\alpha}}$, $\bar{\partial}_{\dot{\alpha}} := \epsilon_{\dot{\alpha}\dot{\beta}} \bar{\partial}^{\dot{\beta}}$, obeying the graded Leibniz rule and

$$\{\partial_\alpha, \theta^\beta\} = \delta_\alpha^\beta, \quad \{\bar{\partial}^{\dot{\alpha}}, \bar{\theta}_{\dot{\beta}}\} = \delta^{\dot{\alpha}}_{\dot{\beta}}, \quad \{\bar{\partial}_{\dot{\alpha}}, \bar{\theta}^{\dot{\beta}}\} = -\delta_{\dot{\alpha}}^{\dot{\beta}}. \quad (2.14)$$

Remark 2.3. The minus sign in the last anticommutator of (2.14) is a standard source of sign errors; it traces back to $\bar{\partial}_{\dot{\alpha}} = \epsilon_{\dot{\alpha}\dot{\beta}} \bar{\partial}^{\dot{\beta}}$ with $\epsilon_{12} = -1$.

A supersymmetry transformation acts on the coordinates as

$$\theta'^\alpha = \theta^\alpha + \varepsilon^\alpha, \quad \bar{\theta}'_{\dot{\alpha}} = \bar{\theta}_{\dot{\alpha}} + \bar{\varepsilon}_{\dot{\alpha}}, \quad x'^\mu = x^\mu + i\varepsilon^\alpha (\sigma^\mu)_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} - i\theta^\alpha (\sigma^\mu)_{\alpha\dot{\beta}} \bar{\varepsilon}^{\dot{\beta}}. \quad (2.15)$$

We use the same symbols for the Noether charges and for their realizations as differential operators on superspace, the difference being clear from context. With $P_\mu = -i\partial_\mu$ the Lorentzian supercharges and supersymmetric covariant derivatives are

$$Q_\alpha = -i\partial_\alpha + (\sigma^\mu)_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_\mu, \quad \bar{Q}_{\dot{\alpha}} = -i\bar{\partial}_{\dot{\alpha}} - \theta^\beta (\sigma^\mu)_{\beta\dot{\alpha}} \partial_\mu, \quad (2.16)$$

$$\mathcal{D}_\alpha = \partial_\alpha - i(\sigma^\mu)_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_\mu, \quad \bar{\mathcal{D}}_{\dot{\alpha}} = \bar{\partial}_{\dot{\alpha}} + i\theta^\beta (\sigma^\mu)_{\beta\dot{\alpha}} \partial_\mu, \quad (2.17)$$

where $\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\alpha}}$ anticommute with all supercharges. Their algebras are

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2i(\sigma^\mu)_{\alpha\dot{\beta}} \partial_\mu = -2(\sigma^\mu)_{\alpha\dot{\beta}} P_\mu, \quad \{\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\beta}}\} = 2i(\sigma^\mu)_{\alpha\dot{\beta}} \partial_\mu = -2(\sigma^\mu)_{\alpha\dot{\beta}} P_\mu, \quad (2.18)$$

with vanishing same-chirality anticommutators. The same relations hold for the Hilbert-space Noether charges; the normalization is fixed by Hamiltonian positivity, $\{Q_1, (Q_1)^\dagger\} + \{Q_2, (Q_2)^\dagger\} = 4H$, and the central charge vanishes. The Lorentzian chiral coordinates are

$$y^\mu := x^\mu - i\theta\sigma^\mu\bar{\theta}, \quad \tilde{y}^\mu := x^\mu + i\theta\sigma^\mu\bar{\theta} (= \bar{y}^\mu), \quad \bar{\mathcal{D}}_{\dot{\alpha}}y^\mu = 0, \quad \mathcal{D}_\alpha\tilde{y}^\mu = 0. \quad (2.19)$$

After Wick rotation the coordinate momentum is $P_m = -\partial_m$ and

$$\begin{aligned} Q_\alpha &= -\partial_\alpha + (\sigma^m)_{\alpha\dot{\beta}}\bar{\theta}^{\dot{\beta}}\partial_m, & \bar{Q}_{\dot{\alpha}} &= -\bar{\partial}_{\dot{\alpha}} - \theta^\beta(\sigma^m)_{\beta\dot{\alpha}}\partial_m, \\ \mathcal{D}_\alpha &= \partial_\alpha + (\sigma^m)_{\alpha\dot{\beta}}\bar{\theta}^{\dot{\beta}}\partial_m, & \bar{\mathcal{D}}_{\dot{\alpha}} &= \bar{\partial}_{\dot{\alpha}} - \theta^\beta(\sigma^m)_{\beta\dot{\alpha}}\partial_m, \end{aligned} \quad (2.20)$$

with algebras

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = +2(\sigma^m)_{\alpha\dot{\beta}}\partial_m = -2(\sigma^m)_{\alpha\dot{\beta}}P_m, \quad \{\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\beta}}\} = -2(\sigma^m)_{\alpha\dot{\beta}}\partial_m = +2(\sigma^m)_{\alpha\dot{\beta}}P_m. \quad (2.21)$$

Same-chirality anticommutators vanish in both signatures, all mixed Q - $\mathcal{D}$ anticommutators vanish, and everything commutes with the momentum. The Euclidean chiral coordinates are

$$y^m := x^m + \theta\sigma^m\bar{\theta}, \quad \tilde{y}^m := x^m - \theta\sigma^m\bar{\theta}, \quad \bar{\mathcal{D}}_{\dot{\alpha}}y^m = 0, \quad \mathcal{D}_\alpha\tilde{y}^m = 0. \quad (2.22)$$

Remark 2.4 (Sign trap). In Lorentzian signature the Q - and $\mathcal{D}$ -algebras coincide, $\{Q, \bar{Q}\} = \{\mathcal{D}, \bar{\mathcal{D}}\} = -2\sigma^\mu P_\mu$; in Euclidean signature they differ by a relative minus sign, $\{Q, \bar{Q}\} = -2\sigma^m P_m$ versus $\{\mathcal{D}, \bar{\mathcal{D}}\} = +2\sigma^m P_m$. All sign conventions of the loop computation in Sections 4–5 are tied to (2.21).

2.4 Theta expansions and superfields

The basic monomials and their normalizations are

$$\theta^2 := \theta^\alpha\theta_\alpha = -2\theta^1\theta^2, \quad \bar{\theta}^2 := \bar{\theta}_{\dot{\alpha}}\bar{\theta}^{\dot{\alpha}} = 2\bar{\theta}_{\dot{1}}\bar{\theta}_{\dot{2}}, \quad \theta^\alpha\theta^\beta = -\frac{1}{2}\epsilon^{\alpha\beta}\theta^2, \quad \bar{\theta}_{\dot{\alpha}}\bar{\theta}_{\dot{\beta}} = -\frac{1}{2}\epsilon_{\dot{\alpha}\dot{\beta}}\bar{\theta}^2. \quad (2.23)$$

With $\mathcal{D}^2 := \mathcal{D}^\alpha\mathcal{D}_\alpha$ and $\bar{\mathcal{D}}^2 := \bar{\mathcal{D}}_{\dot{\alpha}}\bar{\mathcal{D}}^{\dot{\alpha}}$ one has, in either signature,

$$\mathcal{D}^2\theta^2 = -4, \quad \bar{\mathcal{D}}^2\bar{\theta}^2 = -4. \quad (2.24)$$

Writing $X| := X|_{\theta=\bar{\theta}=0}$, the component projectors are

$$[X]_F := -\frac{1}{4}\mathcal{D}^2X|, \quad [\tilde{X}]_{\tilde{F}} := -\frac{1}{4}\bar{\mathcal{D}}^2\tilde{X}|, \quad [Y]_D := \frac{1}{16}\mathcal{D}^2\bar{\mathcal{D}}^2Y| \quad (\bar{\mathcal{D}}^2 \text{ acts first}), \quad (2.25)$$

and the full superspace measure is $\int d^4x d^4\theta Y := \int d^4x [Y]_D$.

Remark 2.5. A general (unconstrained) superfield enters this paper only through the projectors (2.25); its full fourteen-component expansion is never used. The vector multiplet is always reduced to Wess–Zumino gauge, (2.33) below.

A chiral superfield Φ and an antichiral superfield $\tilde{\Phi}$ obey

$$\bar{\mathcal{D}}_{\dot{\alpha}}\Phi = 0, \quad \mathcal{D}_\alpha\tilde{\Phi} = 0, \quad (2.26)$$

and have components (ϕ, ψ_α, F) and $(\tilde{\phi}, \tilde{\psi}_{\dot{\alpha}}, \tilde{F})$ given by

$$\phi := \Phi|, \quad \psi_\alpha := \frac{1}{\sqrt{2}}\mathcal{D}_\alpha\Phi|, \quad F := -\frac{1}{4}\mathcal{D}^2\Phi|; \quad \tilde{\phi} := \tilde{\Phi}|, \quad \tilde{\psi}_{\dot{\alpha}} := \frac{1}{\sqrt{2}}\bar{\mathcal{D}}_{\dot{\alpha}}\tilde{\Phi}|, \quad \tilde{F} := -\frac{1}{4}\bar{\mathcal{D}}^2\tilde{\Phi}|. \quad (2.27)$$

In Lorentzian signature $\tilde{\Phi} = \bar{\Phi}$ and $(\tilde{\phi}, \tilde{\psi}, \tilde{F}) = (\bar{\phi}, \bar{\psi}, \bar{F})$; in Euclidean signature $\tilde{\Phi}$ is independent of Φ (Remark 2.1). Their chiral-coordinate expansions are

$$\Phi(y, \theta) = \phi(y) + \sqrt{2}\theta^\alpha\psi_\alpha(y) + \theta^2F(y), \quad \tilde{\Phi}(\tilde{y}, \bar{\theta}) = \tilde{\phi}(\tilde{y}) + \sqrt{2}\bar{\theta}_{\dot{\alpha}}\tilde{\psi}^{\dot{\alpha}}(\tilde{y}) + \bar{\theta}^2\tilde{F}(\tilde{y}), \quad (2.28)$$

with $y, \tilde{y}$ from (2.19) or (2.22). Expanding in x -space gives, in Euclidean signature,

$$\Phi = \phi + \sqrt{2}\theta\psi + \theta^2 F + \theta\sigma^m\bar{\theta}\partial_m\phi - \frac{1}{\sqrt{2}}\theta^2(\partial_m\psi^\alpha)(\sigma^m)_{\alpha\dot{\beta}}\bar{\theta}^{\dot{\beta}} + \frac{1}{4}\theta^2\bar{\theta}^2\partial^m\partial_m\phi, \quad (2.29)$$

$$\tilde{\Phi} = \tilde{\phi} + \sqrt{2}\bar{\theta}\tilde{\psi} + \bar{\theta}^2\tilde{F} - \theta\sigma^m\bar{\theta}\partial_m\tilde{\phi} + \frac{1}{\sqrt{2}}\theta^\alpha(\sigma^m)_{\alpha\dot{\beta}}(\partial_m\tilde{\psi}^{\dot{\beta}})\bar{\theta}^2 + \frac{1}{4}\theta^2\bar{\theta}^2\partial^m\partial_m\tilde{\phi}, \quad (2.30)$$

where $\partial^m\partial_m$ is the Euclidean Laplacian; the Lorentzian counterparts follow from (2.28) with (2.19).

The vector multiplet is described by the prepotential $\mathcal{V} = \mathcal{V}^A T_A$ with $[T_A, T_B] = if_{AB}{}^C T_C$ in an orthonormal basis, $\kappa_{AB} = \delta_{AB}$. Wess–Zumino gauge is the residual gauge choice

$$\mathcal{V}| = \theta\mathcal{V}| = \bar{\theta}\mathcal{V}| = 0, \quad (2.31)$$

in which the prepotential reduces to

$$\mathcal{V}_{\text{WZ}} = -2\theta\sigma^\mu\bar{\theta}A_\mu + 2i\theta^2\bar{\theta}_{\dot{\alpha}}\bar{\lambda}^{\dot{\alpha}} - 2i\bar{\theta}^2\theta^\alpha\lambda_\alpha + \theta^2\bar{\theta}^2 D \quad (\text{Lorentzian}), \quad (2.32)$$

$$\mathcal{V}_{\text{WZ}} = -2i\theta\sigma^m\bar{\theta}A_m + 2i\theta^2\bar{\theta}_{\dot{\alpha}}\tilde{\lambda}^{\dot{\alpha}} - 2i\bar{\theta}^2\theta^\alpha\lambda_\alpha + \theta^2\bar{\theta}^2 D \quad (\text{Euclidean}), \quad (2.33)$$

with component fields $(A_m, \lambda_\alpha, \tilde{\lambda}_{\dot{\alpha}}, D)$; note that the Euclidean prepotential contains the *independent* antichiral gaugino $\tilde{\lambda}_{\dot{\alpha}}$. The gauge field is the ordered-commutator projection

$$A_\mu = \frac{1}{8}(\bar{\sigma}_\mu)^{\dot{\beta}\alpha}[\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\beta}}]_{\text{ord}}\mathcal{V}| \quad (\text{Lorentzian}), \quad A_m = \frac{i}{8}(\bar{\sigma}_m)^{\dot{\beta}\alpha}[\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\beta}}]_{\text{ord}}\mathcal{V}| \quad (\text{Euclidean}), \quad (2.34)$$

where $[\cdot, \cdot]_{\text{ord}}$ keeps the spinor derivatives in the displayed order. The gaugino field-strength superfields are

$$\mathcal{W}_\alpha := -\frac{1}{8}\bar{\mathcal{D}}^2[e^{-\mathcal{V}}(\mathcal{D}_\alpha e^\mathcal{V})], \quad \widetilde{\mathcal{W}}_{\dot{\alpha}} := +\frac{1}{8}\mathcal{D}^2[e^\mathcal{V}(\bar{\mathcal{D}}_{\dot{\alpha}} e^{-\mathcal{V}})], \quad (2.35)$$

chiral and antichiral respectively, $\bar{\mathcal{D}}_{\dot{\alpha}}\mathcal{W}_\beta = 0$, $\mathcal{D}_\alpha\widetilde{\mathcal{W}}_{\dot{\beta}} = 0$, with component projections

$$\lambda_\alpha := i\mathcal{W}_\alpha|, \quad \tilde{\lambda}_{\dot{\alpha}} := -i\widetilde{\mathcal{W}}_{\dot{\alpha}}|, \quad D := -\frac{1}{2}\mathcal{D}^\alpha\mathcal{W}_\alpha| = +\frac{1}{2}\bar{\mathcal{D}}^{\dot{\alpha}}\widetilde{\mathcal{W}}_{\dot{\alpha}}|. \quad (2.36)$$

In Lorentzian signature $\widetilde{\mathcal{W}}_{\dot{\alpha}} = \bar{\mathcal{W}}_{\dot{\alpha}}$ and $\tilde{\lambda}_{\dot{\alpha}} = \bar{\lambda}_{\dot{\alpha}}$; in Euclidean signature $(\mathcal{W}_\alpha, \widetilde{\mathcal{W}}_{\dot{\alpha}})$ and $(\lambda_\alpha, \tilde{\lambda}_{\dot{\alpha}})$ are independent.

2.5 Chiral and vector representations

Gauge transformations are generated by a chiral parameter Λ and an independent antichiral parameter $\bar{\Lambda}$,

$$h := e^{i\Lambda}, \quad \bar{h} := e^{i\bar{\Lambda}}, \quad \bar{\mathcal{D}}_{\dot{\alpha}}\Lambda = 0, \quad \mathcal{D}_\alpha\bar{\Lambda} = 0. \quad (2.37)$$

In the *chiral representation* superfields are functions of the chiral coordinates (2.19)–(2.22), and

$$\Phi' = h\Phi, \quad \tilde{\Phi}' = \tilde{\Phi}\bar{h}^{-1}, \quad (e^\mathcal{V})' = \bar{h}e^\mathcal{V}h^{-1}. \quad (2.38)$$

Thus the bridge $e^\mathcal{V}$ carries the entire gauge field in this frame: the combination $\tilde{\Phi}e^\mathcal{V}\Phi$ is gauge invariant, the gauge connection appears only through $e^{-\mathcal{V}}(\mathcal{D}_\alpha e^\mathcal{V})$ in the undotted spinor derivative, and the field strengths (2.35) transform homogeneously,

$$\mathcal{W}'_\alpha = h\mathcal{W}_\alpha h^{-1}, \quad \widetilde{\mathcal{W}}'_{\dot{\alpha}} = \bar{h}\widetilde{\mathcal{W}}_{\dot{\alpha}}\bar{h}^{-1}. \quad (2.39)$$

In the *vector representation* one works instead on the full $(x, \theta, \bar{\theta})$ superspace: all superspace derivatives carry connections, the matter fields are covariantly chiral, and gauge transformations act by a single real group element, so that the complexified parameters $h, \bar{h}$ and the explicit bridge disappear from the transformation laws. The two representations are related by a factorization of the bridge $e^\mathcal{V}$ and are physically equivalent; we use the chiral representation in what follows.

Remark 2.6. This subsection states the two frames at the superfield level only. The connection-level definitions of the gauge-covariant derivatives $\nabla_\alpha, \tilde{\nabla}_{\dot{\alpha}}$ of (2.12), the bridge factorization, the gauge action, and the BV–BRST formulation are deferred to Section 3.

3 Supersymmetric gauge theory and BV–BRST quantization

This section collects the classical and quantum input for the one-loop computation of Sections 4–5: the general $\mathcal{N} = 1$ gauge–chiral superfield action in the chiral representation (Section 3.1), its vector-frame reformulation (Section 3.2) and component form (Section 3.3), the $\mathcal{N} = 1, 2, 4$ super–Yang–Mills specializations (Section 3.4), and the Batalin–Vilkovisky (BV) quantization [4] with the Fermi–Feynman supergraph rules and the dimensional-reduction regulator (Sections 3.5–3.7). Two global conventions govern everything below. First, the gauge coupling is absorbed into the connection, $A_M = A_M^A T_A$, $\mathcal{D}_M = \partial_M - iA_M$; the canonical normalization is restored by $A_M = gA_M^{\text{can}}$ (similarly for $\lambda, \phi, \psi, \mathcal{D}, F$), so every non-topological density carries an overall factor g^{-2} : vertices are g -independent up to that common factor, while propagators carry g^2 . The compact gauge algebra is $[T_A, T_B] = if_{AB}^C T_C$ with orthonormal Killing metric $\kappa_{AB} = \delta_{AB}$, and $f^{ABC} := \delta^{CD} f_{AB}^D$ is totally antisymmetric. Second, in Euclidean signature every tilded field is independent of its untilded partner before a middle-dimensional integration contour is chosen; on the Lorentzian contour the tilde becomes the formal conjugate. The computation of this paper is Euclidean; Lorentzian ancestors are quoted where instructive.

3.1 Gauge and chiral superfields in the chiral representation

Chiral and antichiral matter superfields $\Phi^I, \tilde{\Phi}_I$, in a representation $\mathcal{R}_{\text{mat}}$ and its dual, obey $\bar{\mathcal{D}}_{\dot{\alpha}}\Phi^I = 0$, $\mathcal{D}_\alpha\tilde{\Phi}_I = 0$. Gauge transformations are generated by a chiral parameter $\Lambda = \Lambda^A T_A$ and an independent antichiral $\tilde{\Lambda}$ (on the Lorentzian contour $\tilde{\Lambda} = \bar{\Lambda}$):

$$h := e^{i\Lambda}, \quad \tilde{h} := e^{i\tilde{\Lambda}}, \quad \Phi' = h\Phi, \quad \tilde{\Phi}' = \tilde{\Phi}\tilde{h}^{-1}. \quad (3.1)$$

The real prepotential $\mathcal{V} = \mathcal{V}^A T_A$ enters only through the bridge $e^\mathcal{V}$, which dresses the flat spinor derivatives to gauge-covariant ones:

$$(e^\mathcal{V})' = \tilde{h}e^\mathcal{V}h^{-1}, \quad \Gamma_\alpha := e^{-\mathcal{V}}\mathcal{D}_\alpha e^\mathcal{V}, \quad \tilde{\Gamma}_{\dot{\alpha}} := e^\mathcal{V}\bar{\mathcal{D}}_{\dot{\alpha}}e^{-\mathcal{V}}, \quad \nabla_\alpha := \mathcal{D}_\alpha + \Gamma_\alpha, \quad \nabla'_{\dot{\alpha}} = h\nabla_{\dot{\alpha}}h^{-1}. \quad (3.2)$$

The gaugino field-strength superfields are

$$\mathcal{W}_\alpha := -\frac{1}{8}\bar{\mathcal{D}}^2(e^{-\mathcal{V}}\mathcal{D}_\alpha e^\mathcal{V}), \quad \widetilde{\mathcal{W}}_{\dot{\alpha}} := +\frac{1}{8}\mathcal{D}^2(e^\mathcal{V}\bar{\mathcal{D}}_{\dot{\alpha}}e^{-\mathcal{V}}), \quad (3.3)$$

chiral and antichiral respectively and covariant: $\bar{\mathcal{D}}_{\dot{\alpha}}\mathcal{W}_\beta = 0$, $\mathcal{D}_\alpha\widetilde{\mathcal{W}}_{\dot{\beta}} = 0$, $\mathcal{W}'_\alpha = h\mathcal{W}_\alpha h^{-1}$, $\widetilde{\mathcal{W}}'_{\dot{\alpha}} = \tilde{h}\widetilde{\mathcal{W}}_{\dot{\alpha}}\tilde{h}^{-1}$. Engineering dimensions are $[\Phi] = 1$, $[\mathcal{W}_\alpha] = 3/2$, $[\mathcal{V}] = 0$.

With the ordered Berezin projectors of Section 2, $[X]_F := -\frac{1}{4}\mathcal{D}^2 X|$, $[\tilde{X}]_{\tilde{F}} := -\frac{1}{4}\bar{\mathcal{D}}^2 \tilde{X}|$, $[Y]_D := \frac{1}{16}\mathcal{D}^2\bar{\mathcal{D}}^2 Y|$ (bar derivatives acting first), the general local two-derivative action is determined by four independent data [2, 3]: a real gauged Kähler density $\mathcal{K}_g(\tilde{\Phi}, \Phi, e^\mathcal{V})$ of dimension 2, a chiral superpotential $W(\Phi)$ of dimension 3, a symmetric chiral kinetic matrix $f_{AB}(\Phi)$ of dimension 0, and a Fayet–Iliopoulos parameter ξ_A . The exact Euclidean action used throughout this paper is

$$S_E = - \int d^4x \left\{ [\mathcal{K}_g(\tilde{\Phi}, \Phi, e^\mathcal{V})]_D + [W(\Phi)]_F + [\widetilde{W}(\tilde{\Phi})]_{\tilde{F}} + \frac{1}{4}[f_{AB}(\Phi)\mathcal{W}^{A\alpha}\mathcal{W}_\alpha^B]_F + \frac{1}{4}[\tilde{f}_{AB}(\tilde{\Phi})\widetilde{\mathcal{W}}_{\dot{\alpha}}^A\widetilde{\mathcal{W}}^{B\dot{\alpha}}]_{\tilde{F}} + \xi_A[\mathcal{V}^A]_D \right\}, \quad (3.4)$$

the Wick image of its Lorentzian ancestor S_L , with $\mathcal{L}_E = -\mathcal{L}_L|_{\text{Wick}}$. Exact gauge invariance of (3.4) is equivalent to

$$X_A^I \partial_I W = 0, \quad X_C^I \partial_I f_{AB} - f_{CA}^D f_{DB} - f_{CB}^D f_{AD} = 0, \quad \xi_A f_{BC}^A = 0, \quad (3.5)$$

with the holomorphic Killing vectors $X_A^I := i(T_A)^I{}_J \Phi^J$. The canonical slice of (3.4) is

$$\mathcal{K}_g = \tilde{\Phi}_I (e^\nu)^I{}_J \Phi^J, \quad f_{AB} = \left(\frac{1}{g^2} - \frac{i\theta_{\text{YM}}}{8\pi^2} \right) \delta_{AB}, \quad \tilde{f}_{AB} = \left(\frac{1}{g^2} + \frac{i\theta_{\text{YM}}}{8\pi^2} \right) \delta_{AB}. \quad (3.6)$$

3.2 Chiral versus vector representation

The chiral representation is one of three gauge frames related by bridge factorization [3]. Two complexified bridges $\mathcal{B}, \tilde{\mathcal{B}}$ with

$$e^\nu = \tilde{\mathcal{B}} \mathcal{B}, \quad \mathcal{B}' = k \mathcal{B} h^{-1}, \quad \tilde{\mathcal{B}}' = \tilde{h} \tilde{\mathcal{B}} k^{-1} \quad (3.7)$$

interpolate between the chiral frame, in which matter transforms by the complexified parameters $(h, \tilde{h})$, and the vector (real) frame, in which matter $\Phi^\mathbf{V} := \mathcal{B} \Phi$ transforms by the real group element k only, $\Phi^{\mathbf{V}'} = k \Phi^\mathbf{V}$, and chirality is covariant, $\bar{\nabla}_\alpha^\mathbf{V} \Phi^\mathbf{V} = 0$. In the vector frame all superspace derivatives carry connections; imposing the conventional constraints

$$\mathcal{F}_{\alpha\beta}^\mathbf{V} = 0, \quad \mathcal{F}_{\dot{\alpha}\dot{\beta}}^\mathbf{V} = 0, \quad \mathcal{F}_{\alpha\dot{\beta}}^\mathbf{V} = 0 \quad (3.8)$$

solves them in terms of the bridges, $\nabla_\alpha^\mathbf{V} = \tilde{\mathcal{B}}^{-1} \circ \mathcal{D}_\alpha \circ \tilde{\mathcal{B}}$, $\bar{\nabla}_\alpha^\mathbf{V} = \mathcal{B} \circ \bar{\mathcal{D}}_\alpha \circ \mathcal{B}^{-1}$, with the vector derivative defined by $\{\nabla_\alpha^\mathbf{V}, \bar{\nabla}_{\dot{\beta}}^\mathbf{V}\} = -2(\sigma_E^m)_{\alpha\dot{\beta}} \mathcal{D}_m^\mathbf{V}$ in Euclidean signature. The two representations agree identically at the level of the action: the density identity

$$\tilde{\Phi}^\mathbf{V} \Phi^\mathbf{V} = \tilde{\Phi} e^\nu \Phi \quad (3.9)$$

and its field-strength analog imply $S^\mathbf{V} = S^\mathbf{C}$ term by term, the value of (3.4). The symmetric-bridge gauge $\mathcal{B} = \tilde{\mathcal{B}} = e^{\nu/2}$ makes the identification manifest.

Remark 3.1. The perturbative computation of this paper is formulated entirely in the chiral representation. In this frame the gauge orbit is parametrized by the chiral–antichiral pair $(\Lambda, \tilde{\Lambda})$, the Faddeev–Popov ghosts are chiral superfields (Section 3.5), the gauge functions $\mathcal{F}_\pm$ are chiral and antichiral, and every interaction vertex is an ordered word in the single real prepotential $\mathcal{V}$ (Section 3.6); by (3.9) no physics is lost. The vector frame enters only through the background-covariant derivatives used to define $\mathcal{F}_\pm$ and the background-field split.

3.3 Component reconstruction

The Wess–Zumino gauge is the slice in which $\mathcal{V}$ is at most quadratic; the normalized Euclidean representative is

$$\mathcal{V}^{\text{WZ}} = -2i \theta \sigma_E^m \bar{\theta} A_m + 2i \theta^2 \bar{\theta}_{\dot{\alpha}} \tilde{\lambda}^{\dot{\alpha}} - 2i \bar{\theta}^2 \theta^\alpha \lambda_\alpha + \theta^2 \bar{\theta}^2 \mathcal{D}, \quad (3.10)$$

with $\mathcal{D}$ the auxiliary field (not to be confused with the spinor derivatives $\mathcal{D}_\alpha, \bar{\mathcal{D}}_{\dot{\alpha}}$), $(\mathcal{V}^{\text{WZ}})^3 = 0$ so that $e^{\pm\mathcal{V}} = 1 \pm \mathcal{V} + \frac{1}{2} \mathcal{V}^2$, and ordinary gauge transformations as the residual group. Matter component fields are the ordered projections

$$\phi^I := \Phi^I|, \quad \psi_\alpha^I := \frac{1}{\sqrt{2}} \mathcal{D}_\alpha \Phi^I|, \quad F^I := -\frac{1}{4} \mathcal{D}^2 \Phi^I|, \quad (3.11)$$

together with the tilded partners; the vector-multiplet components are

$$\lambda_\alpha := i \mathcal{W}_\alpha|, \quad \tilde{\lambda}_{\dot{\alpha}} := -i \tilde{\mathcal{W}}_{\dot{\alpha}}|, \quad \mathcal{D} := -\frac{1}{2} \mathcal{D}^\alpha \mathcal{W}_\alpha| = +\frac{1}{2} \bar{\mathcal{D}}_{\dot{\alpha}} \tilde{\mathcal{W}}^{\dot{\alpha}}|, \quad \mathcal{D}_{(\alpha} \mathcal{W}_{\beta)}| = -i (\sigma_E^{mn})_{\alpha\beta} F_{mn}, \quad (3.12)$$

the equality of the two $\mathcal{D}$ projections being the superspace Bianchi identity. In the canonical slice (3.6) the Euclidean component density $\mathcal{L}_E = \mathcal{L}_{E,\text{matter}} + \mathcal{L}_{E,W} + \mathcal{L}_{E,\text{gauge}}$ reads

$$\begin{aligned}\mathcal{L}_{E,\text{can}} = & (\mathcal{D}_m \tilde{\phi})_I (\mathcal{D}_m \phi)^I + \tilde{\psi}_{\dot{\alpha},I} (\bar{\sigma}_E^m)^{\dot{\alpha}\alpha} (\mathcal{D}_m \psi_\alpha)^I - \tilde{F}_I F^I - \mu_A^{(0)} \mathcal{D}^A \\ & - i\sqrt{2} [\tilde{\phi}_I (T_A)^I{}_J \lambda^{A\alpha} \psi_\alpha^J - \tilde{\psi}_{\dot{\alpha},I} (T_A)^I{}_J \tilde{\lambda}^{A\dot{\alpha}} \phi^J] \\ & - W_I(\phi) F^I + \frac{1}{2} W_{IJ}(\phi) \psi^{I\alpha} \psi_\alpha^J - \tilde{W}^{,I}(\tilde{\phi}) \tilde{F}_I + \frac{1}{2} \tilde{W}^{,IJ}(\tilde{\phi}) \tilde{\psi}_{\dot{\alpha},I} \tilde{\psi}_J^{\dot{\alpha}} \\ & + \frac{1}{4g^2} F_{mn}^A F_{mn}^A + \frac{1}{g^2} \tilde{\lambda}_\alpha^A (\bar{\sigma}_E^m)^{\dot{\alpha}\alpha} (\mathcal{D}_m \lambda_\alpha)^A - \frac{1}{2g^2} \mathcal{D}^A \mathcal{D}^A + \frac{i\theta_{\text{YM}}}{64\pi^2} \epsilon_E^{mnr s} F_{mn}^A F_{rs}^A,\end{aligned}\tag{3.13}$$

with $\mu_A^{(0)} := \tilde{\phi}_I (T_A)^I{}_J \phi^J + \xi_A$ and $\epsilon_E^{1234} = +1$. The auxiliary fields enter algebraically; their equations of motion

$$F^I = -\tilde{W}^{,I}(\tilde{\phi}), \quad \tilde{F}_I = -W_I(\phi), \quad \mathcal{D}^A = -g^2 \delta^{AB} \mu_B^{(0)}\tag{3.14}$$

complete the squares of the positive bosonic contour.

Remark 3.2. The complete off-shell component map — including the general Kähler geometry of the target, the moment map, and the shifted auxiliaries — has been reconstructed in the source repository and machine-verified coefficient by coefficient (336 independent component coefficients, zero failures); only the canonical slice is displayed here. The component-field route to the anomaly (Section 5) uses this map and is partial in the sense stated there.

3.4 $\mathcal{N} = 1$, $\mathcal{N} = 2$ and $\mathcal{N} = 4$ super-Yang–Mills

Pure $\mathcal{N} = 1$ super-Yang–Mills is the vector multiplet $(A_m^A, \lambda_\alpha^A, \tilde{\lambda}_\alpha^A, \mathcal{D}^A)$ with action

$$S_E^{(1)} = -\frac{1}{4} \int d^4x ([f_{AB} \mathcal{W}^{A\alpha} \mathcal{W}_\alpha^B]_F + [\tilde{f}_{AB} \tilde{\mathcal{W}}_\alpha^A \tilde{\mathcal{W}}^{B\dot{\alpha}}]_{\tilde{F}}).\tag{3.15}$$

$\mathcal{N} = 2$ super-Yang–Mills adds one adjoint chiral multiplet $\Phi = (\phi, \psi_\alpha, F)$; closure of the second supersymmetry at the linearized level, together with gauge invariance, fixes the relative coefficient of the matter kinetic term uniquely, giving

$$S_E^{(2)} = - \int d^4x \left\{ \frac{1}{g^2} \delta_{AB} [\tilde{\Phi}^A (e^{\mathcal{V}_{\text{ad}}})^B{}_C \Phi^C]_D + \frac{1}{4} [f_{AB} \mathcal{W}^{A\alpha} \mathcal{W}_\alpha^B]_F + \frac{1}{4} [\tilde{f}_{AB} \tilde{\mathcal{W}}_\alpha^A \tilde{\mathcal{W}}^{B\dot{\alpha}}]_{\tilde{F}} \right\},\tag{3.16}$$

with $e^{\mathcal{V}_{\text{ad}}}$ the bridge in the adjoint representation. $\mathcal{N} = 4$ super-Yang–Mills, the theory studied in this paper, contains three adjoint chirals Φ_r , $r = 1, 2, 3$, an $SU(3)$ flavor triplet inside $SU(4)_R$, together with the $SU(3)$ -invariant superpotential

$$W = -\frac{\sqrt{2}}{6g^2} \varepsilon_{rst} f^{ABC} \Phi_r^A \Phi_s^B \Phi_t^C,\tag{3.17}$$

whose coefficient $-\sqrt{2}/6$ is fixed uniquely by matching the gauge Yukawa coupling. The action is

$$S_E^{(4)} = - \int d^4x \left\{ \frac{1}{g^2} \delta_{AB} \sum_{r=1}^3 [\tilde{\Phi}_r^A (e^{\mathcal{V}_{\text{ad}}})^B{}_C \Phi_r^C]_D + \frac{1}{4} [f_{AB} \mathcal{W}^{A\alpha} \mathcal{W}_\alpha^B]_F + \frac{1}{4} [\tilde{f}_{AB} \tilde{\mathcal{W}}_\alpha^A \tilde{\mathcal{W}}^{B\dot{\alpha}}]_{\tilde{F}} + [W]_F + [\tilde{W}]_{\tilde{F}} \right\}.\tag{3.18}$$

The unified $\mathcal{N} = 1, 2, 4$ form, with multiplet counts $m_1 = 0$, $m_2 = 1$, $m_4 = 3$, is the starting point of the vertex inventory of Section 3.6.

3.5 BV–BRST quantization

Because the even noncompact gauge-orbit integral diverges and the odd one vanishes, gauge fixing is mandatory, and we use the BV formalism [4]. The minimal sector consists of the physical fields and a chiral–antichiral pair of Faddeev–Popov ghost superfields $c, \tilde{c}$; the non-minimal sector consists of an antighost–Nakanishi–Lautrup doublet $(\bar{c}_+, b_+)$ of chiral superfields and its antichiral partner $(\bar{c}_-, b_-)$, with $s\bar{c}_\pm = b_\pm$, $sb_\pm = 0$. Every field X carries a Grassmann parity ϵ_X , a ghost number and an engineering dimension; its BV antifield X^* (never complex conjugation) obeys

$$\epsilon_{X^*} = \epsilon_X + 1, \quad \text{gh}(X^*) = -1 - \text{gh}(X), \quad [X^*] = d_\Sigma - [X], \quad (3.19)$$

with $d_\Sigma = 2$ for the full superspace and $d_\Sigma = 3$ for the chiral and antichiral subspaces. The complete spectrum is

X	domain	ϵ_X	$\text{gh}(X)$	$[X]$
$\mathcal{V}$	$(x, \theta, \bar{\theta})$	0	0	0
Φ^I	(y, θ)	0	0	1
$\tilde{\Phi}_I$	$(\tilde{y}, \bar{\theta})$	0	0	1
c	(y, θ)	1	+1	0
$\tilde{c}$	$(\tilde{y}, \bar{\theta})$	1	+1	0
$\bar{c}_\pm$	$(y, \theta), (\tilde{y}, \bar{\theta})$	1	−1	2
$b_\pm$	$(y, \theta), (\tilde{y}, \bar{\theta})$	0	0	2

(3.20)

The classical BRST differential s ($\epsilon = 1$, $\text{gh} = 1$) acts as

$$se^\mathcal{V} = i\tilde{c}e^\mathcal{V} - ie^\mathcal{V}c, \quad sc = ic^2, \quad s\tilde{c} = i\tilde{c}^2, \quad s\Phi = ic\Phi, \quad s\tilde{\Phi} = -i\tilde{\Phi}\tilde{c}, \quad (3.21)$$

in components $sc^C = -\frac{1}{2}f_{AB}^C c^A c^B$, and is nilpotent on the untruncated local field algebra: $s^2 = 0$ and $sS_0 = 0$. The exact transformation of the prepotential is the Bernoulli series

$$s\mathcal{V} = \frac{\text{ad}_\mathcal{V}}{1 - e^{-\text{ad}_\mathcal{V}}} (ie^{-\text{ad}_\mathcal{V}}\tilde{c} - ic) = i \sum_{p,q \geq 0} \frac{B_{p+q}}{p!q!} [(-1)^q \mathcal{V}^p \tilde{c} \mathcal{V}^q - (-1)^p \mathcal{V}^p c \mathcal{V}^q], \quad (3.22)$$

with $B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_3 = 0$.

With the ordered pairings $\langle X, Y \rangle_8 := \int d^4x [X_A Y^A]_D$ and their chiral/antichiral analogs $\langle \cdot, \cdot \rangle_\pm$, the antibracket and the minimal master action are

$$(F, G) := \sum_X \int_{\Sigma_X} \left[F \frac{\overleftarrow{\delta}}{\delta X} \frac{\vec{\delta} G}{\delta X^*} - F \frac{\overleftarrow{\delta}}{\delta X^*} \frac{\vec{\delta} G}{\delta X} \right], \quad (3.23)$$

$$S_{\min} := S_0 + \langle \mathcal{V}^*, s\mathcal{V} \rangle_8 + \langle \Phi^*, s\Phi \rangle_+ + \langle \tilde{\Phi}^*, s\tilde{\Phi} \rangle_- - \langle c^*, sc \rangle_+ - \langle \tilde{c}^*, s\tilde{c} \rangle_-, \quad (3.24)$$

supplemented by the non-minimal stabilization $S_{\text{nm}} = S_{\min} + \langle \bar{c}^*, b \rangle$. They obey the classical master equation and generate s as a Hamiltonian vector field:

$$\frac{1}{2}(S_{\min}, S_{\min}) = 0, \quad sF = (S_{\min}, F), \quad s^2 F = \frac{1}{2}((S_{\min}, S_{\min}), F) = 0. \quad (3.25)$$

For the background split $e^{\mathcal{V}_{\text{tot}}} = \tilde{\mathcal{B}}_B e^{\mathcal{V}_q} \mathcal{B}_B$ the gauge functions are the background-covariant chiral and antichiral projections of the quantum prepotential,

$$\mathcal{F}_+ := -\frac{1}{4}(\bar{\nabla}_B^\mathcal{V})^2 \mathcal{V}_q, \quad \bar{\nabla}_{B\dot{\alpha}}^\mathcal{V} \mathcal{F}_+ = 0, \quad \mathcal{F}_- := -\frac{1}{4}(\nabla_B^\mathcal{V})^2 \mathcal{V}_q, \quad \nabla_{B\dot{\alpha}}^\mathcal{V} \mathcal{F}_- = 0, \quad [\mathcal{F}_\pm] = 1. \quad (3.26)$$

With the collective pairing of the chiral/antichiral sectors, the standard gauge fermion and its BRST variation are

$$\Psi := \langle \bar{c}, \mathcal{F} \rangle - \frac{1}{2} \langle \bar{c}, \mathcal{Y}b \rangle, \quad s\Psi = \langle b, \mathcal{F} \rangle - \langle \bar{c}, \mathcal{M}^{\text{FP}} c^{\text{pair}} \rangle - \frac{1}{2} \langle b, \mathcal{Y}b \rangle + \frac{1}{2} \langle \bar{c}, (s\mathcal{Y})b \rangle, \quad (3.27)$$

where $\mathcal{Y}$ is an even, invertible, local, background-covariant kernel of dimension -1 , $c^{\text{pair}} := (c, \bar{c})$ dressed to the background, and the Faddeev–Popov operator $\mathcal{M}^{\text{FP}}$ (even, ghost-number zero, dimension 1) is the gauge tangent of the quantum field, $s\mathcal{F} = \mathcal{M}^{\text{FP}} c^{\text{pair}}$. Integrating out the non-minimal doublets produces the Faddeev–Popov Berezinian

$$\mathfrak{F}^{\text{FP}} = \text{Ber}(\mathcal{M}_0^{-1} \mathcal{M}^{\text{FP}}), \quad (3.28)$$

evaluated on the quotient by the residual stabilizer, and, whenever the kernel $\mathcal{Y}$ is itself gauge averaged, a Nielsen–Kallosh factor fixed by the requirement that the multiplier Gaussian times it reproduce the declared gauge-average normalization. At vanishing external antifield sources the gauge-fixed action is $S_\Psi = S_0 + s\Psi$.

The quantum master action $W = S_{\min} + \sum_{n \geq 1} \hbar^n M_n$ must satisfy the quantum master equation

$$\frac{1}{2}(W, W) - i\hbar\Delta W = 0 \quad (\text{Lorentzian}), \quad \frac{1}{2}(W, W) - \hbar\Delta W = 0 \quad (\text{Euclidean}), \quad (3.29)$$

with Δ the BV Laplacian of the measure; a nonzero left-hand side is the regulated BV anomaly functional. Expanded in $\hbar$ this is a recursion whose first (one-loop) rung in the Euclidean theory reads $sM_1 = \Delta S_{\min}$.

All Ward identities descend from the single Schwinger–Dyson identity of the regulated measure: for every coefficient mode q of the cylindrical expansion, with measure density ϖ and total path-integral exponent $\mathcal{E}_{\text{tot}}$,

$$0 = \int d\mu \, e^{\mathcal{E}_{\text{tot}}} \left[\varpi^{-1} \frac{\vec{\partial} \varpi}{\partial q} + \frac{\vec{\partial} \mathcal{E}_{\text{tot}}}{\partial q} \right]. \quad (3.30)$$

BRST, background and supersymmetry Ward identities are specializations of (3.30) to the corresponding vector fields. The anomalous supersymmetry Ward identity of this paper is the statement that the supersymmetry specialization of (3.30) retains a nonzero defect at one loop; it is constructed in Section 4.

Remark 3.3. At a finite mode cutoff the classical master equation is not automatic: the truncated master action obeys $\frac{1}{2}(S_{\min, \nu}, S_{\min, \nu})_\nu = s_\nu S_{0, \nu} + \sum (-1)^\epsilon Q^* s_\nu^2 Q$, whose right-hand side vanishes only after regulator closure, regulator invariance of the action, and nilpotency of the truncated differential are established. The identities above are therefore always understood with their regulated defect terms retained.

3.6 Gauge fixing and propagators

We now fix the perturbative scheme used in Sections 4–6. The Fermi–Feynman representative is the kernel choice

$$\mathcal{Y}_{\text{FF}}^{-1} = \frac{1}{2g^2} \mathcal{A}, \quad \mathcal{A} := \begin{pmatrix} 0 & -\bar{\mathcal{D}}^2/4 \\ -\mathcal{D}^2/4 & 0 \end{pmatrix}, \quad \mathcal{A}^2 = \square \mathbf{1}, \quad \mathcal{Y}_{\text{FF}} = 2g^2 \frac{\mathcal{A}}{\square}, \quad (3.31)$$

acting on the gauge-function doublet $(\mathcal{F}_+, \mathcal{F}_-)$; it produces the gauge-fixing term $\frac{1}{2} \langle \mathcal{Y}_{\text{FF}}^{-1} \mathcal{F}, \mathcal{F} \rangle$.

Remark 3.4. Two scheme choices are declared here. First, $\mathcal{Y}_{\text{FF}}$ contains $\square^{-1}$ and is therefore nonlocal as a gauge-fermion kernel: perturbation theory is *defined* by Fermi–Feynman Gaussian averaging rather than derived from a local admissible gauge fermion. (The source repository explicitly records this locality obstruction and does not assert a unique propagator set, a selected Nielsen–Kallosh branch, or momentum-space rules; the rules below are the standard Fermi–Feynman supergraph grammar [3], with coefficients that agree with the verified repo expressions.) Second, because $\mathcal{Y}_{\text{FF}}$ is field independent, $s\mathcal{Y}_{\text{FF}} = 0$ and the associated Nielsen–Kallosh factor is a field-independent constant in this slice; it cancels in normalized correlation functions and is set to unity.

With $\square := \delta^{mn} \partial_m \partial_n$ and $z = (x, \theta, \bar{\theta})$, the chiral, antichiral and transverse projectors are

$$\mathcal{P}_+ := \frac{\bar{\mathcal{D}}^2 \mathcal{D}^2}{16\square}, \quad \mathcal{P}_- := \frac{\mathcal{D}^2 \bar{\mathcal{D}}^2}{16\square}, \quad \mathcal{P}_T := -\frac{\mathcal{D}^\alpha \bar{\mathcal{D}}^2 \mathcal{D}_\alpha}{8\square}, \quad \mathcal{P}_0 := \mathcal{P}_+ + \mathcal{P}_-, \quad (3.32)$$

with the projector algebra following from the spinor-derivative identities:

$$\mathcal{P}_i \mathcal{P}_j = \delta_{ij} \mathcal{P}_i \quad (i, j \in \{T, +, -\}), \quad \mathcal{P}_T + \mathcal{P}_+ + \mathcal{P}_- = 1, \quad (3.33)$$

$$\bar{\mathcal{D}}^2 \mathcal{D}^2 \bar{\mathcal{D}}^2 = 16\square \bar{\mathcal{D}}^2, \quad \mathcal{D}^2 \bar{\mathcal{D}}^2 \mathcal{D}^2 = 16\square \mathcal{D}^2, \quad -\mathcal{D}^\alpha \bar{\mathcal{D}}^2 \mathcal{D}_\alpha = 8\square - \frac{1}{2} \{\mathcal{D}^2, \bar{\mathcal{D}}^2\}. \quad (3.34)$$

The Fermi–Feynman quadratic form $S_{E,V}^{(2)} = \frac{1}{4g^2} \int d^4x d^4\theta \mathcal{V}^A \square \mathcal{V}^A$ and the constrained identity kernels $\mathbf{1}_\pm := \mathcal{P}_\pm \delta^8(z - z')$ give the exact vector and matter propagators

$$G^{VV,AB} = 2\hbar g^2 \delta^{AB} \square^{-1}, \quad G^{\Phi\tilde{\Phi},AB} = -\hbar g^2 \delta^{AB} \mathcal{P}_+ \delta^8(z - z'), \quad (3.35)$$

the reversed matter orientation using $\mathcal{P}_-$. In momentum space (phase e^{ipx} , all momenta incoming, $\square \mapsto -p^2$; $\theta_{12} := \theta_1 - \theta_2$):

$$\begin{aligned} \langle \mathcal{V}^A(p, \theta_1) \mathcal{V}^B(p', \theta_2) \rangle &= -(2\pi)^4 \delta^4(p + p') \frac{2\hbar g^2 \delta^{AB}}{p^2} \delta^4(\theta_{12}), \\ \langle \Phi^A(p, \theta_1) \tilde{\Phi}^B(p', \theta_2) \rangle &= (2\pi)^4 \delta^4(p + p') \frac{\hbar g^2 \delta^{AB}}{16p^2} \bar{\mathcal{D}}_1^2 \mathcal{D}_1^2 \delta^4(\theta_{12}). \end{aligned} \quad (3.36)$$

The spinor-derivative algebra collapses products of matter propagators at coincident superspace points through the loop-saturation identity

$$\delta^4(\theta_{12}) \mathcal{D}^2 \bar{\mathcal{D}}^2 \delta^4(\theta_{12}) = 16 \delta^4(\theta_{12}). \quad (3.37)$$

The interaction vertices are ordered superfield words; we abbreviate $\int_8 := \int d^4x d^4\theta$ and $\int_\pm$ for the chiral/antichiral integrals. Expanding (3.3) with the exact exponential derivatives

$$e^{-\mathcal{V}} \mathcal{D}_\alpha e^{\mathcal{V}} = \sum_{p,q \geq 0} \frac{(-1)^p}{p! q! (p+q+1)} \mathcal{V}^p (\mathcal{D}_\alpha \mathcal{V}) \mathcal{V}^q, \quad e^{\mathcal{V}} \bar{\mathcal{D}}_{\dot{\alpha}} e^{-\mathcal{V}} = \sum_{p,q \geq 0} \frac{(-1)^{q+1}}{p! q! (p+q+1)} \mathcal{V}^p (\bar{\mathcal{D}}_{\dot{\alpha}} \mathcal{V}) \mathcal{V}^q, \quad (3.38)$$

the pure-gauge words are

$$S_{E,+}^g = - \sum_{p,q,r,s \geq 0} \frac{(-1)^{p+r} f_{AB}}{256 p! q! r! s! (p+q+1)(r+s+1)} \int_+ [\bar{\mathcal{D}}^2 (\mathcal{V}^p \mathcal{D}^\alpha \mathcal{V} \mathcal{V}^q)]^A [\bar{\mathcal{D}}^2 (\mathcal{V}^r \mathcal{D}_\alpha \mathcal{V} \mathcal{V}^s)]^B, \quad (3.39)$$

the antichiral words carrying $(-1)^{q+s} \tilde{f}_{AB}$ with $\mathcal{D} \leftrightarrow \bar{\mathcal{D}}$; at cubic order this gives the four words with coefficients $\pm f_{AB}/512$. The matter–gauge chain is

$$S_E^{\text{mat}} = -\frac{1}{g^2} \delta_{AB} \sum_{r=1}^{m_N} \sum_{n \geq 0} \frac{1}{n!} \int_8 \tilde{\Phi}_r^A \mathcal{V}^{A_1} \dots \mathcal{V}^{A_n} (T_{A_1} \dots T_{A_n})^B{}_C \Phi_r^C, \quad (3.40)$$

with $(T_{A_1} \cdots T_{A_n})^B{}_C = i^n f_{A_1 D_1}{}^B f_{A_2 D_2}{}^{D_1} \cdots f_{A_n C}{}^{D_{n-1}}$ in the adjoint representation; the labeled-leg vertex carries the color word symmetrized over its n legs, and the cubic vertex is $-\frac{i}{g^2} f_{ABC} \int_8 \tilde{\Phi}^A \mathcal{V}^B \Phi^C$. The $\mathcal{N} = 4$ superpotential contributes the chiral and antichiral cubic primitives

$$\mathcal{V}_{\Phi\Phi\Phi} = \mathcal{V}_{\tilde{\Phi}\tilde{\Phi}\tilde{\Phi}} = \frac{\sqrt{2}}{g^2} \varepsilon_{rst} f^{ABC}. \quad (3.41)$$

The ghost sector is generated by inserting (3.22) into

$$S^{\text{FP}} = \frac{1}{4} \int_+ \bar{c}_+ \bar{\mathcal{D}}^2(s\mathcal{V}) + \frac{1}{4} \int_- \bar{c}_- \mathcal{D}^2(s\mathcal{V}), \quad S_{\text{FP}}^{(2)} = \frac{i}{4} \int_+ \bar{c}_+ \bar{\mathcal{D}}^2 \tilde{c} - \frac{i}{4} \int_- \bar{c}_- \mathcal{D}^2 c, \quad (3.42)$$

with ghost-ghost-vector words of coefficients $\frac{i}{4} \frac{B_{p+q}(-1)^q}{p!q!}$ (for $\tilde{c}$) and $-\frac{i}{4} \frac{B_{p+q}(-1)^p}{p!q!}$ (for c), the lowest being the $B_1 = -\frac{1}{2}$ word $s\mathcal{V}|_{(1)} = -\frac{i}{2}[\mathcal{V}, c + \tilde{c}]$. Supergraph weights follow the standard rule: a factor τ per vertex and $-\tau^{-1}$ per propagator, with $\tau = i/\hbar$ Lorentzian and $\tau = -1/\hbar$ Euclidean, graded permutation signs for labeled legs, and division by the automorphism group of the labeled graph.

3.7 Regularization: dimensional reduction

Loop integrations are regularized by dimensional reduction [5]: loop momenta are d -dimensional, $d = 4 - 2\epsilon$, while the spinor algebra, the superspace derivatives and all numerator spin words remain four-dimensional. Two metric representations coexist and must not be confused. The loop-momentum split is

$$\delta_4^{mn} = \hat{\delta}^{mn} + \check{\delta}^{mn}, \quad \hat{\delta}^m{}_m = d, \quad \check{\delta}^m{}_m = 4 - d = 2\epsilon, \quad \hat{\delta}^m{}_r \check{\delta}^r{}_n = 0, \quad (3.43)$$

with purely hatted loop momenta, $\ell^m = \hat{\delta}^m{}_n \ell^n$; the numerator spinor/vector split is

$$\delta_d^{MN} = \bar{\delta}^{MN} + \tilde{\delta}^{MN}, \quad \text{tr } \bar{\delta} = 4, \quad \text{tr } \tilde{\delta} = d - 4 = -2\epsilon. \quad (3.44)$$

The regulator enters the computation through a single axiom — finite superspace spin words are contracted with the four-dimensional metric $\bar{\delta}$, while propagator inverses use the d -dimensional metric δ_d — and through the evanescent scalar

$$\mu_\ell^2 := \bar{\ell}^2 - \ell_d^2 = -\tilde{\ell}^2, \quad (3.45)$$

the squared length of the loop momentum along the -2ϵ evanescent directions. Fourier and measure conventions are

$$f(x) = \int \frac{d^4 p}{(2\pi)^4} e^{ipx} f(p), \quad \square \mapsto -p^2, \quad \int_\ell := \mu^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d}, \quad (3.46)$$

with all external momenta taken incoming and the superspace delta functions $\delta^4(\theta_{12})$ of Section 3.6.

Remark 3.5. Dimensional reduction is the only scheme-sensitive input of the one-loop computation: the single axiom above, together with the master integrals quoted in Section 5, is the complete regulator data used. Whether evanescent operators are required beyond this axiom is a question about higher-order structure and is deferred to the discussion of Section 4.

4 The superspace Feynman-diagram derivation

This section contains the core computation of the paper: the one-loop anomalous Schwinger–Dyson (Ward) identity for the BPS letters of Euclidean $\mathcal{N} = 4$ super-Yang–Mills theory, quantized in $\mathcal{N} = 1$ superspace [3, 2] and regularized by dimensional reduction (DRED) [5]. We define the letter alphabet (Section 4.1), reduce the tree-level Schwinger–Dyson identity to Euler operators (Section 4.2), isolate the unique regulated cutting failure (Section 4.3), and derive the selection rules that organize the eighty-one ordered channels (Section 4.4). We then fix the supergraph Feynman rules and diagram conventions (Section 4.5) and present the complete one-loop computation of every nonzero channel family, with all supergraphs, Wick contractions, amplitudes, D -algebra and loop integrations shown explicitly: the seed channel (b, b) (Section 4.6), the channels (b, β_r) and (β_r, b) (Section 4.7), (b, γ^r) and (γ^r, b) (Section 4.8), $(b, \partial c)$ and $(\partial c, b)$ (Section 4.9), (β_r, β_s) (Section 4.10), and (β_r, γ^s) with (γ^s, β_r) (Section 4.11). The results are collected in the complete channel table (Section 4.12), and we close with the holomorphic-derivative tower kernel (Section 4.13).

4.1 The letters

We work in the Euclidean plus/minus spin frame of Section 2, with gauge-covariant superspace derivatives whose frame components are $\nabla_\pm$ and the dotted components $\mathcal{D}_{+\dot{\alpha}}$. In Euclidean signature the chiral and antichiral fields are independent: no conjugation relates Φ_r to $\tilde{\Phi}^r$. The semi-chiral fermionic derivative ∇_- is a frame component of the gauge-covariant spinor derivative; it is the operator whose action on composite letters we compute.

Definition 4.1 (BPS letters). The *letters* are the gauge-covariant single-trace composites

$$b := -\frac{i}{\sqrt{2}} \nabla_+ \mathcal{W}_+, \quad \beta_r := \frac{1}{\sqrt{2}} \nabla_+ \Phi_r, \quad \gamma^r := \tilde{\Phi}^r, \quad \partial_{\dot{\alpha}} c := i \widetilde{\mathcal{W}}_{\dot{\alpha}}, \quad r = 1, 2, 3, \quad \dot{\alpha} = \dot{1}, \dot{2}, \quad (4.1)$$

built from the chiral gaugino field strength $\mathcal{W}_+$, the three adjoint chiral superfields Φ_r , their independent Euclidean antichiral partners $\tilde{\Phi}^r$, and the antichiral field strength $\widetilde{\mathcal{W}}_{\dot{\alpha}}$ of Section 2. These are the Euclidean avatars of the 1/16-BPS letters of the holomorphic twist [1].

The letters carry three gradings, summarized here and used throughout:

letter	definition	Grassmann parity	$SU(3)$ flavor	twisted dimension
b	$-\frac{i}{\sqrt{2}} \nabla_+ \mathcal{W}_+$	0	1	2
β_r	$\frac{1}{\sqrt{2}} \nabla_+ \Phi_r$	1	3	$\frac{3}{2}$
γ^r	$\tilde{\Phi}^r$	0	$\bar{3}$	1
$\partial_{\dot{\alpha}} c$	$i \widetilde{\mathcal{W}}_{\dot{\alpha}}$	1	1	$\frac{3}{2}$

The holomorphic (dotted) covariant derivatives are

$$\partial_{\dot{\alpha}} := \mathcal{D}_{+\dot{\alpha}}, \quad \partial^{\dot{\alpha}} := \epsilon^{\dot{\alpha}\dot{\beta}} \partial_{\dot{\beta}}, \quad \epsilon^{\dot{1}\dot{2}} = +1, \quad \epsilon_{\dot{1}\dot{2}} = -1, \quad (4.2)$$

of twisted dimension $[\partial_{\dot{\alpha}}] = 1$, and the elementary output bilinear of the anomaly is the ordered dotted contraction

$$\langle X, Y \rangle := (\partial_{\dot{\alpha}} X)(\partial^{\dot{\alpha}} Y). \quad (4.3)$$

For two odd letters the dotted spinor contraction is symmetric under exchange of the two entries; no symmetry property is needed for mixed-parity pairs.

The alphabet contains nine letters — b (one), β_r (three), γ^r (three), $\partial_{\dot{\alpha}} c$ (two) — and we study the $9 \times 9 = 81$ ordered composite insertions $\nabla_-(L_i^A L_j^B)$. Reversed orderings are independent: (L_i, L_j) and (L_j, L_i) are separate channels, never quotiented. The anomaly operator itself carries twisted dimension $+\frac{1}{2}$, so both sides of the master identity below are homogeneous of dimension $[L_i] + [L_j] + \frac{1}{2}$.

Remark 4.2 (The Q_0 correspondence). The classical supercharge of the BPS sector is identified with the semi-chiral derivative by

$$Q_0 \longleftrightarrow -\frac{1}{2} \nabla_- . \quad (4.4)$$

The tree-level contact terms of the Schwinger–Dyson family are exactly the classical bracket (the Q_0 sector); they are computed and removed first. The one-loop remainder derived below is the one-loop correction to the Q_0 Ward identity. The correspondence (4.4) is used again in Section 7.

4.2 The Schwinger–Dyson identity and the Euler-operator reduction

The regulated path integral obeys a local Schwinger–Dyson identity. Its tree content in the letter sector is a reduction to Euler operators — the functional equation-of-motion operators of the vector and of the antichiral matter fields. Writing $(X \times Y)^A := f^{ABC} X^B Y^C$ for the adjoint commutator, the two Euler operators are

$$\frac{\delta S}{\delta \mathcal{V}} = \nabla^a \mathcal{W}_a - 2i (\Phi_s \times \tilde{\Phi}^s), \quad \frac{\delta S}{\delta \tilde{\Phi}^r} = -\frac{1}{4} \nabla^2 \Phi_r - \frac{1}{\sqrt{2}} \varepsilon_{rst} (\gamma^s \times \gamma^t), \quad (4.5)$$

with a sum over the flavor index s understood, and with the overall g^{-2} of the absorbed frame of Section 3 stripped from the Euler operators throughout this section. Using $\nabla^2 = 2\nabla_- \nabla_+$, the tree-level Schwinger–Dyson reduction of the letters is the exact system

$$\boxed{\begin{aligned} \nabla_- b &= \frac{i}{\sqrt{2}} \nabla_+ \frac{\delta S}{\delta \mathcal{V}} - 2 (\beta_s \times \gamma^s), \\ \nabla_- \beta_r &= -\sqrt{2} \frac{\delta S}{\delta \tilde{\Phi}^r} - \varepsilon_{rst} (\gamma^s \times \gamma^t), \\ \nabla_- \gamma^r &= 0, \quad \nabla_- (\partial_{\dot{a}} c) = 0. \end{aligned}} \quad (4.6)$$

Every ordered pair is then reduced by the graded Leibniz rule

$$\nabla_- (L_i L_j) = (\nabla_- L_i) L_j + (-1)^{|L_i|} L_i (\nabla_- L_j). \quad (4.7)$$

Graphically, an Euler-operator insertion attached to a propagator collapses it, $KK^{-1} = \delta$: this is the *cutting rule*. The surviving tree contact terms are precisely the classical bracket of Remark 4.2. The one-loop content of the same identity is the master result of this paper:

Proposition 4.3 (One-loop anomalous Ward identity). *For each of the eighty-one ordered pairs of letters,*

$$\nabla_- (L_i^A L_j^B) \Big|_{1\text{-loop}} = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \Delta(L_i, L_j)^{DE}, \quad (4.8)$$

where $\Delta(L_i, L_j)$ are the fixed letter bilinears collected in Section 4.12; Δ vanishes for 52 of the 81 ordered pairs.

Remark 4.4 (Classical/quantum separation). If a regulator preserved simultaneously (i) the equation-of-motion/variation correspondence, (ii) the cutting rule $KK^{-1} = \delta$, and (iii) four-dimensional D -algebra, the complete Schwinger–Dyson graph family would cancel and no anomaly could exist. DRED preserves (i) and (iii) but deforms (ii): the quantum remainder is carried solely by the evanescent insertions of Section 4.3.

4.3 The cutting mechanism

The two metric splits of DRED

Dimensional reduction keeps $d = 4 - 2\epsilon$ while splitting the four-dimensional metric and the d -dimensional metric as

$$\delta_4^{mn} = \widehat{\delta}^{mn} + \check{\delta}^{mn}, \quad \widehat{\delta}_m^m = d, \quad \check{\delta}_m^m = 4 - d = 2\epsilon, \quad \widehat{\delta}_r^m \check{\delta}_n^r = 0, \quad (4.9)$$

the quasi-four-dimensional split, and the scalar dimension-shift split

$$\delta_d^{MN} = \bar{\delta}^{MN} + \widetilde{\delta}^{MN}, \quad \text{tr } \bar{\delta} = 4, \quad \text{tr } \widetilde{\delta} = d - 4 = -2\epsilon, \quad \ell_d^2 = \bar{\ell}^2 + \widetilde{\ell}^2. \quad (4.10)$$

The loop momentum has only hatted components, $\ell^m = \widehat{\delta}_n^m \ell^n$, and the measure is $\mu^{2\epsilon} \int d^d \ell / (2\pi)^d$. The evanescent loop-momentum square is

$$\boxed{\mu_\ell^2 := \bar{\ell}^2 - \ell_d^2 = -\widetilde{\ell}^2} \quad (4.11)$$

and the single regulator axiom of the computation is

$$\boxed{\text{finite superspace spin words use } \bar{\delta}, \quad \text{propagator inverses use } \delta_d.} \quad (4.12)$$

The spin algebra is four-dimensional in this representation,

$$\sigma_m \bar{\sigma}_n + \sigma_n \bar{\sigma}_m = 2\bar{\delta}_{mn} \mathbf{1}_2. \quad (4.13)$$

Remark 4.5 (The two splits are never contracted with each other). Combining the two representations (4.9) and (4.10) before the Schwinger family is summed would set the scalar difference (4.11) to zero and erase the anomaly. Our Fourier convention is $X(x) = \int \frac{d^d p}{(2\pi)^d} e^{ip \cdot x} X(p)$ with all momenta incoming, so $\square \mapsto -p^2$.

The exact cutting-failure identity

On an internal edge e the momentum is $r_e = \ell + Q_e$, and the external shifts Q_e carry no evanescent components, so $\bar{r}_e^2 = r_{e,d}^2 + \mu_\ell^2$ and

$$\frac{\bar{r}_e^2}{r_{e,d}^2} = 1 + \frac{\mu_\ell^2}{r_{e,d}^2}. \quad (4.14)$$

The first term is exactly the Schwinger cut. For a triangle with denominators $D_i = r_{i,d}^2$ one therefore has, for *every marked line e of every one-loop graph*, the exact cutting-failure identity

$$\boxed{\frac{\bar{r}_e^2}{D_0 D_1 D_2} - \frac{1}{\prod_{j \neq e} D_j} = \frac{\mu_\ell^2}{D_0 D_1 D_2}} \quad (4.15)$$

— one universal insertion, with no graph-specific $(4 - d)$ factor. The entire anomaly is the sum of these remainders. Replacing $\bar{r}_e^2$ by $r_{e,d}^2$ before the subtraction makes the expression identically zero and removes the anomaly.

The evanescent master integral

Finiteness and locality of the anomaly follow from the master integral

$$\boxed{\lim_{\epsilon \rightarrow 0} \mu^{2\epsilon} \int \frac{d^{4-2\epsilon} \ell}{(2\pi)^{4-2\epsilon}} \frac{\mu_\ell^2}{(\ell_d^2 + \Delta)^3} = \frac{1}{32\pi^2}} \quad (4.16)$$

which is independent of the Feynman-parameter combination Δ . The proof uses two ingredients. First, by d -dimensional rotational invariance inside a rotationally invariant integral,

$$\int \tilde{\ell}^2 f(\ell^2) = \frac{d-4}{d} \int \ell^2 f(\ell^2). \quad (4.17)$$

Second, the scalar master family

$$J_n(\Delta) := \mu^{2\epsilon} \int \frac{d^d r}{(2\pi)^d} \frac{1}{(r^2 + \Delta)^n} = \frac{\mu^{2\epsilon}}{(4\pi)^{d/2}} \frac{\Gamma(n - d/2)}{\Gamma(n)} \Delta^{d/2-n} \quad (4.18)$$

obeys

$$\Delta J_3 = \frac{\epsilon}{2} J_2, \quad \text{Res}_{\epsilon=0} J_2 = \frac{1}{16\pi^2}, \quad J_2 - \Delta J_3 = \left(1 - \frac{\epsilon}{2}\right) J_2. \quad (4.19)$$

Combining (4.11), (4.17) and (4.19),

$$\int_\ell \frac{\mu_\ell^2}{(\ell_d^2 + \Delta)^3} = -\frac{d-4}{d} (J_2 - \Delta J_3) = \frac{2\epsilon}{4-2\epsilon} \cdot \frac{4-2\epsilon}{4} J_2 = \frac{\epsilon}{2} J_2 \xrightarrow{\epsilon \rightarrow 0} \frac{1}{2} \cdot \frac{1}{16\pi^2} = \frac{1}{32\pi^2}, \quad (4.20)$$

using $\epsilon \Gamma(\epsilon) \rightarrow 1$. The would-be nonlocal Δ -dependence enters only at $O(\epsilon)$ through $\Delta^{-\epsilon}$, so the anomaly is a polynomial in external momenta: a local operator.

Remark 4.6 (Pole bookkeeping as a cross-check principle). For each channel the isolated parent graphs are UV divergent; the complete Schwinger–Dyson family is finite. The $1/\epsilon$ poles cancel between the parent tensor reduction, which involves the physical metric $\hat{\delta}$, and the Schwinger contact rows, which involve the four-dimensional metric δ_4 , leaving $\hat{\delta} - \delta_4 = -\check{\delta}$: a pure evanescent numerator whose trace supplies the finite answer. Equivalently, every anomaly factorizes as (evanescent trace $\propto 2\epsilon$) $\times$ (pole residue $1/(16\pi^2\epsilon)$). Any error in a single vertex factor or sign would break this cancellation, so the finiteness of each channel is itself a consistency proof of its factor chain.

4.4 Selection rules: the census of the eighty-one ordered pairs

Before any loop integral, conservation laws restrict the possible uncontracted outputs. The gradings that survive the construction are:

- (i) $SU(3)$ flavor: b and $\partial_{\dot{\alpha}} c$ are singlets, $\beta_r \in \mathbf{3}$, $\gamma^r \in \bar{\mathbf{3}}$;
- (ii) Grassmann parity: $|b| = |\gamma| = 0$, $|\beta_r| = |\partial_{\dot{\alpha}} c| = 1$;
- (iii) twisted dimension: $[\gamma] = 1$, $[\beta] = [\partial c] = \frac{3}{2}$, $[b] = 2$, $[\partial_{\dot{\alpha}}] = 1$, with the anomaly operator carrying $+\frac{1}{2}$;
- (iv) the output structure $\langle X, Y \rangle$ of equation (4.3) — exactly one holomorphic derivative on each of two letters — forced by the σ -chain trace of Section 4.6.

Checking each ordered family against these gradings and against the cubic-vertex spectator structure yields exactly

$$29 \text{ nonzero} = 1_{(b,b)} + 6_{(b,\beta),(\beta,b)} + 6_{(b,\gamma),(\gamma,b)} + 4_{(b,\partial c),(\partial c,b)} + 6_{(\beta,\gamma)} \delta\text{-diagonal} + 6_{(\beta,\beta)} \varepsilon\text{-offdiagonal}. \quad (4.21)$$

The remaining 52 ordered pairs vanish exactly, in four structural classes:

1. **25 no-descendant pairs.** Since $\nabla_- \gamma = \nabla_- (\partial c) = 0$, the Leibniz rule (4.7) makes $\nabla_- (L_i L_j)$ vanish identically when both letters are drawn from $\{\gamma^1, \gamma^2, \gamma^3, \partial_1 c, \partial_2 c\}$ — all $5 \times 5 = 25$ pairs of this set.
2. **12 flavor-off-diagonal** $(\beta_r, \gamma^s)/(\gamma^s, \beta_r)$, $r \neq s$. Exact zeros: the available singlet contraction $\delta_{rs} \langle \partial c, \partial c \rangle$ has no off-diagonal support.
3. **3 diagonal** (β_r, β_r) . Exact zeros: the candidate D -word and the repeated-index ε routes both vanish.
4. **12** $(\beta_r, \partial_{\dot{\alpha}} c)/(\partial_{\dot{\alpha}} c, \beta_r)$. Killed exactly by the charge and cubic-vertex basis analysis.

Remark 4.7 (Three independent routes to the same census). The 29/52 census is obtained independently by (a) the selection-rule analysis above, (b) the explicit evaluation of all eighty-one ordered Schwinger–Dyson families, and (c) the vanishing list of the holomorphic-twist calculation [1] (Section 7). All three routes agree.

Remark 4.8 (Status). The eighty-one-row result — twenty-nine exact nonzero channels and fifty-two exact zeros — is the accepted physical one-loop anomaly sector of the project, with the physical verifier passing all thirty-three checks. An adversarial census review (seventeen candidate structures) remains open; it does not alter the accepted coefficients stated here.

4.5 Supergraph Feynman rules and diagram conventions

This subsection fixes the Feynman rules, insertion operators, graph weights, and loop-integral kit from which every one-loop channel of Section 4 is assembled. We write the flat superspace spinor derivatives in the plus/minus frame as $\mathcal{D}_{\pm}$, $\bar{\mathcal{D}}_{\pm}$, with

$$\mathcal{D}^2 = 2 \mathcal{D}_- \mathcal{D}_+, \quad \bar{\mathcal{D}}^2 = 2 \bar{\mathcal{D}}_+ \bar{\mathcal{D}}_-, \quad \theta_{12} := \theta_1 - \theta_2. \quad (4.22)$$

The Fourier convention is e^{ipx} with all vertex momenta incoming, so $\square \mapsto -p^2$; the overall momentum-conservation factor $(2\pi)^4 \delta^4(p + p')$ is suppressed in every propagator. The adjoint Killing form is $\kappa_{AB} = \delta_{AB}$ throughout.

Fermi–Feynman propagators

In the absorbed-coupling normalization of Section 3 the quadratic kernels invert to the Fermi–Feynman propagators

$$\langle \mathcal{V}^A(p, 1) \mathcal{V}^B(-p, 2) \rangle = -\frac{2 \hbar g^2 \delta^{AB}}{p^2} \delta^4(\theta_{12}), \quad (4.23)$$

$$\langle \Phi_r^A(p, 1) \tilde{\Phi}_s^B(-p, 2) \rangle = \delta_{rs} \frac{\hbar g^2 \delta^{AB}}{16 p^2} \bar{\mathcal{D}}_1^2 \mathcal{D}_1^2 \delta^4(\theta_{12}), \quad (4.24)$$

and the reversed matter orientation uses the opposite chirality projector. The canonical-normalization audits underlying this section instead use the rescaled vector prepotential $u := \mathcal{V}/(\sqrt{2} g)$, for which the vector line has the unit propagator $-\hbar \delta^{AB} p^{-2} \delta^4(\theta_{12})$; the translation between the two frames is the single map

$$\mathcal{V} = \sqrt{2} g u, \quad (4.25)$$

applied to every field, letter, and propagator at once, and no anomalous factor of g or $\sqrt{2}$ is generated by the change of frame.

Remark 4.9 (Status of the propagator slice). The propagators (4.23)–(4.24) are the recorded rules on a local, residual-free Fermi–Feynman slice: a unique propagator set, a Nielsen–Kallosh branch, and a complete momentum-space rule set are *not* asserted by the underlying audits, whose stated scope is partial. The downstream computations of this paper use this conditional slice and say so; the complete graph census remains open (Remark 4.10).

Vertex inventory

The one-loop channels use four classes of vertices. (a) *Cubic gauge words*. The undotted and dotted cubic gauge functionals give the exponent vertices

$$-\frac{ig}{4\hbar} f^{UCE} \mathcal{W}^{E\gamma} (\mathcal{D}_{C\gamma} - \mathcal{D}_{U\gamma}), \quad +\frac{ig}{4\hbar} f^{UCD} \widetilde{\mathcal{W}}_{\dot{\gamma}}^D (\bar{\mathcal{D}}_C^{\dot{\gamma}} - \bar{\mathcal{D}}_U^{\dot{\gamma}}), \quad (4.26)$$

where the two slot labels record the marked quantum ports; the second variation already contains the two labeled port assignments, so no $2!$ remains. Two bookkeeping frames appear in the channel computations, and they are never mixed within one graph weight. In the *canonical frame* of (4.25) the vertices carry $\mp ig/(4\hbar)$, the internal lines carry $-\hbar\delta^{AB}p^{-2}$, and the insertion product is $(-1/(8\sqrt{2}))(-1/(4\sqrt{2})) = 1/64$. In the *unit-insertion frame* used in Sections 4.6–4.11, the insertions are the unit-normalized field strengths $\nabla_+\mathcal{W}_+$: relative to the canonical frame, each of the two quantum legs of every cubic vertex rescales by $\sqrt{2}$, so the effective ordered vertex factors are $\mp ig/2$; each insertion rescales by $\sqrt{2}$, so the insertion product is $1/32$; and each internal line carries the rescaled propagator $-\hbar/(2p^2)\delta^4(\theta_{12})$. The three effects compensate exactly,

$$\underbrace{(\sqrt{2})^2(\sqrt{2})^2}_{\text{two vertices}} \underbrace{(\sqrt{2})(\sqrt{2})}_{\text{two insertions}} \underbrace{\left(\frac{1}{2}\right)^3}_{\text{three lines}} = 1, \quad (4.27)$$

so every channel coefficient is frame-independent. In the unit-insertion frame each ordered pair of cubic gauge vertices therefore contributes

$$\left(+\frac{ig}{2}\right)\left(-\frac{ig}{2}\right) = \frac{g^2}{4}, \quad (4.28)$$

as used in equation (4.62). (b) *Matter–gauge chain*. In the canonical frame ($u, \Phi_c = g^{-1}\Phi$) the first link of the matter chain is

$$S_{\text{mat}}^{(1)} = -\sqrt{2}g \int d^8z \tilde{\Phi}_r^A \mathcal{V}^U (T_U)^B{}_C \Phi_r^C \quad (4.29)$$

(in the absorbed frame of Section 3 the same link reads $-g^{-2} \int d^8z \tilde{\Phi} \mathcal{V} \Phi$), with the ordered adjoint color chain $(T_{A_1} \cdots T_{A_n})^B{}_C = i^n f^{A_1 D_1 B} f^{A_2 D_2 D_1} \cdots f^{A_n C D_{n-1}}$; the marked matter vertex carries port coefficient $+\sqrt{2}g/\hbar$ per generator, so the two matter vertices of a marked matter word give $2g^2/\hbar^2$, the three propagators give $\hbar^3$, and the ordered adjoint color route gives $f^{ACD}f^{BCE}$. (c) *Superpotential cubics*. With $W = -\frac{\sqrt{2}}{6g^2} \varepsilon_{rst} f^{ABC} \Phi_r^A \Phi_s^B \Phi_t^C$, the chiral cubic vertex primitive is

$$\frac{\sqrt{2}}{g^2} \varepsilon_{rst} f^{ABC}, \quad (4.30)$$

with the tilded analog on the antichiral side (the Euclidean weight $\eta_E = -1$ is included in (4.30)). (d) *Ghost words*. The BV spectrum contains the ghosts $c, \tilde{c}$ and the antighost $\bar{c}$; their role in the accepted channels is stated in the following remark.

Remark 4.10 (Ghost and gauge-fixing sector: open census). No accepted ghost, Nielsen–Kallosh, gauge-fixing, counterterm, or evanescent-mixing graph rule enters any of the twenty-nine nonzero channels of this paper: the accepted anomaly-sector audits record no ghost-loop contribution to those channels, and the complete graph census of these sectors is explicitly flagged open in the underlying audits. Ghost lines appear in the legend of Figure 1 for completeness only.

External-letter insertion operators

A marked external letter leg is not a propagator endpoint: it carries its own insertion operator, while every uncontracted pair of fields is Wick-replaced by (4.23)–(4.24). The linearized port operator and its marked derivative are

$$K_+^c := -\frac{1}{4\sqrt{2}} \mathcal{D}_+ \bar{\mathcal{D}}^2 \mathcal{D}_+, \quad A_{(1)} = K_+^c u, \quad (4.31)$$

$$\mathcal{D}_- A_{(1)} = -\frac{1}{8\sqrt{2}} \mathcal{D}^2 \bar{\mathcal{D}}^2 \mathcal{D}_+ u, \quad (4.32)$$

where the second line uses $\mathcal{D}^2 = 2\mathcal{D}_- \mathcal{D}_+$ of (4.22). Translated by (4.25) to the paper's absorbed fields and letters, the linearized field-strength insertion and its semi-chiral derivative are

$$\nabla_+ \mathcal{W}_+|_{\text{lin}} = g K_+^c u = -\frac{1}{8} \mathcal{D}_+ \bar{\mathcal{D}}^2 \mathcal{D}_+ \mathcal{V}, \quad \mathcal{D}_- \nabla_+ \mathcal{W}_+|_{\text{lin}} = -\frac{1}{16} \mathcal{D}^2 \bar{\mathcal{D}}^2 \mathcal{D}_+ \mathcal{V}, \quad (4.33)$$

which sit on the external legs of the letter $b = -(i/\sqrt{2}) \nabla_+ \mathcal{W}_+$. The external bottom projectors of the matter letters are

$$\Phi_r(p, \theta, \bar{\theta}) = e^{i\theta p \bar{\theta}} \theta^+ \sqrt{2} \beta_r(p), \quad \tilde{\Phi}^r(q, \theta, \bar{\theta}) = e^{-i\theta q \bar{\theta}} \gamma^r(q), \quad (4.34)$$

obeying $\bar{\mathcal{D}}_\alpha \Phi_r = 0$ and $\mathcal{D}_\alpha \tilde{\Phi}^r = 0$.

Remark 4.11 (Matter and antichiral ports: what the sources do not name). The sources define *no* separately named matter insertion operator and *no* antichiral port K_-^c : the same K_+^c and $\mathcal{D}_- K_+^c$ of (4.31)–(4.32) act on the matter lines of a marked matter word, and the matter letters enter through the bottom projectors (4.34). We do not introduce any operator beyond this recorded content.

Graph weights and Wick symmetry

Left functional differentiation of a homogeneous ordered word is graded: removing slot j of $X_1 \cdots X_n$ costs $(-1)^{|X_j| \sum_{k < j} |X_k|}$, and a permutation π of labeled legs costs the parity sign $(-1)^{\epsilon(\pi)}$. Every connected coordinate-superspace graph then carries the labeled-graph weight

$$w(G) = \tau_E^{|V(G)|} (-\tau_E^{-1})^{|E(G)|} \left(\prod_v \mathcal{C}_v \right) \left(\prod_e K_e^{-1} \right) (-1)^{\epsilon(G)}, \quad \tau_E = -\frac{1}{\hbar}, \quad -\tau_E^{-1} = \hbar, \quad (4.35)$$

i.e. one factor τ_E per vertex and one factor $\hbar$ per propagator, with $\mathcal{C}_v$ the ordered vertex coefficient and K_e^{-1} the propagator kernel. An unlabeled graph is divided once by the automorphism group preserving field type, chirality, edge orientation, derivative slot, and ordered color word; no general per-graph symmetry-factor table beyond this rule is recorded in the sources. For the mixed-chirality exponential the action-order factor is exactly one,

$$\frac{1}{2!} (S_1 S_2 + S_2 S_1) = S_1 S_2, \quad (4.36)$$

and for one fixed directed source attachment the port-preserving Wick factor is likewise one, as in equation (4.55). The closed superspace loop is saturated by the identity of Section 3,

$$\delta^4(\theta_{12}) \mathcal{D}^2 \bar{\mathcal{D}}^2 \delta^4(\theta_{12}) = 16 \delta^4(\theta_{12}), \quad \text{i.e.} \quad \left[\mathcal{D}^2 \bar{\mathcal{D}}^2 \delta^4(\theta) \right]_{\theta=0} = 16, \quad (4.37)$$

equivalently $\mathcal{D}^2 \bar{\mathcal{D}}^2 (\theta^2 \bar{\theta}^2) = 16$ by direct left-Berezin differentiation with (4.22). The two marked source factors of (4.31)–(4.32) multiply to

$$\left(-\frac{1}{8\sqrt{2}} \right) \left(-\frac{1}{4\sqrt{2}} \right) = \frac{1}{64}, \quad \left(\frac{1}{64} \right) (16)(2)(2) = 1, \quad (4.38)$$

where the two factors of 2 are the mixed anticommutators $\{\mathcal{D}, \bar{\mathcal{D}}\} = 2\mathbf{p}$ on the closed loop, $\mathbf{p}$ the loop momentum. This product is a D -algebra factor; it is not four additional Wick contractions.

The ordered-triangle integral kit

Route the ordered triangle by

$$r_0 = k, \quad r_1 = k + q, \quad r_2 = k + p + q, \quad Q := p + q, \quad (4.39)$$

(the matter sector uses $r_1 = k - q$, $r_2 = k - p - q$). The two-simplex parametrization of the three denominators is

$$\frac{1}{r_0^2 r_1^2 r_2^2} = 2 \int_{\Delta_2} \frac{dx dy dz \delta(1-x-y-z)}{(\ell^2 + \Delta)^3}, \quad \ell = k + yq + zQ, \quad \Delta = xyq^2 + xzQ^2 + yzp^2, \quad (4.40)$$

with simplex volume $2 \int_{\Delta_2} 1 = 2 \cdot \frac{1}{2} = 1$. With $\int_k := \mu^{2\epsilon} \int d^d k / (2\pi)^d$, $d = 4 - 2\epsilon$, define

$$I_2(\Delta) := \int_{\ell} \frac{1}{(\ell^2 + \Delta)^2}, \quad I_3(\Delta) := \int_{\ell} \frac{1}{(\ell^2 + \Delta)^3}, \quad I_2(\Delta) \Big|_{1/\epsilon} = \frac{1}{16\pi^2 \epsilon}. \quad (4.41)$$

Hatted rotational reduction and the master-family identity $\Delta I_3 = \frac{\epsilon}{2} I_2$ of equation (4.19) give

$$\int_{\ell} \frac{\ell^m \ell^n}{(\ell^2 + \Delta)^3} = \frac{\widehat{\delta}^{mn}}{d} [I_2(\Delta) - \Delta I_3(\Delta)] = \frac{\widehat{\delta}^{mn}}{4} I_2(\Delta), \quad \frac{1}{4 - 2\epsilon} \left(1 - \frac{\epsilon}{2}\right) = \frac{1}{4}, \quad (4.42)$$

the quarter being exact in ϵ . For the numerator vectors $L_1 = 2k + q$, $L_2 = 2k + p + 2q$ the pole part of the ordered triangle is therefore

$$\text{Pole}_{\epsilon=0} \int_k \frac{L_1^m L_2^n}{r_0^2 r_1^2 r_2^2} = \frac{\widehat{\delta}^{mn}}{16\pi^2 \epsilon}. \quad (4.43)$$

The finite anomaly rests on the evanescent master integral

$$\lim_{\epsilon \rightarrow 0} \mu^{2\epsilon} \int \frac{d^{4-2\epsilon} \ell}{(2\pi)^{4-2\epsilon}} \frac{\mu_{\ell}^2}{(\ell^2 + \Delta)^3} = \frac{1}{32\pi^2}, \quad (4.44)$$

proved as equation (4.16) of Section 4.3; it is independent of Δ .

Line-style legend

The rules of this subsection feed every ordered channel: assembled channel by channel, they produce the seven nonzero families (4.268)–(4.274), each with the common prefactor $-\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE}$, and the fifty-two structural zeros of the census. The comparison of these results with the holomorphic-twist alphabet is deferred to Section 7 [1].

4.6 The seed channel (b, b) : the complete one-loop computation

We now compute the seed channel (b, b) , the simplest nonzero channel and the scale anchor of the whole table of Section 4.12, at full granularity: the graph inventory with orientations, the Wick contraction enumeration, the amplitude assembly, the D -algebra numerator chain, the evanescent extraction, the pole cancellation, the loop integral, and the restoration of the letter normalization, closing with the word-by-word comparison against the holomorphic-twist result of [1]. We fix once and for all an orientation of the two field-strength insertions and a marked internal line. The graph weights below are computed with unit-normalized field-strength insertions $\nabla_+ \mathcal{W}_+$; the letter normalization $b = -(i/\sqrt{2})\nabla_+ \mathcal{W}_+$ is restored at the end.

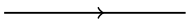
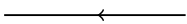
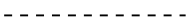
	vector superfield $\mathcal{V}$, propagator (4.23)
	chiral matter Φ_r , propagator (4.24)
	antichiral matter $\tilde{\Phi}^r$ (reversed arrow)
	ghost $c, \tilde{c}, \bar{c}$ (Remark 4.10)
	letter insertion: $\nabla_+ \mathcal{W}_+, \nabla_+ \Phi_r, \tilde{\Phi}^r, \tilde{\mathcal{W}}_{\dot{\alpha}}$
	Schwinger-cut (Euler-collapsed) line

Figure 1: Line-style legend for the supergraphs of this section. Wavy lines are vector-superfield propagators; solid lines with forward (reversed) arrows are chiral (antichiral) matter propagators; dashed lines are ghosts. A filled dot on an external leg is a letter insertion carrying its operator, while every uncontracted field pair becomes a propagator. A wavy line with an $\times$ mark is a Schwinger-cut (Euler-collapsed) line of the cutting mechanism of Section 4.3.

1. Setup: the channel, the letters, and the marking conventions. The channel is the one-loop Schwinger–Dyson descent of the ordered composite of two field-strength insertions,

$$\nabla_- \left(\nabla_+ \mathcal{W}_+^A \nabla_+ \mathcal{W}_+^B \right), \quad b^A = -\frac{i}{\sqrt{2}} \nabla_+ \mathcal{W}_+^A, \quad (4.45)$$

with the letter b of Definition 4.1. Both insertions are Grassmann-even, so the Leibniz rule (4.7) contributes no sign and splits the probe into two marked occurrences,

$$\nabla_- \left(\nabla_+ \mathcal{W}_+^A \nabla_+ \mathcal{W}_+^B \right) = (\nabla_- \nabla_+ \mathcal{W}_+^A) \nabla_+ \mathcal{W}_+^B + \nabla_+ \mathcal{W}_+^A (\nabla_- \nabla_+ \mathcal{W}_+^B). \quad (4.46)$$

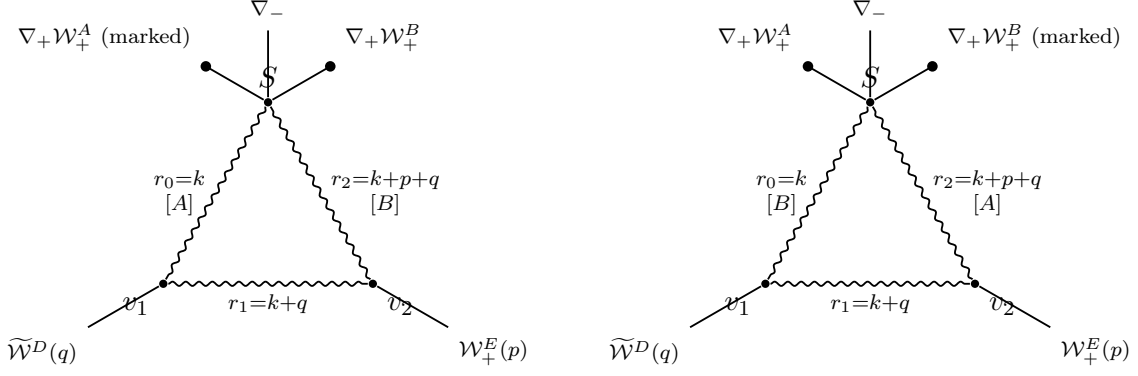
These are two different marked occurrences, not a multiplicity of either one; there is no factor 2 from the source. We fix the ordered external data once and for all: an antichiral background leg $\tilde{\mathcal{W}}^D(q)$ at the antichiral cubic vertex v_1 , a chiral background leg $\mathcal{W}_+^E(p)$ at the chiral cubic vertex v_2 , and the composite source vertex S carrying both insertions and the ∇_- probe. All momenta are incoming (Remark 4.5), the loop momentum is k , and the three internal edges are routed as

$$r_0 = k \quad (S \rightarrow v_1), \quad r_1 = k + q \quad (v_1 \rightarrow v_2), \quad r_2 = k + p + q \quad (v_2 \rightarrow S). \quad (4.47)$$

There are exactly two ordered source attachments: the *directed* attachment, in which the insertion of color A contracts toward the antichiral vertex v_1 and the insertion of color B toward the chiral vertex v_2 , and the *reflected* attachment, in which they are interchanged. The complete set of source/mark rows is

source attachment	marked occurrence	rank-two evanescent numerator	ordered output
$A \rightarrow \tilde{\mathcal{W}}, B \rightarrow \mathcal{W}$	A	yes	$\langle \tilde{\mathcal{W}}, \nabla_+ \mathcal{W}_+ \rangle$
$A \rightarrow \tilde{\mathcal{W}}, B \rightarrow \mathcal{W}$	B	no	none
$A \rightarrow \mathcal{W}, B \rightarrow \tilde{\mathcal{W}}$	A	no	none
$A \rightarrow \mathcal{W}, B \rightarrow \tilde{\mathcal{W}}$	B	yes	$\langle \nabla_+ \mathcal{W}_+, \tilde{\mathcal{W}} \rangle$

The two anomaly-bearing rows are the directed and the reflected term; they are not two copies of either term. Reflection is a distinct ordered output with weight one.



(a) directed: $(A \rightarrow \widetilde{W}, B \rightarrow W)$, probe on A (b) reflected: $(A \rightarrow W, B \rightarrow \widetilde{W})$, probe on B

Figure 2: The two ordered parent triangles of the seed channel (b, b) . Each triangle has the composite source vertex S carrying the two letter insertions (filled dots) and the ∇_- probe leg, the antichiral cubic vertex v_1 with external leg $\widetilde{W}^D(q)$, the chiral cubic vertex v_2 with external leg $W_+^E(p)$, and three internal vector-superfield lines (wavy) with edge momenta $r_0 = k$, $r_1 = k + q$, $r_2 = k + p + q$. The bracket $[A]/[B]$ on an edge records which source color contracts toward that vertex. (a) The directed orientation, with the probe on the A occurrence. (b) The reflected orientation, with the probe on the B occurrence.

2. Graph inventory. There is exactly one gauge-sector graph topology at order g^2 : the mixed-chirality pure-gauge triangle, with one chiral cubic vertex, one antichiral cubic vertex, the composite source vertex, and three internal vector-superfield lines. With the canonically normalized prepotential $u := \mathcal{V}/(\sqrt{2}g)$ and $[X, u]^A := f^{ABC}X^B u^C$, the two cubic actions generating the vertices are

$$S_{+,3} = -\frac{g}{4} \int d^8z \, \mathcal{W}^{A\alpha} [\mathcal{D}_\alpha u, u]^A, \quad S_{-,3} = +\frac{g}{4} \int d^8z \, \widetilde{\mathcal{W}}_\alpha^A [\bar{\mathcal{D}}^\alpha u, u]^A. \quad (4.48)$$

The two ordered chiral assignments of the interaction expansion collapse:

$$\frac{1}{2!} (S_{+,3} S_{-,3} + S_{-,3} S_{+,3}) = S_{+,3} S_{-,3}, \quad (4.49)$$

so the action-order factor is exactly 1. The two ordered parent triangles are displayed in Figure 2.

For each marked occurrence, the Schwinger cut of Section 4.3 collapses the internal edge carrying the marked line: the occurrence probed at the v_1 -side port collapses the edge e_0 with momentum r_0 , and the occurrence probed at the v_2 -side port collapses the edge e_2 with momentum r_2 . For one marked edge the row count is

$$\frac{1}{2!} \cdot 2_{\text{action orderings}} \cdot 2_{\text{Wick endpoints}} = 2_{\text{endpoint rows}}, \quad (4.50)$$

so there are exactly four occurrence-resolved Schwinger contact rows in total (two cut edges times two Wick endpoints). The contact topology is displayed in Figure 3.

Every remaining order- g^2 orbit vanishes exactly in the anomaly sector. The spectator self-energy insertions, the one-cubic-vertex bubbles, and the quartic seagull obey

$$\begin{aligned} \langle I_2 \rangle_0 &= 0 \quad (\text{source-quadratic self-energy: its only denominator is } k^2, \text{ hence a scaleless tadpole}), \\ -\frac{1}{\hbar} \langle I_1 S_{g3} \rangle_0 &= 0 \quad (\text{one-cubic bubble: numerator trace } \text{tr}_4 H = 2 - 2 = 0), \\ -\frac{1}{\hbar} \langle I_0 S_{g4} \rangle_0 &= 0 \quad (\text{quartic seagull: } \text{tr}_4 H_4 = -4 + 4 = 0), \end{aligned} \quad (4.51)$$

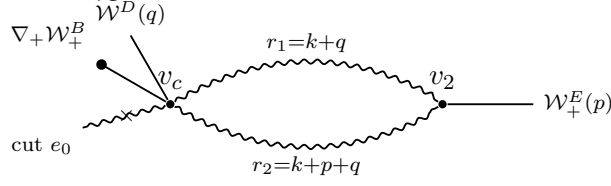


Figure 3: The Schwinger contact (cut) topology of the seed channel. Cutting the marked edge e_0 collapses the source vertex S and the antichiral vertex v_1 into the single contact vertex v_c : the probed occurrence $\nabla_-(\nabla_+ W_+^A)$ has become the Euler-operator insertion that collapses the e_0 line (wavy with $\times$), while the unmarked insertion $\nabla_+ W_+^B$ and the external leg $\widetilde{W}^D(q)$ remain at v_c . Two internal vector lines with momenta $r_1 = k + q$ and $r_2 = k + p + q$ run from v_c to the chiral vertex v_2 . The e_2 family is the mirror image, with contact vertex $v_c = S \sqcup v_2$ and internal lines r_0 and r_1 . With the two Wick port rows per edge this gives the four occurrence-resolved contact rows of paragraph 7.

so only the two-cubic triangle $(2\hbar^2)^{-1} \langle I_0 S_{g3} S_{g3} \rangle_0$ of Figure 2 and its Schwinger cuts survive. The matter orbits I_2 , $I_1 S_{m3}$, $I_0 S_{m4}$ vanish in the same way; the matter triangle of the same channel is evaluated in paragraph 9.

3. Wick contractions. The three internal lines are vector-superfield contractions with propagator

$$\langle u^A(p, 1) u^B(-p, 2) \rangle = -\frac{\hbar \delta^{AB}}{p^2} \delta^4(\theta_1 - \theta_2), \quad (4.52)$$

with the Killing form normalized to $\kappa_{AB} = \delta_{AB}$. The two cubic vertices each supply two quantum-port assignments through the derivative differences of their second variations, the Hessians

$$H_W = +\frac{ig}{4} f^{UCE} \mathcal{W}^{E\gamma} (\mathcal{D}_{C\gamma} - \mathcal{D}_{U\gamma}), \quad H_{\widetilde{W}} = -\frac{ig}{4} f^{UCD} \widetilde{\mathcal{W}}_{\dot{\gamma}}^D (\bar{\mathcal{D}}_C^{\dot{\gamma}} - \bar{\mathcal{D}}_U^{\dot{\gamma}}). \quad (4.53)$$

The labeled double variation is

$$\delta_1 \delta_2 [\mathcal{D}u, u] = [\mathcal{D}u_1, u_2] + [\mathcal{D}u_2, u_1], \quad (4.54)$$

whose two summands are the port rows (S, B) and (B, S) — the row in which the first, respectively the second, quantum port carries the source role. The second variation already contains the two labeled quantum-port assignments, so no $2!$ from the equal Hessians remains. For one fixed directed source attachment the port-preserving Wick factor is likewise 1, the two rows being equal by relabeling of the Gaussian integral. With the two labeled action-expansion assignments of paragraph 2 and their Taylor factor, the Wick weight is

$$w_{\text{Wick}} = \frac{1}{2!} (1 + 1) = 1, \quad (4.55)$$

the same arithmetic as the action-order factor of (4.49).

Remark 4.12 (Granularity of the contraction enumeration). The enumeration above is organized by the quantum-role rows (S, B) , (B, S) and by the derivative endpoint pairs (r_i, r_j) ; a field-by-field pairing picture of the individual u fields is not used at any step. The endpoint content is not multiplied after the two Hessians are expanded: no endpoint row acquires an additional factor.

4. Amplitude assembly. The linearized unit insertion and its probed descendant are

$$K_+^c := -\frac{1}{4\sqrt{2}} \mathcal{D}_+ \bar{\mathcal{D}}^2 \mathcal{D}_+, \quad \nabla_- K_+^c = -\frac{1}{4\sqrt{2}} \mathcal{D}_- \mathcal{D}_+ \bar{\mathcal{D}}^2 \mathcal{D}_+ = -\frac{1}{8\sqrt{2}} \mathcal{D}^2 \bar{\mathcal{D}}^2 \mathcal{D}_+, \quad (4.56)$$

where on the internal lines the frame derivatives act as their flat counterparts $\mathcal{D}_\pm$, $\bar{\mathcal{D}}_\pm$ and we used the spinor-derivative square move

$$\mathcal{D}^2 = 2 \mathcal{D}_- \mathcal{D}_+, \quad \bar{\mathcal{D}}^2 = 2 \bar{\mathcal{D}}_+ \bar{\mathcal{D}}_-. \quad (4.57)$$

After the three contractions, the unmarked leg retains the residual operator K_+^c and the probed leg retains $\nabla_- K_+^c$; their numerical product is

$$\left(-\frac{1}{8\sqrt{2}}\right) \left(-\frac{1}{4\sqrt{2}}\right) = \frac{1}{64}. \quad (4.58)$$

The exponent Hessians are the second variations of $e^{-S/\hbar}$, i.e. $\hat{H} = -H/\hbar$ of (4.53) including the $1/\hbar$ of each interaction vertex:

$$\hat{H}_{\mathcal{W}} = -\frac{ig}{4\hbar} f^{UCE} \mathcal{W}^{E\gamma} (\mathcal{D}_{C\gamma} - \mathcal{D}_{U\gamma}), \quad \hat{H}_{\widetilde{\mathcal{W}}} = +\frac{ig}{4\hbar} f^{UCD} \widetilde{\mathcal{W}}_{\dot{\gamma}}^D (\bar{\mathcal{D}}_C^{\dot{\gamma}} - \bar{\mathcal{D}}_U^{\dot{\gamma}}), \quad (4.59)$$

and assembling them with the three propagators (4.52) gives

$$\left(-\frac{ig}{4\hbar}\right) \left(+\frac{ig}{4\hbar}\right) (-\hbar)^3 = \frac{g^2}{16\hbar^2} (-\hbar^3) = -\frac{\hbar g^2}{16}. \quad (4.60)$$

The three Killing deltas of the propagators reduce the color factor of the two Hessians, with the source colors attached per the fixed orientation, to

$$f^{UCD} f^{U'C'E} \delta_{UA} \delta_{U'B} \delta_{CC'} \longrightarrow f^{ACD} f^{BCE} \quad (\text{sum over } C). \quad (4.61)$$

Remark 4.13 (Two equivalent normalization frames). Equation (4.60) is the canonical-frame bookkeeping: vertex magnitudes $g/(4\hbar)$, propagators $-\hbar \delta^{AB} p^{-2}$, and the explicit insertion factor $1/64$ of (4.58). In the unit-insertion frame used for the remaining steps, the same graph weight is organized with the ordered cubic vertex factors

$$\left(+\frac{ig}{2}\right) \left(-\frac{ig}{2}\right) = \frac{g^2}{4} \quad (4.62)$$

and the D -algebra weight w_D of (4.67) below. The two frames are related by the rescaling of (4.27): each of the two vertices gains $(\sqrt{2})^2$, each of the two insertions gains $\sqrt{2}$ (the product $1/64 \rightarrow 1/32$, the $1/32$ appearing in (4.67)), and each of the three internal lines loses $1/2$, a net factor 1. Each cubic vertex carries $\hbar^{-1}$ and each of the three propagators carries $\hbar$, leaving the net one-loop power $\hbar$ in either frame. The signed endpoint-row weight of (4.69) is a property of the closed triangle word and applies to either completion route.

5. The D -algebra numerator chain. For the fixed orientation the covariant spinor numerator is the chain

$$N_{G,+}^{\dot{\alpha}} = (r_0 + r_1)_{+\dot{\beta}} (ip)^{\dot{\beta}\gamma} (r_1 + r_2)_{\gamma}^{\dot{\alpha}}, \quad T_{m\rho n,+}^{\dot{\alpha}} := (\sigma_m \bar{\sigma}_\rho \sigma_n)_{+}^{\dot{\alpha}}, \quad (4.63)$$

where the explicit i is tied to the locked dotted-index convention $k_+^{\dot{0}} = \sigma_E(k)_{0\dot{1}}$, $k_+^{\dot{1}} = -\sigma_E(k)_{0\dot{0}}$; in the opposite convention the same covariant chain is written with $p^{\dot{\beta}\gamma}$ and no explicit i . The

$D^2/\bar{D}^2$ moves of (4.57) put the marked insertion into the form $-\frac{1}{8\sqrt{2}}\mathcal{D}^2\bar{\mathcal{D}}^2\mathcal{D}_+$ displayed in (4.56). The loop is saturated by the identity

$$\delta^4(\theta_{12})\mathcal{D}^2\bar{\mathcal{D}}^2\delta^4(\theta_{12}) = 16\delta^4(\theta_{12}) \quad (4.64)$$

of Section 3, and the two mixed anticommutators

$$\{\mathcal{D}, \bar{\mathcal{D}}\} = 2\sigma \cdot p \quad (4.65)$$

contribute one factor 2 each, so the D -algebra content of the closed loop is

$$\left(\frac{1}{64}\right)(16)(2)(2) = 1. \quad (4.66)$$

This is a D -algebra factor; it is not four additional Wick contractions. Equivalently, in the unit-insertion frame the closed-loop mixed D -algebra weight is

$$w_D = \frac{1}{32} \cdot 16 \cdot 2 \cdot 2 = 2, \quad (4.67)$$

where the $1/32$ is the insertion normalization of that frame and the 16 is the loop-saturation identity (4.64). The total prefactor before the loop integral is therefore

$$w_{\text{Wick}} \frac{g^2}{4} w_D = 1 \cdot \frac{g^2}{4} \cdot 2 = \frac{g^2}{2}. \quad (4.68)$$

The four derivative endpoint rows of the closed (parent) triangle word, with their raw and transport signs, are

$$\begin{aligned} (r_0, r_1) : & \quad (-1)(-1) = +1, \\ (r_0, r_2) : & \quad (+1)(+1) = +1, \\ (r_1, r_1) : & \quad (+1)(+1) = +1, \\ (r_1, r_2) : & \quad (-1)(-1) = +1, \end{aligned} \quad (4.69)$$

and the closed word carries in addition the single odd source-loop integration-by-parts sign $s_{\text{global}} = -1$, so the signed row weight is

$$w_{T,\text{rows}} = s_{\text{global}} \left(\frac{1}{64}\right)(16)(2)(2) = (-1)(1) = -1, \quad \left(-\frac{\hbar g^2}{16}\right) w_{T,\text{rows}} = +\frac{\hbar g^2}{16}. \quad (4.70)$$

The rank-two finite spinor word is reduced by the four-dimensional identity

$$\sigma \cdot L \bar{\sigma} \cdot p \sigma \cdot L = 2(L \cdot p) \sigma \cdot L - \bar{L}^2 \sigma \cdot p, \quad (4.71)$$

whose two factors of $2L$ give $4\sigma \cdot L \bar{\sigma} \cdot p \sigma \cdot L = 8(L \cdot p) \sigma \cdot L - 4\bar{L}^2 \sigma \cdot p$; the first term belongs to the full- d Schwinger-exact tensor reduction, and the selected inverse-square parent and its cut differ by

$$-4(\bar{L}^2 - L_d^2) \sigma \cdot p = -4\mu_\ell^2 \sigma \cdot p, \quad (4.72)$$

which vanishes identically if $\bar{L}^2$ is replaced by L_d^2 before the cut subtraction. For the reflected ordering the dotted-index reordering gives

$$(\partial^{\dot{\alpha}} \nabla_+ \mathcal{W}_+^D) \widetilde{\mathcal{W}}_{\dot{\alpha}}^E = -(\partial_{\dot{\alpha}} \nabla_+ \mathcal{W}_+^D) \widetilde{\mathcal{W}}^{E\dot{\alpha}}, \quad (4.73)$$

one minus sign per reordering, which is the origin of the antisymmetric ordered pair of outputs. The exact marked words behind the endpoint table are

$$\mathcal{G}_1 = (b_{0+} + b_{1+}) W_{02}, \quad \mathcal{G}_2 = (b_{0+} - b_{1+}) W_{02}, \quad W_{02} = r_{0+} \wedge r_{2+}, \quad (4.74)$$

with $b_{0+} - b_{1+} = q_{+\perp}$ purely external, so $\mathcal{G}_2$ has no rank-two loop component (the rows marked “no” in the table of paragraph 1), while $b_{0+} + b_{1+} = 2L_{+\perp} + \text{external}$ is rank two (the rows marked “yes”).

Remark 4.14 (Status of the sign table). The endpoint-row assignment (4.69) and the global sign $s_{\text{global}} = -1$ are recorded primitives of the computation: no independent human-readable noncommutative D -algebra trace emitting them is quoted here. They are superseded in effect by the exact exterior-algebra words (4.74) and by the machine replays of Remark 4.16, which confirm the full factor chain with zero failures.

6. Evanescent extraction and tensor reduction. On every marked line e of every one-loop graph the cutting-failure identity (4.15) isolates one universal insertion:

$$\frac{\bar{r}_e^2}{D_0 D_1 D_2} - \frac{1}{\prod_{j \neq e} D_j} = \frac{\mu_\ell^2}{D_0 D_1 D_2}, \quad (4.75)$$

with $D_i = r_{i,d}^2$; replacing $\bar{r}_e^2$ by $r_{e,d}^2$ before the subtraction makes the expression identically zero and removes the anomaly. With $D_0 = \ell^2$, $D_1 = (\ell + q)^2$, $D_2 = (\ell + p + q)^2$, the ordered Feynman parametrization is

$$\frac{1}{D_0 D_1 D_2} = 2 \int_{x,y,z \geq 0} dx dy dz \delta(1-x-y-z) \frac{1}{(r^2 + \Delta)^3}, \quad \int_{x,y,z \geq 0} dx dy dz \delta(1-x-y-z) = \frac{1}{2}, \quad (4.76)$$

with the shift, writing $Q = p + q$,

$$\ell = k + yq + zQ, \quad \Delta = xyq^2 + xzQ^2 + yzp^2. \quad (4.77)$$

The two numerator legs $L_1 = 2k + q$ and $L_2 = 2k + p + 2q$ become

$$L_1 = 2\ell + (1 - 2y)q - 2zQ, \quad L_2 = 2\ell + (1 - 2y)q + (1 - 2z)Q, \quad (4.78)$$

so the odd- ℓ terms integrate to zero and only the $\ell_m \ell_n$ piece survives. Hatted rotational reduction and (4.19) give

$$\int_r \frac{r_m r_n}{(r^2 + \Delta)^3} = \frac{\widehat{\delta}_{mn}}{d} (J_2 - \Delta J_3) = \frac{\widehat{\delta}_{mn}}{4} J_2, \quad (4.79)$$

the $\frac{1}{4}$ being exact since $(1 - \frac{\epsilon}{2})/(4 - 2\epsilon) = \frac{1}{4}$ identically in ϵ . Consequently the pole part of the parent triangle with numerator legs $L_1^m L_2^n$ is

$$\text{Pole}_{\epsilon=0} \int_\ell \frac{L_1^m L_2^n}{D_0 D_1 D_2} = \frac{\widehat{\delta}^{mn}}{16\pi^2 \epsilon}. \quad (4.80)$$

Assembling the prefactor (4.68), the simplex factor (4.76), the residue (4.80), and the spinor chain $\sigma_m \bar{\sigma}_\rho \sigma_n$ contracted with the hatted external momentum p^ρ , the parent triangle pole is

$$\Gamma_T = \frac{\hbar g^2}{32\pi^2 \epsilon} f^{ACD} f^{BCE} \widehat{\delta}^{mn} \sigma_m \bar{\sigma}_\rho \sigma_n p^\rho. \quad (4.81)$$

7. Schwinger contact rows. The four occurrence-resolved contact rows of Figure 3 are

row	cut edge	marked occurrence	Wick port row
$j = 1$	e_0	$\nabla_- (\nabla_+ \mathcal{W}_+^A)$	(S, B)
$j = 2$	e_0	$\nabla_- (\nabla_+ \mathcal{W}_+^A)$	(B, S)
$j = 3$	e_2	$\nabla_+ \mathcal{W}_+^A \nabla_- (\nabla_+ \mathcal{W}_+^B)$	(S, B)
$j = 4$	e_2	$\nabla_+ \mathcal{W}_+^A \nabla_- (\nabla_+ \mathcal{W}_+^B)$	(B, S)

and each row carries the same sign chain

$$\left(-\frac{1}{4}\right)_{\text{cut}} (-1)_{\text{endpoint}} \left(\frac{1}{2}\right)_{\nabla_- \nabla_+ = \nabla^2/2} (2)_{\text{mixed}} (2)_{\text{Wick}} \left(-\frac{1}{2}\right)_{p_+} = -\frac{1}{4}, \quad (4.82)$$

whose factors are, in order: the cut/insertion coefficient, the endpoint transport sign, the spinor-derivative square move (4.57), the mixed anticommutator factor (4.65), the two Wick endpoints, and the plus-frame momentum projection. Like the endpoint-row signs of Remark 4.14, this per-row chain is a recorded primitive of the underlying ledger, independently machine-verified row by row. There is no extra edge factor 2 and no metric-trace factor d or 4 on the contact side. A single row gives

$$\Gamma_{C,j} = -\frac{1}{4} \cdot \frac{\hbar g^2}{2} \cdot \frac{1}{16\pi^2\epsilon} f^{ACD} f^{BCE} \delta_4^{mn} \sigma_m \bar{\sigma}_\rho \sigma_n p^\rho = -\frac{\hbar g^2}{128\pi^2\epsilon} f^{ACD} f^{BCE} \delta_4^{mn} \sigma_m \bar{\sigma}_\rho \sigma_n p^\rho, \quad (4.83)$$

and the four occurrence-resolved rows sum to

$$\Gamma_C = \sum_{j=1}^4 \Gamma_{C,j} = -\frac{\hbar g^2}{32\pi^2\epsilon} f^{ACD} f^{BCE} \delta_4^{mn} \sigma_m \bar{\sigma}_\rho \sigma_n p^\rho. \quad (4.84)$$

8. Pole cancellation and the -2ϵ trace. The four-dimensional metric parts of (4.81) and (4.84) cancel exactly: since $\hat{\delta}^{mn} - \delta_4^{mn} = -\check{\delta}^{mn}$,

$$\Gamma_T + \Gamma_C = -\frac{\hbar g^2}{32\pi^2\epsilon} f^{ACD} f^{BCE} \check{\delta}^{mn} \sigma_m \bar{\sigma}_\rho \sigma_n p^\rho, \quad (4.85)$$

a pure evanescent numerator, precisely as in the bookkeeping principle of Remark 4.6. The external momentum is hatted, so contracting with p^ρ first and using $\check{\delta} p = 0$ and $\check{\delta}^{mn} \bar{\sigma}_m \sigma_n = \check{\delta}^m_m = 2\epsilon$ with the four-dimensional spin algebra (4.13),

$$p^\rho \check{\delta}^{mn} \sigma_m \bar{\sigma}_\rho \sigma_n = 2p^\rho \check{\delta}_\rho^n \sigma_n - p^\rho \sigma_\rho \check{\delta}^{mn} \bar{\sigma}_m \sigma_n = 0 - (\check{\delta}^m_m) p^\rho \sigma_\rho = -2\epsilon p^\rho \sigma_\rho. \quad (4.86)$$

9. The loop integral and the complete unit-insertion result. The shifted coefficient is independent of the simplex parameters, so $2 \int_{\Sigma_2} 1 = 2(\frac{1}{2}) = 1$ by (4.76). Substituting (4.86) into (4.85) and using the evanescent master integral (4.16),

$$\Gamma_T + \Gamma_C = -\frac{\hbar g^2}{32\pi^2\epsilon} (-2\epsilon) f^{ACD} f^{BCE} \sigma_\rho p^\rho = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \sigma \cdot p. \quad (4.87)$$

The σ -chain trace forces the position-space output to be a dotted bilinear of the form (4.3): this is the output structure quoted as grading (iv) in Section 4.4. The directed triangle emits the ordered word $\langle \widetilde{\mathcal{W}}^D, \nabla_+ \mathcal{W}_+^E \rangle$; the reflected triangle emits the reordered word with the minus sign of (4.73); the two words in which the holomorphic momentum acts along the q -leg are total derivatives (equation-of-motion carriers) and drop out of the physical output. The gauge sector of the channel is therefore

$$\Gamma_{(b,b),\text{gauge}}^{(1)} = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \{ \langle \widetilde{\mathcal{W}}, \nabla_+ \mathcal{W}_+ \rangle - \langle \nabla_+ \mathcal{W}_+, \widetilde{\mathcal{W}} \rangle \}^{DE}. \quad (4.88)$$

The matter sector of the same channel is the two- $\widetilde{\Phi} \mathcal{V} \Phi$ marked triangle. Its two marked placements give, with the same master integral (4.16),

$$4\hbar g^2 \left(\frac{2}{3}\right) \left(\frac{1}{32\pi^2}\right) = \frac{\hbar g^2}{12\pi^2}, \quad (4.89)$$

$$4\hbar g^2 \left(-\frac{1}{6}\right) \left(\frac{1}{32\pi^2}\right) = -\frac{\hbar g^2}{48\pi^2}, \quad (4.90)$$

and the two placements sum to

$$\frac{\hbar g^2}{12\pi^2} - \frac{\hbar g^2}{48\pi^2} = \frac{3\hbar g^2}{48\pi^2} = \frac{\hbar g^2}{16\pi^2}, \quad (4.91)$$

carried by the matter words $\sum_r (\langle \nabla_+ \Phi_r, \tilde{\Phi}^r \rangle - \langle \tilde{\Phi}^r, \nabla_+ \Phi_r \rangle)$. Collecting the gauge sector (4.88) and the matter sector (4.91), the complete channel in unit insertions is

$$\begin{aligned} \nabla_- \left(\nabla_+ \mathcal{W}_+^A \nabla_+ \mathcal{W}_+^B \right) \Big|_{1\text{-loop}} &= \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \left\{ \langle \widetilde{\mathcal{W}}, \nabla_+ \mathcal{W}_+ \rangle - \langle \nabla_+ \mathcal{W}_+, \widetilde{\mathcal{W}} \rangle \right. \\ &\quad \left. + \sum_{r=1}^3 \left(\langle \nabla_+ \Phi_r, \tilde{\Phi}^r \rangle - \langle \tilde{\Phi}^r, \nabla_+ \Phi_r \rangle \right) \right\}^{DE}, \end{aligned} \quad (4.92)$$

with the dotted contraction of (4.3) understood in each bilinear.

Remark 4.15 (The matter source-orientation sign). The relative size of the two matter placements in (4.89) and (4.90) is exact; the overall matter source-orientation sign is fixed by the target-blind primitive-sign chain ($\eta_{\text{src}} = +1$), giving the word order displayed in (4.92). The same sign is consistent with the accepted component bottom projection of Section 5 (an exact projection of the superfield result, not an independent derivation) and is checked independently by the holomorphic-twist comparison of paragraph 11.

10. Restoring the letter normalization: the boxed result. The letters of Definition 4.1 give the substitutions

$$\nabla_+ \mathcal{W}_+ = i\sqrt{2}b, \quad (\nabla_+ \mathcal{W}_+^A)(\nabla_+ \mathcal{W}_+^B) = (i\sqrt{2})^2 b^A b^B = -2b^A b^B, \quad (4.93)$$

$$\langle \widetilde{\mathcal{W}}, \nabla_+ \mathcal{W}_+ \rangle = \langle -i\partial c, i\sqrt{2}b \rangle = \sqrt{2} \langle \partial c, b \rangle, \quad \langle \nabla_+ \Phi_r, \tilde{\Phi}^r \rangle = \langle \sqrt{2}\beta_r, \gamma^r \rangle = \sqrt{2} \langle \beta_r, \gamma^r \rangle, \quad (4.94)$$

so that dividing (4.92) by the insertion factor -2 yields

$$\nabla_- (b^A b^B) = \frac{1}{-2} \cdot \sqrt{2} \cdot \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \{ \dots \} = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \{ \dots \}. \quad (4.95)$$

Equivalently, in the bookkeeping of the unit-insertion frame: the two insertions contribute a factor $(-i/\sqrt{2})^2 = -\frac{1}{2}$ relative to the unit field-strength normalization used above, while the output bilinears of the seed channel contribute $\sqrt{2}$ in the alphabet of Definition 4.1, giving the net prefactor $-\frac{1}{2} \cdot \sqrt{2} \cdot \frac{\hbar g^2}{16\pi^2} = -\frac{\hbar g^2}{16\sqrt{2}\pi^2}$. The position-space identity reads

$$\boxed{\nabla_- (b^A b^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \left\{ \langle \partial c, b \rangle - \langle b, \partial c \rangle + \sum_{r=1}^3 (\langle \beta_r, \gamma^r \rangle - \langle \gamma^r, \beta_r \rangle) \right\}^{DE}.} \quad (4.96)$$

Remark 4.16 (Independent machine cross-checks). The factor chain of this subsection was replayed externally on 269 external slot assignments with zero failures, and on the full color mask (9216 entries, with a sparse replay of 2048) with zero equality failures.

11. The holomorphic-twist comparison for this channel. Budzik and Kulp compute the same channel in the holomorphic twist; their zero-derivative display [1, eq. (3.64)] is

$$Q_1(b^A b^B) = \kappa^2 f_{ACD} f_{BCE} [\partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} b^E - \partial_{\dot{\alpha}} b^D \partial^{\dot{\alpha}} c^E + \partial_{\dot{\alpha}} (\beta_I)^D \partial^{\dot{\alpha}} (\gamma^I)^E - \partial_{\dot{\alpha}} (\gamma^I)^D \partial^{\dot{\alpha}} (\beta_I)^E], \quad (4.97)$$

with κ the dual Coxeter normalization $\text{Tr } t_A t_B = \kappa \delta_{AB}$. The dictionary from the twist letters to the unit-normalized composites of this section is

$$b \mapsto -\frac{i\rho}{\sqrt{2}} \nabla_+ \mathcal{W}_+, \quad \beta_r \mapsto \frac{\rho}{\sqrt{2}} \nabla_+ \Phi_r, \quad \gamma^r \mapsto \rho \tilde{\Phi}^r, \quad \partial_{\dot{\alpha}} c \mapsto i\rho \tilde{\mathcal{W}}_{\dot{\alpha}}, \quad (4.98)$$

with color, flavor, and dotted-spin frames R_A^P , U_I^r , $S_{\dot{\alpha}}^{\dot{\beta}}$ and a scale $\rho \neq 0$, all set to the identity after tracking. Applied word by word, the dictionary gives

$$\begin{aligned} \partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} b^E &\mapsto (i\rho) \left(-\frac{i\rho}{\sqrt{2}} \right) \langle \tilde{\mathcal{W}}^D, \nabla_+ \mathcal{W}_+^E \rangle = \frac{\rho^2}{\sqrt{2}} \langle \tilde{\mathcal{W}}^D, \nabla_+ \mathcal{W}_+^E \rangle, \\ -\partial_{\dot{\alpha}} b^D \partial^{\dot{\alpha}} c^E &\mapsto -\frac{\rho^2}{\sqrt{2}} \langle \nabla_+ \mathcal{W}_+^D, \tilde{\mathcal{W}}^E \rangle, \\ \partial_{\dot{\alpha}} (\beta_I)^D \partial^{\dot{\alpha}} (\gamma^I)^E &\mapsto \frac{\rho^2}{\sqrt{2}} U_I^r (U^{-1})_s^I \langle \nabla_+ \Phi_r^D, \tilde{\Phi}^{sE} \rangle = \frac{\rho^2}{\sqrt{2}} \sum_r \langle \nabla_+ \Phi_r^D, \tilde{\Phi}^{rE} \rangle, \\ -\partial_{\dot{\alpha}} (\gamma^I)^D \partial^{\dot{\alpha}} (\beta_I)^E &\mapsto -\frac{\rho^2}{\sqrt{2}} \sum_r \langle \tilde{\Phi}^{rD}, \nabla_+ \Phi_r^E \rangle. \end{aligned} \quad (4.99)$$

The input pair contributes $(-i\rho/\sqrt{2})^2 = -\rho^2/2$, undoing it contributes $-2/\rho^2$, and the total-differential scale $s_1 = -1/2$ contributes $s_1^{-1} = -2$; the net prefactor on every one of the eight words is therefore

$$\left(-\frac{2}{\rho^2} \right) (-2) \kappa_H^2 \hbar_{\text{HT}} \frac{\rho^2}{\sqrt{2}} = 2\sqrt{2} \kappa_H^2 \hbar_{\text{HT}} = \frac{\hbar g^2}{16\pi^2}, \quad (4.100)$$

with coefficient vector, in the ordered basis of (4.92),

$$\left(\langle \tilde{\mathcal{W}}, \nabla_+ \mathcal{W}_+ \rangle, \langle \nabla_+ \mathcal{W}_+, \tilde{\mathcal{W}} \rangle, \langle \nabla_+ \Phi_1, \tilde{\Phi}^1 \rangle, \langle \tilde{\Phi}^1, \nabla_+ \Phi_1 \rangle, \dots \right) = (1, -1, 1, -1, 1, -1, 1, -1), \quad (4.101)$$

matching (4.92) entry by entry. Equivalently, the loop-scale relation

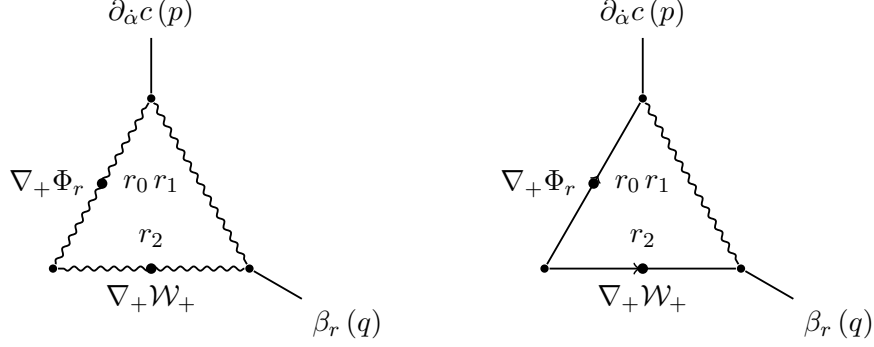
$$-2 \kappa_H^2 \hbar_{\text{HT}} = -\frac{\hbar g^2}{16\sqrt{2} \pi^2}, \quad \text{i.e.} \quad \kappa_H^2 \hbar_{\text{HT}} = \frac{\hbar g^2}{32\sqrt{2} \pi^2}, \quad (4.102)$$

identifies the twist computation with the boxed result (4.96): the matter words of (4.97), which [1] obtains from the superpotential vertex, are the same matter triangle as in paragraph 9. Only the zero-jet kernel entry $K_{0,0;0,0} = 1$ enters this channel; Budzik and Kulp's printed master-triangle value is $\frac{1}{2}$, exactly one half, the factor 2 being the Feynman-parameter prefactor of (4.76). The full tower comparison is Section 7.

Remark 4.17 (Status of the comparison). The match of this paragraph is an external cross-check against [1], not a derivation input and not a completeness proof of the graph census: the factor chain of paragraphs 1–10 was sealed before the twist formulas were consulted. The loop-scale relation (4.102) uses the conditional total-differential scale $s_1 = -1/2$ of the dictionary. The adversarial census review remains open; it does not alter the accepted coefficient (4.96).

4.7 The channels (b, β_r) and (β_r, b)

We now compute the mixed family $\nabla_-(b^A \beta_r^B)$ and $\nabla_-(\beta_r^A b^B)$ in full, taking $r = 1$ without loss (flavor transport gives $r = 2, 3$). It is the only family whose output word is not read off a cut parent graph directly: its raw graph carrier contains an equation-of-motion (total-derivative) layer that must be quotiented, and a finite normal-product term that must be added, before the renormalized answer lands on the channel-table row (4.272). Graph weights use unit-normalized insertions $\nabla_+ \mathcal{W}_+$, $\nabla_+ \Phi_r$; the letter normalizations (4.1) are restored at the end.



G1: three vector internal lines G2/G3: two matter lines, one vector line

Figure 4: The ordered triangles of the $(b, \beta_r)/(\beta_r, b)$ family in the routing (4.103): one $\nabla_+ \mathcal{W}_+$ and one $\nabla_+ \Phi_r$ insertion (filled dots), residual external legs $\partial c(p)$, $\beta_r(q)$. The right panel shows the G2 line content (vector edge r_1 , legs $\partial c, \beta_r$); the two G3 routings share the vertex structure with the vector edge on r_0 and the antichiral legs γ^s, γ^t in place of $\partial c, \beta_r$, one matter-cubic vertex replaced by the antichiral superpotential vertex.

1. Graph inventory. Three ordered-triangle families contribute, with routing, external slot map, and denominators

$$r_0 = \ell, \quad r_1 = \ell - p, \quad r_2 = \ell - p - q, \quad \partial c = \partial c(p), \quad \beta_r = \beta_r(q), \quad D_i = r_{i,d}^2 \quad (4.103)$$

(the reflected slot map $\beta_r(p), \partial c(q)$ is false): **G1**, the gauge triangle (three vector internal lines, one matter-cubic and one gauge-cubic vertex; the $\nabla_+ \mathcal{W}_+$ mark tags r_2 , the $\nabla_+ \Phi_r$ mark tags r_0), output carrier $\langle \partial c, \beta_r \rangle$ for (b, β_r) ; **G2**, the mixed triangle (one vector and two matter internal lines, two matter-cubic vertices), output $\langle \beta_r, \partial c \rangle$ for (b, β_r) ; **G3**, two ordered triangles $G_{3,2}, G_{3,3}$ (one matter-cubic and one antichiral superpotential vertex; internal flavors $(1, 2), (1, 3)$), outputs $\langle \gamma^2, \gamma^3 \rangle, \langle \gamma^3, \gamma^2 \rangle$. Each ordering is reflected for (β_r, b) ; the line contents are displayed in Figure 4. Excluded from the graph sum: the zero-square current bubbles JE, JX, PE, PX , which cancel in exact pairs ($I_{JE} + I_{JX} = 0, I_{PE} + I_{PX} = 0$) and contain no four-dimensional inverse square, so each DRED defect is zero; and the separately drawn source-resolvent bubbles, which are presentations of the same induced derivative-of-action contact and are not independent anomaly graphs.

2. Wick contractions. The Euclidean propagators in the canonically normalized frames are

$$\langle \mathcal{V}^A(p, 1) \mathcal{V}^B(-p, 2) \rangle_E = -\frac{2\hbar g^2 \kappa^{AB}}{p^2} \delta^4(\theta_{12}), \quad \langle \Phi_r^A(p, 1) \tilde{\Phi}^{sB}(-p, 2) \rangle_E = \delta_r^s \frac{\hbar g^2 \kappa^{AB}}{16 p^2} \bar{\mathcal{D}}_1^2 \mathcal{D}_1^2 \delta^4(\theta_{12}). \quad (4.104)$$

For G3 the flavor- and chirality-preserving Wick pairing is unique: the vector legs of the source and matter vertices pair along r_0 , the matter leg of the matter vertex pairs with an antichiral leg of the superpotential vertex along r_1 , and the source matter leg pairs with the remaining antichiral leg along r_2 ; the two uncontracted legs are the external legs. There is no second preserving pairing and no graph automorphism division, so

$$C_{\text{prop}} = (-\hbar) \left(\frac{\hbar}{16} \right)^2 = -\frac{\hbar^3}{256}. \quad (4.105)$$

Every primitive weight is one:

$$\left[\frac{1}{2} (2)_{\text{cyc. source}} \right] \left[\frac{1}{2!} (2)_{\text{vertex orders}} \right] \left[\frac{1}{3!} (6)_{\text{gauge-vertex labels}} \right] \left[\frac{1}{3!} (6)_{\mathcal{V}\Phi\tilde{\Phi}} \right] (1)_{\text{Wick}} = 1 \text{ (G1)}, \quad \left[\frac{1}{2} (2) \right] \left[\frac{1}{2!} (2) \right] \left[\frac{1}{3!} (6)_{\text{sup}} \right] \quad (4.106)$$

3. Amplitude assembly. The mixed source Hessian, the matter-cubic vertex, and the antichiral superpotential vertex (from W of Section 2) contribute, with (4.105) and the unit factorial weights (4.106),

$$C_I = -\frac{g^2}{4\sqrt{2}}, \quad C_M = \frac{\sqrt{2}g}{\hbar}, \quad C_{H_-} = -\frac{\sqrt{2}g}{\hbar}; \quad C_{\text{preD}}^{G2} = C_I C_M^2 C_{\text{prop}} = +\frac{\sqrt{2}\hbar g^4}{1024}, \quad C_{\text{preD}}^{32} = C_I C_M C_{\text{prop}} \quad (4.107)$$

the relative sign of $G_{3,2}, G_{3,3}$ being the ordered flavor/color reflection sign of the superpotential vertex. The residual external-leg operators are: for G1, the pairing $X_{\partial c \beta} := \langle \partial c, \beta_r \rangle$ and the equation-of-motion composite $E := \beta_r (P^{\dot{\alpha}} \partial_{\dot{\alpha}} c)$ ($P^{\dot{\alpha}}$ the holomorphic momentum of the ∂c leg in Fourier space); for G2, the single word $\frac{1}{3} \beta_r (4p - q)^{\dot{\alpha}} \partial_{\dot{\alpha}} c$; for G3, the uncontracted γ^s, γ^t legs, contracted by the wedge $W_{ij} := r_{i,+} \wedge r_{j,+} = r_{i,+ \dot{\alpha}} r_{j,+}^{\dot{\alpha}}$, $W_{ji} = -W_{ij}$.

4. The D -algebra chain. Discarding a longitudinal word at edge e against the neighboring projector at edge n transports the occurrence by superspace integration by parts; the endpoint sign, the corner identity, and the loop-saturation 16 give the universal transport factor and its full- d companion

$$\left(-\frac{1}{2}\right) (-1)(16) = 8, \quad 8 \bar{r}_n^2 \mathcal{T}_{e|n} - 8 r_{n,d}^2 \mathcal{T}_{e|n} = 8 \mu_\ell^2 \mathcal{T}_{e|n}. \quad (4.108)$$

For G3 the raw Grassmann-measure monomial, converted to the full measure, settles the unit

$$32768 \left(\frac{1}{4}\right)_{d^4 \theta_M} \left(\frac{1}{2}\right)_{d^2 \bar{\theta}_H} \left(-\frac{1}{2}\right)_{\mu_\ell^2 \text{ extraction}} (2)_{\text{trace/metric}} = -4096, \quad (4.109)$$

the trace/metric factor 2 being algebraic and canceled by the $-\frac{1}{2}$ rank extraction, not a graph multiplicity. The three kinetic cuts and their parents are, with $W_0 = W_{12}$, $W_1 = W_{P0}$, $W_2 = W_{p0}$,

$$K_0 = +4096 W_0, \quad K_1 = -4096 W_1, \quad K_2 = +4096 W_2, \quad (4.110)$$

$$P_0 = -4096 (D_0 + \mu_\ell^2) W_0, \quad P_1 = +4096 (D_1 + \mu_\ell^2) W_1, \quad P_2 = -4096 (D_2 + \mu_\ell^2) W_2. \quad (4.111)$$

For G2 the selected and transported occurrences give

$$-\frac{8(D_2 + \mu_\ell^2) \mathcal{R}}{D_0 D_1 D_2} + \frac{8 \mathcal{R}}{D_0 D_1} = -\frac{8 \mu_\ell^2 \mathcal{R}}{D_0 D_1 D_2}, \quad -\frac{1}{2} (D_1 + \mu_\ell^2) W + \frac{1}{2} D_1 W = -\frac{1}{2} \mu_\ell^2 W, \quad (4.112)$$

and every G1 row obeys $F_A + C_{A,d} = -8R_A \mu_\ell^2$, $F_B + C_{B,d} = -8R_B \mu_\ell^2$, with $C_{A,d} = -L_A - \alpha_A r_{2,d}^2$, $\alpha_A = -8R_A$, $L_A = -\frac{1}{2} \mathcal{D}_+ \mathcal{D}^2 \mathcal{D}^2$. The wedge algebra of the three G3 remainders closes pointwise: with $r_1 = r_0 - p$, $r_2 = r_0 - p - q$,

$$-W_{12} + W_{P0} - W_{p0} = -p_+ \wedge q_+ \implies \mathcal{R}_{G3}^{\text{full}} = -\frac{4096 \mu_\ell^2 (p_+ \wedge q_+)}{D_0 D_1 D_2}. \quad (4.113)$$

5. Evanescent extraction and the rank-one moments. Every remainder of step 4 is governed by the cutting-failure identity (4.15) and the master integral (4.16). The rank-one moments are the ordered-simplex averages

$$\lim_{\epsilon \rightarrow 0} \int_\ell \frac{\mu_\ell^2 \bar{r}_0^\mu}{D_0 D_1 D_2} = \frac{2p^\mu + q^\mu}{96\pi^2}, \quad \lim_{\epsilon \rightarrow 0} \int_\ell \frac{\mu_\ell^2 \bar{r}_1^\mu}{D_0 D_1 D_2} = \frac{q^\mu - p^\mu}{96\pi^2}, \quad \lim_{\epsilon \rightarrow 0} \int_\ell \frac{\mu_\ell^2 \bar{r}_2^\mu}{D_0 D_1 D_2} = -\frac{p^\mu + 2q^\mu}{96\pi^2}, \quad \lim_{\epsilon \rightarrow 0} \int_\ell \frac{\mu_\ell^2}{D_0} \quad (4.114)$$

from $2 \int_{\Sigma_2} x_i dx_1 dx_2 = \frac{1}{3}$ per coordinate. Applied graph by graph: G1 integrates to the mark coefficients $(\frac{4}{3}, \frac{2}{3})$, $(\frac{2}{3}, \frac{4}{3})$ in the two slot orderings, i.e. the word $2(p+q)^{\dot{\alpha}} \beta_r \partial_{\dot{\alpha}} c = 2 P^{\dot{\alpha}} (\beta_r \partial_{\dot{\alpha}} c)$, a pure total derivative vanishing in the quotient; G2 integrates to the selected word $\frac{2}{3} \beta_r (2p +$

$q)^{\dot{\alpha}} \partial_{\dot{\alpha}} c$ plus the transported word $-\beta_r q^{\dot{\alpha}} \partial_{\dot{\alpha}} c$, i.e. $\frac{1}{3} \beta_r (4p - q)^{\dot{\alpha}} \partial_{\dot{\alpha}} c$ with pair/equation-of-motion coefficients $(-\frac{1}{3}, \frac{4}{3})$ on the graph-word placement (Remark 4.20); G3 carries the marked simplex weights $\frac{2}{3}, \frac{1}{3}$ per mark and sums to the raw wedge coefficients $+2\sqrt{2}i$ ($G_{3,2}$) and $-2\sqrt{2}i$ ($G_{3,3}$) in units of $\hbar g^2/(16\pi^2)$. Passing from the raw momentum wedge to the dotted bilinear flips the sign once, since with the locked phase e^{ipx} , $(ip)_+ \wedge (iq)_+ = -(p_+ \wedge q_+)$.

6. Pole cancellation. No $1/\epsilon$ pole occurs anywhere in this family: every parent-plus-contact pair vanishes identically at $\mu_\ell^2 = 0$,

$$\frac{P_0}{D_0 D_1 D_2} + \frac{K_0}{D_1 D_2} = -\frac{4096 \mu_\ell^2 W_0}{D_0 D_1 D_2}, \quad \frac{P_1}{D_0 D_1 D_2} + \frac{K_1}{D_0 D_2} = +\frac{4096 \mu_\ell^2 W_1}{D_0 D_1 D_2}, \quad \frac{P_2}{D_0 D_1 D_2} + \frac{K_2}{D_0 D_1} = -\frac{4096 \mu_\ell^2 W_2}{D_0 D_1 D_2}, \quad (4.115)$$

and likewise for the G2 pairs (4.112) and the G1 rows. Only the evanescent remainders survive. (Contrast the seed channel (b, b) , where literal $1/\epsilon$ poles cancel between the parent triangle and the Schwinger contact rows.)

7. Raw carrier, total-derivative quotient, finite shift. Collect the integrated outputs as coefficient vectors of unit-normalized carriers in the ordered basis of candidate bilinears

$$(X_{\partial c \beta}, E, X_{\beta \partial c}, E, C_{st}, C_{ts}), \quad X_{\beta \partial c} := \langle \beta_r, \partial c \rangle, \quad C_{st} := \langle \gamma^s, \gamma^t \rangle, \quad C_{ts} := \langle \gamma^t, \gamma^s \rangle, \quad (4.116)$$

with $\varepsilon_{rst} = +1$ and $E := \beta_r (P^{\dot{\alpha}} \partial_{\dot{\alpha}} c)$ the equation-of-motion composite of step 3, entering in both orderings. The two pairings are a single carrier: the odd-odd Koszul swap ($|\partial c| = |\beta_r| = 1$) and the dotted-index flip $\psi^{\dot{\alpha}} \chi_{\dot{\alpha}} = -\psi_{\dot{\alpha}} \chi^{\dot{\alpha}}$ of the locked convention cancel,

$$X_{\partial c \beta} = (\partial_{\dot{\alpha}} c)(\partial^{\dot{\alpha}} \beta_r) = -(\partial^{\dot{\alpha}} \beta_r)(\partial_{\dot{\alpha}} c) = (\partial_{\dot{\alpha}} \beta_r)(\partial^{\dot{\alpha}} c) = X_{\beta \partial c} =: X. \quad (4.117)$$

There is accordingly one total-derivative identity, not two: the even derivation $P^{\dot{\alpha}}$ acting on the odd-odd product gives no Koszul sign, and the pair term enters with the index flip $(P^{\dot{\alpha}} \beta_r) \partial_{\dot{\alpha}} c = -\langle \beta_r, \partial c \rangle = -X$,

$$T := P^{\dot{\alpha}}(\beta_r \partial_{\dot{\alpha}} c) = (P^{\dot{\alpha}} \beta_r) \partial_{\dot{\alpha}} c + \beta_r (P^{\dot{\alpha}} \partial_{\dot{\alpha}} c) = -X + E, \quad (4.118)$$

so $aX + bE = (a+b)X + bT$ in either ordering. (Equivalently $P^{\dot{\alpha}}(\partial_{\dot{\alpha}} c \beta_r) = X - E = -T$: a single ray. The combination $X + E$ is not a total derivative, and X itself is not one either.) The raw pair coefficients displayed below are quoted on the graph-word placement $X_{>} := (\partial^{\dot{\alpha}} \beta_r) \partial_{\dot{\alpha}} c = -X$, where the same identity reads $T = X_{>} + E$, i.e. $aX_{>} + bE = (a - b)X_{>} + bT$: thus the G1 entries $(2, 2)_{(X_{>}, E)} = (0, 2)_{(X_{>}, T)}$ quotient to zero, and the G2 split and quotient are the machine-sealed replay values of Remark 4.20. The coefficient flow of the family is the aligned display

$$\text{raw graph carrier: } (2, 2, -\frac{1}{3}, \frac{4}{3}, -2i\sqrt{2}, +2i\sqrt{2}) \quad \text{on the six-carrier basis (4.116),} \quad (4.119)$$

$$\text{total-derivative quotient: } (0, 1, -2i\sqrt{2}, +2i\sqrt{2}) \quad \text{on } (X_{\partial c \beta}, X_{\beta \partial c}, C_{st}, C_{ts}), \quad (4.120)$$

$$\text{finite normal-product shift: } + (1, 0, +i\sqrt{2}, -i\sqrt{2}), \quad (4.121)$$

$$\text{renormalized vector: } (1, 1, -i\sqrt{2}, +i\sqrt{2}). \quad (4.122)$$

The shift is fixed by residual supersymmetry: acting with the q_r of (4.275) (stated with the channel table in Section 4.12) on a general vector $x = (x_1, x_2, x_3, x_4)$ in the four-component basis, the Ward conditions of the descendant system $q_s[x \cdot (\text{basis})] = i\delta_{sr} \Delta(b, b)$ of (4.277) —

the homogeneous $s = 2, 3$ mixing plus the $s = 1$ matching against the anchored (b, b) singlet, restricted to this flavor ansatz — read

$$x_1 - x_2 = 0, \quad i\sqrt{2}x_1 + x_3 = 0, \quad -i\sqrt{2}x_1 + x_4 = 0, \quad (4.123)$$

a rank-three system whose solution space is the ray $x = t(1, 1, -i\sqrt{2}, +i\sqrt{2})$: the renormalized vector (4.122) is exactly annihilated by (4.123), and the descendant identity (4.277) anchors the ray to the (b, b) scale, fixing $t = 1$.

Remark 4.18 (The finite shift is a scheme term, not a new anomaly graph). The vector (4.121) is a finite composite-source normal-product (scheme) term. It is not a triangle, collapsed graph, Schwinger cut, inverse-square occurrence, or new cutting failure: all bare-graph anomaly of this family enters through the μ_ℓ^2 remainders of steps 4–6, and the shift is precisely what removes the residual factor of two between the raw G3 $\gamma\gamma$ coefficients and the Ward-compatible ray.

Remark 4.19 (Granularity of the machine replays). Three ingredients are sealed at machine level and displayed here at the granularity of the extracted formulas: the full 192-row G1 D -word table (only test-point rows admit an analytic display; the row identities $F_A + C_{A,d} = -8R_A\mu_\ell^2$, $F_B + C_{B,d} = -8R_B\mu_\ell^2$ hold for every row), the replay producing the G2 raw residual, and the color-route phase $+i$ of the G3 coefficients. All quoted coefficients are the machine-checked ones.

Remark 4.20 (Placement sensitivity of the G2 slot map). The decomposition of the G2 word $\frac{1}{3}\beta_r(4p - q)\partial_{\dot{\alpha}}\partial_{\dot{\alpha}}c$ is placement-sensitive: on the graph-word placement $(\partial_{\dot{\alpha}}\beta_r)\partial_{\dot{\alpha}}c$ the pair coefficient is $-\frac{1}{3}$, while on the canonical pairing $\langle\beta_r, \partial c\rangle = -(\partial_{\dot{\alpha}}\beta_r)\partial_{\dot{\alpha}}c$ of (4.117) it is $+\frac{1}{3}$, and the consistent md-level total-derivative quotient of the displayed word on the canonical carrier is $\frac{5}{3}$. The intermediate slot-map displays mix the two placements (the typed slot map of the Omega-routing audit versus the identity $(P^{\dot{a}}B_1)D_{\dot{a}} = -\langle B_1, D\rangle$ of the external-divergence audit). The displayed G2 raw pair/EOM entries and the quotient value $+1$ in (4.120) are therefore machine-sealed by the typed replay rather than derived at md level. The renormalized endpoint does not depend on this link: the Ward ray and its unit scale are anchored by the descendant identity (4.277), and the full renormalized vector matches the holomorphic-twist target word by word in the 81/81 machine round-trip.

8. The final result. Restoring the letter normalizations (4.1) — the insertions contribute $(-i/\sqrt{2})(1/\sqrt{2}) = -i/2$, each dotted output pair contributes $-i\sqrt{2}$, and the γ pairs contribute 1, so the unit-normalized vector is $-i\sqrt{2}$ times the bracket and $(-i/2)(-i\sqrt{2}) = -1/\sqrt{2}$ — the renormalized vector (4.122) becomes the channel-table row

$$\nabla_-(b^A\beta_r^B), \quad \nabla_-(\beta_r^Ab^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD}f^{BCE} \{ \langle\beta_r, \partial c\rangle + \langle\partial c, \beta_r\rangle + \varepsilon_{rst}\langle\gamma^s, \gamma^t\rangle \}^{DE}, \quad (4.124)$$

in exact agreement with (4.272). The two orderings are equal: the source-order Koszul sign is $(-1)^{|b||\beta|} = (-1)^{0 \cdot 1} = +1$, the superpotential flavor/color reflection contributes $+1$, the simplex moments are invariant under the reflection relabeling, and the reflected vector maps word by word onto (4.122).

9. Comparison with the holomorphic twist. Budzik and Kulp print, for this family, [1, eq. (3.68)]

$$Q_1((\beta_I)^Ab^B) = \kappa^2 f_{ACD}f_{BCE} \left[\partial_{\dot{\alpha}}(\beta_I)^D \partial^{\dot{\alpha}}c^E + \partial_{\dot{\alpha}}c^D \partial^{\dot{\alpha}}(\beta_I)^E + \varepsilon_{IJK} \partial_{\dot{\alpha}}(\gamma^J)^D \partial^{\dot{\alpha}}(\gamma^K)^E \right]. \quad (4.125)$$

Under the letter dictionary of Section 7, the three words of (4.125) map exactly onto $\langle\beta_r, \partial c\rangle + \langle\partial c, \beta_r\rangle$ and $\varepsilon_{rst}\langle\gamma^s, \gamma^t\rangle$, and BK's prefactor κ^2 maps onto $-\hbar g^2/(16\sqrt{2}\pi^2)$: the match is word

by word, for both orderings. The reversed ordering $Q_1(b^A(\beta_I)^B)$ is not printed as a final display in [1]; it is fixed there by the descent relations, and the machine round-trip confirms the identical canonical word for both directions. Two source inconsistencies of [1] are recorded in Section 7: the missing $\Gamma(3) = 2$ in their printed triangle kernel, and the $-\frac{1}{4}$ versus $-\kappa^2$ clash, adjudicated to the main branch $-\kappa^2$.

Remark 4.21 (Status of the comparison). The match with (4.125) is a cross-check against an external target, not a completeness proof of the graph census: the adversarial seventeen-candidate census review remains open, and no holomorphic-twist coefficient enters the derivation of (4.124). Before the finite shift (4.121), the raw $G_{3,2}, G_{3,3}$ coefficients are exactly twice their twist counterparts in every common basis (since $\varepsilon_{1JK}X_{JK} = X_{23} - X_{32}$); the shift is what removes this factor two.

4.8 The channels (b, γ^r) and (γ^r, b)

We now compute the mixed vector–antichiral family: the six ordered channels (b, γ^r) and (γ^r, b) , $r = 1, 2, 3$, of the master identity (4.8). The two ordered insertions are

$$I_{b\gamma^r} = \nabla_- (b^A \gamma^{rB}), \quad I_{\gamma^r b} = \nabla_- (\gamma^{rA} b^B). \quad (4.126)$$

Both letters are Grassmann-even and $\nabla_- \gamma^r = 0$, so in each insertion the outer derivative marks only the b line; every nonlinear occurrence of the covariantized source is nevertheless retained. The two directions are computed independently below — no graph coefficient is copied from one ordering to the other. As in the seed channel, the graph weights are assembled with the unit-normalized field-strength insertion $X := \nabla_+ \mathcal{W}_+$; the letter normalization $b = -(i/\sqrt{2})X$ is restored only in the final step.

1. Graph inventory. The covariant completion of the insertion and its $\mathcal{D}_-$ -words are

$$X_c = X_1 + gX_2 + g^2X_3, \quad N_0 = \mathcal{D}_-X_1, \quad N_1 = \mathcal{D}_-X_2 + [\Gamma_1, X_1], \quad N_2 = \mathcal{D}_-X_3 + [\Gamma_1, X_2] + [\Gamma_2, X_1], \quad (4.127)$$

where $\nabla_- = \mathcal{D}_- + g\Gamma_1 + g^2\Gamma_2$ is the connection expansion of the semi-chiral derivative, while $\gamma^r = \tilde{\Phi}^r$ has no nonlinear completion. The connected order- g^2 orbit is

$$\Gamma_{g^2}^{(1)} = \langle I_2 \rangle_0 - \frac{1}{\hbar} \langle I_1 S_3 \rangle_{0,c} - \frac{1}{\hbar} \langle I_0 S_4 \rangle_{0,c} + \frac{1}{2\hbar^2} \langle I_0 S_3 S_3 \rangle_{0,c}, \quad (4.128)$$

with I_n the source term at order g^n , S_3 and S_4 the cubic and quartic action vertices, and $\langle \rangle_{0,c}$ the connected Gaussian average. The admitted parent triangle skeletons are the gauge–matter class $T_{GM}^{(r)} = (I, G, M_r)$ and the two-matter class $T_{MM}^{(r,r)} = (I, M_r, M_r)$, with $G = \mathcal{V}_{VVV}$ the cubic gauge vertex and $M_r = \mathcal{V}_{\tilde{\Phi}_r V \Phi_r}$ the matter–gauge cubic; the superpotential class $T_{MH} = (I, M_r, H_{\tilde{\Phi}_3})$ contributes two complement-flavor routes per direction, both exactly zero by (4.145) below, and there is no all-gauge parent for this letter pair. Per ordered direction the route census is

$$N_{TGM} = 6, \quad N_{TMM} = 1, \quad N_{TMH} = 2, \quad (4.129)$$

the six gauge–matter routes being the six permutations of the three internal edges (r_0, r_1, r_2) , each with two derivative placements times two Euclidean chiralities of the matter line. Figure 5 shows the ordered triangle in the two-matter topology.

2. Wick contractions. Every labeled Wick contraction carries weight 1: the $1/2!$ of the two cubic action copies is canceled by the two labeled action orderings,

$$w_{\text{Wick}} = \frac{1}{2!} (1 + 1) = 1, \quad (4.130)$$

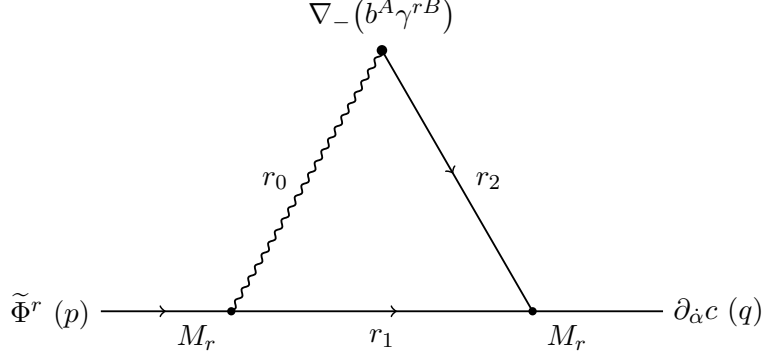


Figure 5: The ordered triangle of the (b, γ^r) family in the two-matter topology $T_{MM}^{(r,r)}$. The filled insertion node carries the letter operator $\nabla_-(b^A \gamma^{rB})$ with $b = -(i/\sqrt{2})\nabla_+ \mathcal{W}_+$; the internal edges are the source- b vector line r_0 (wavy), the chiral matter bridge r_1 , and the source- γ reversed-chiral line r_2 , with $r_1 - r_0 = p$. The external antichiral letter $\gamma^r = \tilde{\Phi}^r$ (momentum p) enters at the left matter vertex, the vertex where the vector edge r_0 attaches (each matter cubic vertex has exactly one vector slot); the vector output letter $\partial_{\dot{\alpha}c}$ (momentum q) enters at the right matter vertex. The gauge-matter topology $T_{GM}^{(r)}$ replaces the left matter vertex by the cubic gauge vertex G (the edges incident to G are vector lines); its six ordered routings of (r_0, r_1, r_2) are the six permutation routes of the text, and the reversed channel (γ^r, b) has the same figure with the source attachments exchanged.

and no residual symmetry factor occurs anywhere in the family. In the triangle term $\langle I_0 S_3 S_3 \rangle_{0,c}$ the two external ports attach to two distinct cubic vertices and the three internal edges are paired among the vertex slots, giving the six T_{GM} routings, the single T_{MM} routing (two matter propagators, hence two chiral projectors), and the two T_{MH} routings. In the nonlinear-source bubble $\langle I_1 S_3 \rangle_{0,c}$ the two internal legs of the connection word $[\Gamma_1, X_1]$ pair with each other through the loop, and the cubic matter vertex ties the loop to the external γ^r leg. The quartic seagull $\langle I_0 S_4 \rangle_{0,c}$ is a single graph: differentiating

$$[\mathcal{V}, [\mathcal{V}, \Phi]] = \mathcal{V}^2 \Phi - 2\mathcal{V} \Phi \mathcal{V} + \Phi \mathcal{V}^2 \quad (4.131)$$

with respect to one internal and one external $\mathcal{V}$ places both orderings inside one seagull vertex; they are not two Feynman graphs. The remaining source tadpole $\langle I_2 \rangle_0$ is a single coincident vector loop. The full color contraction of the family — two matter vertices, two matter inverse metrics, and the complete vector covariance — reduces, with $[T_A, T_B] = i f_{AB}^C T_C$ and $\text{tr}(T_A T_B) = \delta_{AB}/2$, to

$$C_{\text{full}} \propto \sum_X \text{tr}(T_D [T_A, T_X]) \text{tr}(T_X [T_E, T_B]) \longrightarrow + f^{ACD} f^{BCE}, \quad (4.132)$$

the numerical proportionality being carried into the assembled prefactors of paragraph 3. This is the adjudicated signed color word of the family: the same tensor unit as in the master identity (4.8).

3. Amplitude assembly. The propagators entering the orbit are the vector line and the two matter orientations,

$$-\frac{\hbar}{\square}, \quad +\frac{\hbar \bar{\mathcal{D}}^2 \mathcal{D}^2}{16 \square}, \quad +\frac{\hbar \mathcal{D}^2 \bar{\mathcal{D}}^2}{16 \square}; \quad (4.133)$$

each matter vertex contributes $+\sqrt{2}g/\hbar$ with measure $1/4$ per vertex, and the canonical source normalization is $-1/(4\sqrt{2})$. The two-matter orbit therefore assembles to the exact pre- D factor

$$\left(-\frac{1}{4\sqrt{2}}\right)\left(\frac{\sqrt{2}g}{\hbar}\right)^2\left[-\hbar\left(\frac{\hbar}{16}\right)^2\right] = \frac{\sqrt{2}\hbar g^2}{1024}, \quad (4.134)$$

while each gauge-matter route carries, per Euclidean chirality,

$$\text{preD}_- = +\frac{\hbar g^2}{8192\sqrt{2}}, \quad \text{preD}_+ = -\frac{\hbar g^2}{8192\sqrt{2}}. \quad (4.135)$$

After loop saturation the external legs carry residual operator words: the γ^r leg is a plain antichiral field (factor 1), and the vector output leg collapses as

$$\mathcal{D}^2\bar{\mathcal{D}}_{\dot{\alpha}}u = -4\sqrt{2}D_{\dot{\alpha}}, \quad \mathcal{D}^2\bar{\mathcal{D}}_{\dot{\alpha}}(\theta + \bar{\theta}\theta_{\dot{\alpha}}) = -2, \quad (4.136)$$

where u is the formal leg slot and $D_{\dot{\alpha}}$ the unit vector output slot (the antichiral field strength $\widetilde{\mathcal{W}}_{\dot{\alpha}}$; the letter $\partial_{\dot{\alpha}}c = i\widetilde{\mathcal{W}}_{\dot{\alpha}}$ is restored in paragraph 7). The ordered output kernels are $(\partial_{\dot{\alpha}}Y)^D D^{E\dot{\alpha}}$ for the two-matter orbit and $D_{\dot{\alpha}}^D(\partial^{\dot{\alpha}}Y)^E$ for the gauge-matter orbit, with $Y = \widetilde{\Phi}^r$.

4. The D -algebra chain. The master D -word behind every route is

$$\mathcal{D}_-\mathcal{D}_+\bar{\mathcal{D}}^2\mathcal{D}_+ = 8\bar{\square}\mathcal{D}_+ - \frac{1}{2}\mathcal{D}_+\bar{\mathcal{D}}^2\mathcal{D}^2, \quad (4.137)$$

with $\bar{\square}$ the d'Alembertian of the barred spin metric of (4.13); loop saturation uses $\delta^4(\theta_{12})\mathcal{D}^2\bar{\mathcal{D}}^2\delta^4(\theta_{12}) = 16\delta^4(\theta_{12})$ as in the seed channel. For the two-matter route the four-dimensional numerator before the metric split is

$$\mathcal{F}_{b>\gamma}^{(4)} = 8(\det r_0)\mathcal{R}_0 - 8(\det r_1)\mathcal{J}_1, \quad \mathcal{R}_0 = -128\mathfrak{b}(r_1), \quad \mathcal{J}_1 = -128\mathfrak{b}(r_0), \quad (4.138)$$

where r_0 is the source- b vector edge, r_1 the chiral bridge, $\det r_e$ the 2×2 determinant of the edge momentum (kept symbolic; its index-level definition is not needed below), and $\mathfrak{b}(\cdot)$ the external bilinear carrier. The collapsed neighbor of the same cut orbit, its migrated and transverse residuals, and the projector-collapse check are

$$K_{\text{neighbor}} = (64r_{0\dot{2}}, -64r_{0\dot{1}}), \quad (-128r_{0\dot{2}}, +128r_{0\dot{1}}), \quad (+128(p_{\dot{2}}-r_{0\dot{2}}), -128(p_{\dot{1}}-r_{0\dot{1}})), \quad 16\det(r_1)K_{\text{neighbor}} = \quad (4.139)$$

with the longitudinal tag migrating the source vector r_0 into the adjacent chiral projector r_1 ; for the reversed direction the inverse-edge rows read $+8\det(r_2)\mathcal{R}_1$ (primary) and $-8\det(r_1)\mathcal{R}_2$ (neighbor after longitudinal integration by parts). The finite-Grassmann replay of the nonlinear contact words returns the same polynomials in the reversed ports for the two directions; no reflected coefficient is assumed. Route by route, the full- d Schwinger contact identity

$$\mathcal{F}_{\text{route}}^{(4)} + \mathcal{C}_{\text{source}}^{(d)} + \mathcal{C}_{\text{neighbor}}^{(d)} = 0 \quad (\mu_{\ell}^2 = 0) \quad (4.140)$$

holds before the evanescent remainder is extracted. The six gauge-matter routes give the following signed traces (full, longitudinal, transverse; per Euclidean chirality $\pm$), reproduced identically at two independent external-momentum samples:

route	permutation	− full	− long.	− trans.	+ full	+ long.	+ trans.
001	(0, 1, 2)	−512	1024	−1536	1024	1024	0
002	(0, 2, 1)	−1024	−1024	0	0	−512	512
003	(1, 0, 2)	−1024	−1024	0	512	−512	1024
004	(2, 0, 1)	−512	512	−1024	1024	1024	0
005	(1, 2, 0)	−1024	−512	−512	0	0	0
006	(2, 1, 0)	0	0	0	1536	0	1536

The per-route coefficients of the master unit, $-3/8, -1/4, -3/8, -3/8, -1/4, -3/8$ in the order of the table, sum to -2 ; combined with the signed color word $-f^{ACD}f^{BCE}$ carried by the gauge-matter orbit before assembly, this supplies the $+2$ of paragraph 6.

5. Evanescent extraction. On every marked edge e of every route the two metric splits of Section 4.3 give

$$\frac{r_{e,d}^2 \mathcal{R}_e}{D_0 D_1 D_2} - \frac{\mathcal{R}_e}{\prod_{j \neq e} D_j} = 0, \quad \frac{\bar{r}_e^2 \mathcal{R}_e}{D_0 D_1 D_2} - \frac{\mathcal{R}_e}{\prod_{j \neq e} D_j} = \frac{\mu_\ell^2 \mathcal{R}_e}{D_0 D_1 D_2}, \quad (4.141)$$

the numerator-weighted form of the exact cutting-failure identity (4.15); no graph-specific $(4-d)$ factor is inserted. With the ordered Feynman parametrization of (4.76), the numerator integral is

$$I_{\ell^2}(\Delta) := \mu^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d} \frac{\ell_d^2}{(\ell_d^2 + \Delta)^3} = J_2 - \Delta J_3 \quad (4.142)$$

in the notation of (4.18), and the evanescent master integral of (4.16)–(4.20) reads

$$J_{\mu^2} := \mu^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d} \frac{\mu_\ell^2}{(\ell_d^2 + \Delta)^3} = \frac{4-d}{d} I_{\ell^2} = \frac{1}{32\pi^2}. \quad (4.143)$$

For the two-matter route the contact sum rule and evanescent probe are

$$\mathcal{F}_{b>\gamma}^{\text{DRED}} + \mathcal{C}_0^{(d)} + \mathcal{C}_1^{(d)} = 8\mu_\ell^2 (\mathcal{R}_0 - \mathcal{J}_1) = -1024 \mu_\ell^2 \mathfrak{b}(p), \quad (4.144)$$

collapsing to $(-1024 p_2, +1024 p_1)$ on the two dotted spinor components (the ordering convention of the recorded orbit replay), with $r_1 - r_0 = p$ by momentum conservation, so that $\mathfrak{b}(r_1) - \mathfrak{b}(r_0) = \mathfrak{b}(p)$. The reversed direction gives $8\mu_\ell^2 (\mathcal{R}_2 - \mathcal{R}_1)$ with probe $(-1024 q_2, +1024 q_1)$, and the zero-source-momentum check at $q = -p$ returns the same magnitude. The two super-potential routes are exact zeros,

$$T_{MH}^{(1)} = T_{MH}^{(2)} = 0, \quad (4.145)$$

each with vanishing longitudinal, transverse, and full projected polynomials.

6. Pole cancellation. In the two-matter orbit the parent triangle and the seagull contribute, in their I_{ℓ^2} parts,

$$\Gamma_{TMM}^\Delta \Big|_{I_{\ell^2}} = -\frac{8}{d} I_{\ell^2}, \quad \Gamma_{TMM}^{I_0 S_{m4}} \Big|_{I_{\ell^2}} = +2 I_{\ell^2} \implies \left(-\frac{8}{d} + 2\right) I_{\ell^2} = 0 \quad (d=4). \quad (4.146)$$

In DRED the same pair leaves

$$\Gamma_{TMM}^{\text{DRED}} \Big|_{\text{local}} = \left(2 - \frac{8}{d}\right) I_{\ell^2} = \frac{2d-8}{d} I_{\ell^2} = -2 \frac{4-d}{d} I_{\ell^2} = -2 J_{\mu^2} \implies \hbar g^2 (-2) J_{\mu^2} = -\frac{\hbar g^2}{16\pi^2}. \quad (4.147)$$

The gauge-matter orbit pairs its parent triangle with both the nonlinear-source bubble and the seagull:

$$\Gamma_{TGM}^\Delta \Big|_{I_{\ell^2}} = +\frac{8}{d} I_{\ell^2}, \quad \Gamma_{TGM}^{I_1 S_{m3}} \Big|_{I_{\ell^2}} = -I_{\ell^2}, \quad \Gamma_{TGM}^{I_0 S_{m4}} \Big|_{I_{\ell^2}} = -I_{\ell^2} \implies \left(\frac{8}{d} - 1 - 1\right) I_{\ell^2} = 0 \quad (d=4), \quad (4.148)$$

$$\Gamma_{TGM}^{\text{DRED}} \Big|_{\text{local}} = \left(\frac{8}{d} - 2\right) I_{\ell^2} = \frac{8-2d}{d} I_{\ell^2} = +2 \frac{4-d}{d} I_{\ell^2} = +2 J_{\mu^2} \implies \hbar g^2 (+2) J_{\mu^2} = +\frac{\hbar g^2}{16\pi^2}. \quad (4.149)$$

The remaining order- g^2 rows vanish: the X_3 source tadpole is scaleless,

$$\langle I_2 \rangle_0 \supset \int \frac{d^d k}{(2\pi)^d} \frac{(k^2)^n}{k^2} = 0; \quad (4.150)$$

the superpotential routes vanish by (4.145); and the gauge, gauge-fixing, Faddeev–Popov, and Nielsen–Kallosh rows have zero color trace in this sector and contain no J_{μ^2} remainder. The Euler-current and explicit-current rows of the outer reduction of the unit insertion X , $\nabla_- X = -\nabla_+(\delta S/\delta \mathcal{V}) - 2i\sqrt{2}(\beta_s \times \gamma^s)$ (flavor-summed), remain separate until their ports, denominators, and edge tags are matched:

$$E_{\text{cur}} : +2i \text{ on } D_0 D_1, \quad X_{\text{cur}} : -2i \text{ on } D_0 D_1, \quad (+2i - 2i) K_{\text{cur}}(k) = 0, \quad (4.151)$$

(the $\pm 2i$ are quoted in the recorded machine units of the current rows, i.e. before the $\sqrt{2}$ of the letter normalization in the reduction above), with no marked inverse edge on either row (selected square $\emptyset$); this pairing vanishes before loop integration and does not produce the factor 1/2 below.

Remark 4.22 (The occurrence decomposition). An earlier replay of this family transported both inverse-square rows $\bar{r}_0^2 - r_{0,d}^2 = \mu_\ell^2$ and $\bar{r}_1^2 - r_{1,d}^2 = \mu_\ell^2$ to independent nonlinear descendants and recorded $|\Gamma_{\text{old}}| = 4J_{\mu^2}$, twice the accepted value. The complete orbit shows that these are two decompositions of one metric pole orbit, so the complete value is $|\Gamma_{\text{complete}}| = 2J_{\mu^2}$: the factor 1/2 is the occurrence-decomposition correction, not the current-pair cancellation of (4.151). An external review of this family proposing a different repair was rejected for an arithmetic gap; the occurrence decomposition above is the adjudicated mechanism.

7. Final result. In the ordered basis

$$(e_1, e_2)^{DE} := \left((\partial_{\dot{\alpha}} \gamma^r)^D (\partial^{\dot{\alpha}} c)^E, (\partial_{\dot{\alpha}} c)^D (\partial^{\dot{\alpha}} \gamma^r)^E \right) = \left(\langle \gamma^r, \partial c \rangle, \langle \partial c, \gamma^r \rangle \right)^{DE} \quad (4.152)$$

the two surviving orbits of paragraph 6 sum to the independent coefficient pairs

$$\mathbf{c}_{b>\gamma^r} = (-1, +1), \quad \mathbf{c}_{\gamma^r>b} = (-1, +1) \quad (4.153)$$

in units of $(\hbar g^2/16\pi^2) f^{ACD} f^{BCE}$: for the forward direction the two-matter orbit contributes -1 on e_1 and the gauge–matter orbit $+1$ on e_2 ; the reversed direction exchanges the orbit–word assignment and returns the same pair. Equivalently, in the unit normalization $X = \nabla_+ \mathcal{W}_+$, $Y = \tilde{\Phi}^r$, $D_{\dot{\alpha}} = \tilde{\mathcal{W}}_{\dot{\alpha}}$ of paragraphs 1–6,

$$\nabla_-(X^A Y^B) \Big|_{1\text{-loop}} = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \left[D_{\dot{\alpha}}^D (\partial^{\dot{\alpha}} Y)^E - (\partial_{\dot{\alpha}} Y)^D D^{E\dot{\alpha}} \right]. \quad (4.154)$$

Restoring the letters of Definition 4.1 — one insertion factor $-i/\sqrt{2}$ from $b = -(i/\sqrt{2})X$, none from $\gamma^r = Y$, and $D_{\dot{\alpha}} = -i \partial_{\dot{\alpha}} c$ from $\partial_{\dot{\alpha}} c = i\tilde{\mathcal{W}}_{\dot{\alpha}}$ on each output word —

$$\nabla_-(b^A \gamma^r B) = -\frac{i}{\sqrt{2}} \cdot \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} (-i) \{ \langle \partial c, \gamma^r \rangle - \langle \gamma^r, \partial c \rangle \}^{DE}, \quad (4.155)$$

and $(-i/\sqrt{2})(-i) = -1/\sqrt{2}$ gives the channel formula

$$\boxed{\nabla_-(b^A \gamma^r B) = \nabla_-(\gamma^r A b^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \{ \langle \partial c, \gamma^r \rangle - \langle \gamma^r, \partial c \rangle \}^{DE}.} \quad (4.156)$$

This is row (4.269) of the channel table. The two orderings are equal for a one-line reason: both letters are Grassmann-even, so the graded Leibniz rule gives no Koszul sign under exchange, and $\nabla_- \gamma^r = 0$, so both insertions expand into the same source orbit with reversed attachments — and the two independent replays return the same ordered bracket, as (4.153) records.

8. Holomorphic-twist comparison. Budzik and Kulp compute the one-loop shift of the holomorphic supercharge on the same letter pair (their equation (3.67)) [1]:

$$Q_1(b^A(\gamma^I)^B) = \kappa^2 f_{ACD} f_{BCE} [\partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} (\gamma^I)^E - \partial_{\dot{\alpha}} (\gamma^I)^D \partial^{\dot{\alpha}} c^E]. \quad (4.157)$$

The output word is exactly $\langle \partial c, \gamma^r \rangle - \langle \gamma^r, \partial c \rangle$: translating (4.157) into the present alphabet, the coefficient pair in the ordered basis (4.152) is $(-1, +1)$, in word-by-word agreement with (4.153), and the raw ratio to the printed zero-shift row is 1. On the holomorphic-twist side the row is obtained by the $\mathfrak{psl}(3|3)$ descent $Q_1(b\gamma^I) = -Q_+^I Q_1(bc)$ from the (b, c) channel.

Remark 4.23 (Status of the comparison). The reversed row $Q_1((\gamma^I)^A b^B)$ is not printed in [1]; the (γ^r, b) channel above is an independent computation, not a quotient of the forward row. The comparison of this paragraph is an after-check only: no coefficient of [1] enters the derivation in paragraphs 1–7, and the agreement is a cross-check, not an internal completeness proof of the graph census.

Remark 4.24 (Recorded granularity). The sources record the stage outputs of the D -algebra — traces, residuals, probes, and the table above — but not the index-level intermediate spinor manipulations of the seven surviving routes; accordingly (4.138)–(4.140) display the recorded outputs of each step rather than unattributed intermediate lines, and Figure 5 is drawn from the recorded edge routings and the vertex slot structure.

4.9 The channels $(b, \partial_{\dot{\alpha}} c)$ and $(\partial_{\dot{\alpha}} c, b)$

We now compute the mixed-chirality family in full: the ordered channels $(b, \partial_{\dot{\alpha}} c)$ and $(\partial_{\dot{\alpha}} c, b)$, $\dot{\alpha} = \dot{1}, \dot{2}$, the block $4_{(b, \partial c), (\partial c, b)}$ of the census (4.21). The computation repeats the seed-channel pattern of Section 4.6 with three new features: the cubic vertices have opposite chirality, the D -algebra numerator is rank one in the loop momentum, and the ordered source admits two color allocations. Graph weights are computed with unit-normalized insertions $\nabla_+ \mathcal{W}_+$ and $\widetilde{\mathcal{W}}_{\dot{\alpha}}$; the letter normalization of Definition 4.1 is restored in step 7. The dotted transport $\dot{1} \mapsto \dot{2}$ leaves $f^{ACD} f^{BCE}$ and the sign fixed, so we display $\dot{\alpha} = \dot{1}$.

1. Graph inventory and the occurrence census

Each ordering is carried by *one ordered triangle plus one Schwinger cut/contact graph*, paired by the cutting rule of Section 4.3. The triangle (Figure 6) has the ordered bilocal insertion I and two cubic background vertices of opposite chirality, the same mixed pair as the seed chain,

$$V_{\widetilde{\mathcal{W}}} : +\frac{ig}{2} f^{UCD} \widetilde{\mathcal{W}}_{\dot{\gamma}}^D (\bar{\mathcal{D}}_{\dot{C}}^{\dot{\gamma}} - \bar{\mathcal{D}}_{\dot{U}}^{\dot{\gamma}}), \quad V_{\mathcal{W}} : -\frac{ig}{2} f^{VC'E} \mathcal{W}_{\dot{\gamma}}^E (\mathcal{D}_{\dot{C}'}^{\dot{\gamma}} - \mathcal{D}_{\dot{V}}^{\dot{\gamma}}). \quad (4.158)$$

$V_{\mathcal{W}}$ emits the $\nabla_+ \mathcal{W}_+$ insertion (the b leg), $V_{\widetilde{\mathcal{W}}}$ emits the $\widetilde{\mathcal{W}}_{\dot{\alpha}}$ insertion (the $\partial_{\dot{\alpha}} c$ leg), and the three internal vector lines carry the raw routing $r_0 = \ell$, $r_1 = \ell + p$, $r_2 = \ell + p + q$, so $D_0 = \ell^2$, $D_1 = (\ell + p)^2$, $D_2 = (\ell + p + q)^2$ as in the seed channel. The marked line is e_0 for $(b, \partial_{\dot{\alpha}} c)$ and e_2 for $(\partial_{\dot{\alpha}} c, b)$.

The raw engine census per ordering is $36 \text{ TGG} + 6 \text{ TMM} + 42 \text{ spectator} = 84$ connected $= 42_{\text{split source}} + 42_{\text{spectator}}$, where TGG (two gauge cubic vertices) is one ordered triangle of Figure 6 with a distinguished marked edge, and TMM has two matter vertices. The occurrence resolution: the 36 TGG rows per ordering, with their same-edge Schwinger cuts, carry the entire anomaly (steps 2–6). The 6 TMM rows vanish exactly (wrong external field type for two gauge letters). The matter-current Euler and explicit-contact pair shares the word $K_1 = ir_{+2}$, $K_2 = -ir_{+1}$ with weights $\pm 2iK_{\dot{\alpha}}$ and cancels before regulation; the word is affine, so its $\mu_{\dot{\ell}}^2$ coefficient vanishes. The source-tagged bubbles have affine numerators and zero standalone anomaly; their two-denominator rows belong to the inverse-kernel brackets of step 4 and are not additional finite diagrams. The tadpole is scaleless, $\int d^d \ell P(\ell)/\ell^2 = 0$ in DRED [5]. The

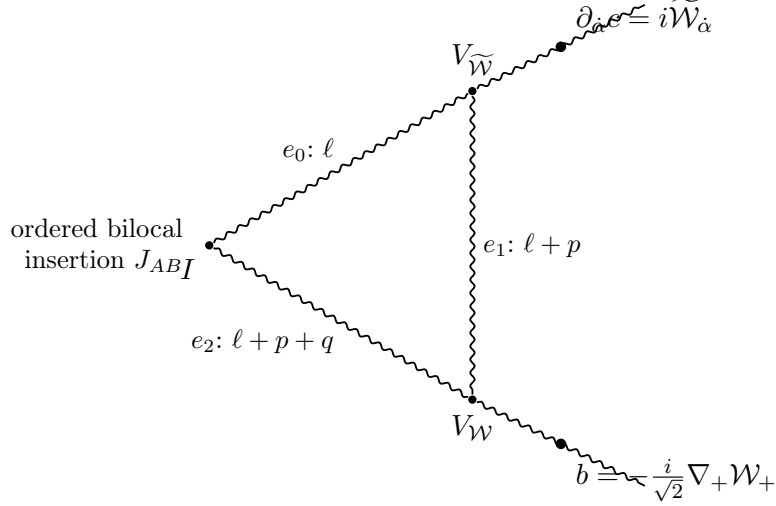


Figure 6: The ordered triangle of $(b, \partial_{\dot{\alpha}} c)$ and $(\partial_{\dot{\alpha}} c, b)$. The bilocal insertion I carries the two letters; $V_{\mathcal{W}}$ emits the $\nabla_+ \mathcal{W}_+$ insertion (the b leg) and $V_{\widetilde{\mathcal{W}}}$ the $\widetilde{\mathcal{W}}_{\dot{\alpha}}$ insertion (the $\partial_{\dot{\alpha}} c$ leg); all internal lines are vector-superfield propagators. The marked line is e_0 for $(b, \partial_{\dot{\alpha}} c)$ and e_2 for $(\partial_{\dot{\alpha}} c, b)$; the Schwinger cut/contact partner collapses the marked line with the same two vertices.

gauge-fixing Euler sector cancels the longitudinal source row (step 4) on 32 component masks. In the ghost sector, the Euler bubble and its full- d cut are exact negatives and the ghost kernel is the full- d $\square$, so no μ_ℓ^2 residue arises; the ghost triangle with linear source is absent at one loop by the edge/vertex count, and the Nielsen–Kallosh factor is absent from the selected graph set. The 42 spectator routes per ordering are connected but one-particle reducible (external-leg self-energies), and the amputated 1PI projector annihilates them.

Remark 4.25 (Status of the auxiliary sectors). The local Fermi–Feynman proper slice is not derived from first principles; it is fixed by the scheme decisions of Section 3 (Gaussian averaging; the Nielsen–Kallosh factor kept external). Every contributing graph is accounted for by the triangle and its same-edge cut; all other occurrences are exactly zero, cancel in pairs, or are absent.

2. Wick contractions

The two vertices of (4.158) are distinct (one chiral, one antichiral), so the interaction expansion contributes once at second order:

$$w_{\text{Wick}} = \frac{1}{2!} (1 + 1) = 1, \quad (4.159)$$

the $(1+1)$ counting the two equivalent Wick pairings, and the labeled source Hessian contributes $1!1! = 1$. There is *no reflection multiplicity*: $(b, \partial_{\dot{\alpha}} c)$ (marked edge e_0) and $(\partial_{\dot{\alpha}} c, b)$ (marked edge e_2) are distinct ordered descendants, stored separately; the compact output word is already oriented. The three internal lines contract against the vertices and the source with the vector propagators of the Fermi–Feynman slice ($\kappa_{AB} = \delta_{AB}$, Section 3):

$$\langle \mathcal{V}^A \mathcal{V}^U \rangle_{e_0} = \frac{\hbar \delta^{AU} \delta^4(\theta_{12})}{\ell^2}, \quad \langle \mathcal{V}^C \mathcal{V}^{C'} \rangle_{e_1} = \frac{\hbar \delta^{CC'} \delta^4(\theta_{12})}{(\ell + p)^2}, \quad \langle \mathcal{V}^V \mathcal{V}^B \rangle_{e_2} = \frac{\hbar \delta^{VB} \delta^4(\theta_{12})}{(\ell + p + q)^2}. \quad (4.160)$$

Remark 4.26 (Edge-weight sign convention). The edge weights displayed in (4.160) are recorded in the graph-IR convention of the underlying computation, in which the overall minus sign of

the Feynman propagator $\langle uu \rangle = -\hbar\kappa^{AB}/p^2$ of Section 4.5 is folded into the vertex signs of the closed word; the displayed $+\hbar\kappa^{AB}/p^2$ is the edge weight per line, not an independent propagator sign. All downstream factors of this subsection use the same convention consistently, and the final coefficients are machine-verified against the propagator-level replay.

Contraction with the vertex structure constants and the source yields the common color tensor in its two allocations, $f^{ACD}f^{BCE}$ (direct) and $f^{BCD}f^{ACE}$ (crossed), exchanged by $A \leftrightarrow B$ and independent for a general gauge algebra. The surviving direct ordered source Hessians on the fixed odd component directions are $i/4$ for $(b, \partial_{\dot{\alpha}}c)$ and $1/4$ for $(\partial_{\dot{\alpha}}c, b)$, with $1!1! = 1$; each external $\partial_{\dot{\alpha}}c$ leg carries the component preimage $-\frac{\sqrt{2}}{8}\eta_{\dot{\alpha}}$. Reversing the functional order of the two odd directions changes the sign of the crossed Hessian, which is why both color allocations must be kept.

3. Amplitude assembly

The two ordered cubic vertices contribute $(+\frac{ig}{2})(-\frac{ig}{2}) = \frac{g^2}{4}$ as in (4.62), and the closed loop is saturated by $\delta^4(\theta_{12})\mathcal{D}^2\bar{\mathcal{D}}^2\delta^4(\theta_{12}) = 16\delta^4(\theta_{12})$. The expansion runs over the four chirality sectors $(++)$, $(+-)$, $(-+)$, $(--)$ of the two vertices, each of weight 1; the four contractions coincide polynomially, so the full action contribution is $4_{\text{chirality}} \times (\text{one sector})$. For each occurrence (ω, e) the frozen source identity splits the marked D -algebra word as

$$\mathcal{D}_-\mathcal{D}_+\bar{\mathcal{D}}^2\mathcal{D}_+ = -8\bar{r}_e^2\mathcal{D}_+ - \frac{1}{2}\mathcal{D}_+\bar{\mathcal{D}}^2\mathcal{D}^2, \quad (4.161)$$

so that, writing $\mathcal{W}_{\omega,e}^K$ and $\mathcal{W}_{\omega,e}^L$ for the kinetic and longitudinal spin words,

$$K_{\omega,e} = -8\bar{r}_e^2\mathcal{W}_{\omega,e}^K, \quad L_{\omega,e}^{\text{source}} = -\frac{1}{2}\mathcal{W}_{\omega,e}^L, \quad L_{\omega,e}^{\text{gf}} = +\frac{1}{2}\mathcal{W}_{\omega,e}^L, \quad L_{\omega,e}^{\text{source}} + L_{\omega,e}^{\text{gf}} = 0. \quad (4.162)$$

With $R_{\omega,e}$ the reduced numerator and $C_{\omega,e}^{\text{SD}} = -R_{\omega,e}/\prod_{j \neq e} D_j$ the same-edge Schwinger contact, the full- d and four-dimensional identities are

$$K_{\omega,e}^{(d)} + C_{\omega,e}^{\text{SD}} = 0, \quad K_{\omega,e}^{(4)} + C_{\omega,e}^{\text{SD}} = \frac{\mu_\ell^2 R_{\omega,e}}{D_0 D_1 D_2}, \quad (4.163)$$

the edgewise parent-minus-cut of (4.15).

Remark 4.27 (The factored weight chain is recorded, not displayed). Unlike the seed channel, no single displayed chain $w_{\text{Wick}} \frac{g^2}{4} w_D$ exists for this family; the total is fixed by the recorded conversions: a per-chirality conditional factor -8 , the absolute conversion $4_{\text{chirality}} \times (\frac{1}{2})_{\text{rank/trace}} \times (-8)_{\text{legacy}} = -16$, and the isolating color value -2 (the $SU(2)$ sample $(A, B; D, E) = (0, 1; 1, 0)$ gives $(f^{ACD}f^{BCE}, f^{BCD}f^{ACE}) = (-2, 0)$). The rank/trace factor $\frac{1}{2}$ is inserted exactly once; $m_{\text{contact}} = 1$ of step 6 supplies no second factor.

4. The D -algebra numerator chain

With the conditional Fermi–Feynman projector $\mathcal{P}_0 = (\bar{\mathcal{D}}^2\mathcal{D}^2 + \mathcal{D}^2\bar{\mathcal{D}}^2)/(16\Box)$ and $\mathcal{D}_+\mathcal{D}^2\bar{\mathcal{D}}^2 = 0$,

$$8\mathcal{D}_+\Box\mathcal{P}_0 = \frac{1}{2}\mathcal{D}_+\bar{\mathcal{D}}^2\mathcal{D}^2 + \frac{1}{2}\mathcal{D}_+\mathcal{D}^2\bar{\mathcal{D}}^2 = \frac{1}{2}\mathcal{D}_+\bar{\mathcal{D}}^2\mathcal{D}^2, \quad (4.164)$$

so the longitudinal term of (4.161) is pure gauge and cancels the gauge-fixing Euler row, $-\frac{1}{2}\mathcal{D}_+\bar{\mathcal{D}}^2\mathcal{D}^2 + \frac{1}{2}\mathcal{D}_+\mathcal{D}^2\bar{\mathcal{D}}^2 = 0$, verified sparsely on $32 + 32$ component masks. What survives is the first term of (4.161): per marked edge the parent-minus-cut of (4.163) leaves exactly the evanescent remainder (4.15). Summing the 36 TGG routes per ordering gives the two rank-one route sums

$$R_1 = \frac{i}{16}\mu_\ell^2\left(\ell + p + \frac{1}{2}q\right)_{+i}, \quad R_2 = -\frac{i}{16}\mu_\ell^2\ell_{+i}, \quad (4.165)$$

with X_{+i} the plus-frame dotted spinor component, and the evanescent numerators of the two orderings carry the color mix

$$N_{(b,\partial c)}^{\text{ev}} = f^{ACD} f^{BCE} R_1 + f^{BCD} f^{ACE} R_2, \quad N_{(\partial c,b)}^{\text{ev}} = f^{ACD} f^{BCE} R_2 + f^{BCD} f^{ACE} R_1, \quad (4.166)$$

the marked edge being e_0 for the direct $(b, \partial c)$ and crossed $(\partial c, b)$ rows and e_2 for the other two. Occurrence by occurrence, the aggregated two-denominator contact rows and the same-edge remainders obey

$$\begin{aligned} (b, \partial c), e_0: & \quad 7iq_{+i} + \left[i\left(\ell + p + \frac{1}{2}q\right)_{+i} - 7iq_{+i} \right] = i\left(\ell + p + \frac{1}{2}q\right)_{+i}, \\ (b, \partial c), e_1: & \quad 0 + 0 = 0, \quad (b, \partial c), e_2: \quad 2iq_{+i} - 2iq_{+i} = 0, \\ (\partial c, b), e_0: & \quad -\frac{i}{2}q_{+i} + \frac{i}{2}q_{+i} = 0, \quad (\partial c, b), e_1: \quad i(p+q)_{+i} - i(p+q)_{+i} = 0, \\ (\partial c, b), e_2: & \quad i(p+q)_{+i} + \left[-i\ell_{+i} - i(p+q)_{+i} \right] = -i\ell_{+i}, \end{aligned} \quad (4.167)$$

and the crossed-color replay exchanges the two orderings edge by edge. The two-denominator members of (4.167) belong to the inverse-kernel brackets and must not be added again as finite diagrams once N^{ev} is formed.

5. Tensor integrals and the ordered simplex moments

The two DRED masters are the scalar master (4.16) and its rank-one partner,

$$\int \frac{d^d \ell}{(2\pi)^d} \frac{\mu_\ell^2}{D_0 D_1 D_2} = \frac{1}{32\pi^2}, \quad \int \frac{d^d \ell}{(2\pi)^d} \frac{\mu_\ell^2 \ell_{+i}}{D_0 D_1 D_2} = -\frac{(2p+q)_{+i}}{96\pi^2}. \quad (4.168)$$

Combining denominators,

$$\frac{1}{D_0 D_1 D_2} = 2 \int_0^1 dy \int_0^{1-y} dz \frac{1}{(L^2 + \Delta)^3}, \quad L = \ell + (y+z)p + zq, \quad (4.169)$$

with exact simplex moments, writing $x = 1 - y - z$,

$$2 \int_{\Delta_2} 1 = 1, \quad 2 \int_{\Delta_2} x = \frac{1}{3}, \quad 2 \int_{\Delta_2} (y+z) = \frac{2}{3}, \quad 2 \int_{\Delta_2} z = \frac{1}{3}. \quad (4.170)$$

For $(b, \partial_{\dot{\alpha}} c)$ the shifted numerator is $\ell + p + \frac{1}{2}q = L + xp + (\frac{1}{2} - z)q$; the term odd in L integrates to zero, and $2 \int_{\Delta_2} (\frac{1}{2} - z) = 2(\frac{1}{4} - \frac{1}{6}) = \frac{1}{6}$, so

$$\int_\ell \frac{N_{(b,\partial c)}^{\text{ev}}}{D_0 D_1 D_2} \Big|_{R_1} = \frac{i}{16} \cdot \frac{1}{32\pi^2} \left(\frac{1}{3}p + \frac{1}{6}q \right)_{+i} = \frac{i}{512\pi^2} \left(\frac{1}{3}p + \frac{1}{6}q \right)_{+i}. \quad (4.171)$$

For $(\partial_{\dot{\alpha}} c, b)$, $-\ell = -L + (y+z)p + zq$ and the moments (4.170) give

$$\int_\ell \frac{N_{(\partial c,b)}^{\text{ev}}}{D_0 D_1 D_2} \Big|_{R_2} = -\frac{i}{16} \left(-\frac{1}{32\pi^2} \right) \left(\frac{2}{3}p + \frac{1}{3}q \right)_{+i} = \frac{i}{512\pi^2} \left(\frac{2}{3}p + \frac{1}{3}q \right)_{+i}, \quad (4.172)$$

in agreement with the direct evaluation of (4.168), $-\frac{i}{16} \left(-\frac{(2p+q)_{+i}}{96\pi^2} \right) = \frac{i}{512\pi^2} \left(\frac{2}{3}p + \frac{1}{3}q \right)_{+i}$. In the authority routing the raw external momenta are the ordered action ports, $p = k_L$ (left output leg) and $q = k_R$ (right output leg). The jet weights of the final bilinear are the ordered-simplex moments of the tower parametrization (4.279) at $(m, n) = (1, 0)$:

$$\begin{aligned} 2 \int_0^1 db \int_0^b da \, a &= 2 \int_0^1 \frac{b^2}{2} db = \frac{1}{3} \quad (\text{jet on the first letter}), \\ 2 \int_0^1 db \int_0^b da \, b &= 2 \int_0^1 b^2 db = \frac{2}{3} \quad (\text{jet on the second letter}), \end{aligned} \quad (4.173)$$

precisely $K_{1,0;1,0} = \frac{2}{3,2}$ and $K_{1,0;0,0} = \frac{2}{3,1}$ of (4.280). Assembling the four chirality sectors, the two color allocations and the contact rows, the target-blind momentum shapes are

$$\begin{aligned}\Gamma_{(b,\partial c)} &= -i \frac{\hbar g^2}{16\pi^2} \left[f^{ACD} f^{BCE} \left(\frac{1}{3}k_L + \frac{1}{6}k_R \right) + f^{BCD} f^{ACE} \left(\frac{2}{3}k_L + \frac{1}{3}k_R \right) \right]_{+i}, \\ \Gamma_{(\partial c,b)} &= -i \frac{\hbar g^2}{16\pi^2} \left[f^{ACD} f^{BCE} \left(\frac{2}{3}k_L + \frac{1}{3}k_R \right) + f^{BCD} f^{ACE} \left(\frac{1}{3}k_L + \frac{1}{6}k_R \right) \right]_{+i}.\end{aligned}\quad (4.174)$$

Remark 4.28 (The jet readout is the assembled output, not a Fourier map). A naive identification of momentum shape with jet weights fails: the direct-color shape in (4.174) is $(\frac{1}{3}, \frac{1}{6})$, not $(\frac{1}{3}, \frac{2}{3})$. The canonical bilinear coefficients of step 7 are the recorded output of the full four-chirality, two-color, contact-inclusive assembly, not of a one-line Fourier identification; (4.173) displays the intrinsic jet weights of that assembled result.

6. Pole cancellation

The triangle parent and the aggregate Schwinger contact row carry the UV-pole pair

$$\begin{aligned}\Gamma_T &= + \frac{\hbar g^2}{32\pi^2\epsilon} \times (\text{edge-tagged tensor with } \hat{\delta}^{mn}), \\ \Gamma_C &= - \frac{\hbar g^2}{32\pi^2\epsilon} \times (\text{the identical tensor with } \delta_4^{mn}),\end{aligned}\quad (4.175)$$

so the four-dimensional metric parts cancel exactly and the remainder is the pure evanescent numerator $\hat{\delta}^{mn} - \delta_4^{mn} = -\check{\delta}^{mn}$, whose sigma-trace supplies the -2ϵ of (4.86): finiteness holds precisely as in Remark 4.6. At the occurrence level, the local contact kernel $K_{AB} = -\kappa_{AB}r_e^2$ with $\kappa_{AB} = \frac{1}{2}\delta_{AB}$ obeys $r_e^2 P^A = -2P^A K_A$, and

$$\begin{aligned}c_{\text{parent}} &= (-2)^3 = -8, \quad c_{\text{contact}} = (-1)_{\text{SD}} (-2)_{r^2 P = -2PK} (-2)_{\text{uncut}}^2 = +8, \\ m_{\text{contact}} &= \left(\frac{1}{2!} 2_{\text{action}} \right) \left(\frac{1}{2!} 2_{\text{cut}} \right) \left(\frac{1}{2!} 2_{\text{Wick}} \right) = 1.\end{aligned}\quad (4.176)$$

Consequently, occurrence by occurrence and in each chirality sector,

$$N_{d,\omega}^{\text{parent}} + K_{\text{raw},\omega}^{\text{contact}} = 0, \quad N_{4,\omega}^{\text{parent}} + K_{\text{raw},\omega}^{\text{contact}} = \frac{R_\omega \mu_\ell^2}{D_0 D_1 D_2}.\quad (4.177)$$

The finite- w link sector cancels before loop integration: with $x = iwr$, $y = iwr'$ and $(x - y) \int_0^1 ds e^{sx + (1-s)y} = e^x - e^y$, the one-link term and the two endpoint terms cancel, $C_{\omega,e}(e^x - e^y) - C_{\omega,e}(e^x - e^y) = 0$; the sector is longitudinal in arbitrary d , so its μ_ℓ^2 coefficient is exactly zero.

7. The final coefficients and the tower-kernel instance

With unit-normalized insertions the assembled channel results are

$$\nabla_- ((\nabla_+ \mathcal{W}_+)^A \widetilde{\mathcal{W}}_\alpha^B) = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \left\{ \frac{1}{3} \langle \partial_\alpha \widetilde{\mathcal{W}}, \widetilde{\mathcal{W}} \rangle + \frac{2}{3} \langle \widetilde{\mathcal{W}}, \partial_\alpha \widetilde{\mathcal{W}} \rangle \right\}^{DE}, \quad (4.178)$$

$$\nabla_- (\widetilde{\mathcal{W}}_\alpha^A (\nabla_+ \mathcal{W}_+)^B) = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \left\{ \frac{2}{3} \langle \partial_\alpha \widetilde{\mathcal{W}}, \widetilde{\mathcal{W}} \rangle + \frac{1}{3} \langle \widetilde{\mathcal{W}}, \partial_\alpha \widetilde{\mathcal{W}} \rangle \right\}^{DE}. \quad (4.179)$$

Restoring the letters of Definition 4.1 is exact arithmetic: the insertions and the output letters give

$$\begin{aligned}b^A \partial_\alpha c^B &= \left(-\frac{i}{\sqrt{2}} \right) (i) (\nabla_+ \mathcal{W}_+)^A \widetilde{\mathcal{W}}_\alpha^B = \frac{1}{\sqrt{2}} (\nabla_+ \mathcal{W}_+)^A \widetilde{\mathcal{W}}_\alpha^B, \\ \langle \partial_\alpha \widetilde{\mathcal{W}}, \widetilde{\mathcal{W}} \rangle &= (-i)^2 \langle \partial_\alpha (\partial c), \partial c \rangle,\end{aligned}\quad (4.180)$$

and $\frac{1}{\sqrt{2}} \times (-1) \times \frac{\hbar g^2}{16\pi^2} = -\frac{\hbar g^2}{16\sqrt{2}\pi^2}$. Substituting into (4.178)–(4.179),

$$\boxed{\nabla_-(b^A \partial_{\dot{\alpha}} c^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \times \left\{ \frac{1}{3} \langle \partial_{\dot{\alpha}}(\partial c), \partial c \rangle + \frac{2}{3} \langle \partial c, \partial_{\dot{\alpha}}(\partial c) \rangle \right\}^{DE}} \quad (4.181)$$

$$\boxed{\nabla_-(\partial_{\dot{\alpha}} c^A b^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \times \left\{ \frac{2}{3} \langle \partial_{\dot{\alpha}}(\partial c), \partial c \rangle + \frac{1}{3} \langle \partial c, \partial_{\dot{\alpha}}(\partial c) \rangle \right\}^{DE}} \quad (4.182)$$

— exactly the rows (4.273) and (4.274) of the channel table. The weights are the $(m, n) = (1, 0)$ instance of the tower kernel (4.280): by (4.173), $K_{1,0;1,0} = \frac{2}{3 \cdot 2} = \frac{1}{3}$ (jet on the first letter) and $K_{1,0;0,0} = \frac{2}{3 \cdot 1} = \frac{2}{3}$ (jet on the second), and the two orderings are exchanged by the Leibniz distribution $(1, 1) - (\frac{1}{3}, \frac{2}{3}) = (\frac{2}{3}, \frac{1}{3})$.

Remark 4.29 (No 1 : 2 ratio between the two orderings). An earlier conditional proposal assigned the two orderings the ratio $\mathcal{N}_{(b,\partial c)}/\mathcal{N}_{(\partial c,b)} = 1/2$ through a rejected momentum map; it was explicitly *not* promoted to a coefficient of this paper. The final results (4.181)–(4.182) share the single prefactor $-\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE}$ and differ only in the jet weights $(\frac{1}{3}, \frac{2}{3}) \leftrightarrow (\frac{2}{3}, \frac{1}{3})$.

Remark 4.30 (Machine cross-checks for this family). The full chain of this subsection was replayed externally: 94/94 checks on the ordered-port crossed-Hessian assembly, 112/112 on each of the two polarized D -algebra replays, and 255/255 on the raw contact and finite- w link replay, with zero failures.

8. Comparison with the holomorphic twist

In the holomorphic twist of [1] this family has no zero-derivative display; it exists only through the derivative tower applied to the seed pair $Q_1(b^A c^B) = \kappa_H^2 f_{ACD} f_{BCE} \partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} c^E$ (their eq. (3.63)). Their Appendix-B tower (their eq. (B.6)),

$$Q_1(b^A \partial_1^m \partial_2^n c^B) = \kappa_H^2 f_{ACD} f_{BCE} \frac{1}{m+n+2} \sum_{k=0}^m \sum_{\ell=0}^n \frac{1}{k+\ell+1} \binom{m}{k} \binom{n}{\ell} \times [\partial_1^k \partial_2^\ell \partial_{\dot{\alpha}} c^D \partial_1^{m-k} \partial_2^{n-\ell} \partial^{\dot{\alpha}} c^E], \quad (4.183)$$

evaluated at $(m, n) = (1, 0)$, gives the printed weights

$$\begin{aligned} k=0 : & \quad \frac{1}{3} \text{ on } \partial_{\dot{\alpha}} c^D \partial_1 \partial^{\dot{\alpha}} c^E \text{ (jet on the second letter),} \\ k=1 : & \quad \frac{1}{6} \text{ on } \partial_1 \partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} c^E \text{ (jet on the first letter),} \end{aligned} \quad (4.184)$$

exactly one half of our coefficients on both words. After the factor-two adjudication of Section 7 — the printed Appendix-B branch of [1] is uniformly smaller by the factor $2 = \Gamma(3)$ of (4.76) — the corrected weights $(\frac{1}{3}, \frac{2}{3})$ coincide with (4.181). The reversed ordering $(\partial_{\dot{\alpha}} c, b)$ is not printed in [1] at all; it follows from their recursion (their eq. (B.1)), $Q_1(\partial_{\dot{\alpha}} f g) = \partial_{\dot{\alpha}} Q_1(f g) - Q_1(f \partial_{\dot{\alpha}} g)$, applied to $(f, g) = (c, b)$, the unprinted orderings being identified with the printed ones by supercommutativity of the letters (no sign, since $|c||b| = 0$). On our side the same law is the Leibniz distribution on the formal ghost slot: the seed distributes one unit of weight on each output word, $(1, 1)$, and $(1, 1) - (\frac{1}{3}, \frac{2}{3}) = (\frac{2}{3}, \frac{1}{3})$, exactly the swap between (4.181) and (4.182). Both dotted rows agree word by word with the twist computation (Section 7).

Remark 4.31 (Status of the ghost-slot recursion). The recursion anchor involves the formal odd slot of the twist dictionary (the ghost placeholder, of parity 1), whose identification with project quantities is conditional; the physical rows (4.181)–(4.182) are derived independently in steps 1–7 and are unaffected. The comparison of this subsection is a cross-check, not a derivation input, and not a completeness proof of the graph census (Section 7).

4.10 The channel (β_r, β_s)

We now compute the (β_r, β_s) family in full: the six ordered off-diagonal channels $r \neq s$ of the census (4.21) and the three diagonal zeros. The result is the row (4.271) of the channel table, derived here graph by graph. As in the seed channel (Section 4.6), all graph weights are evaluated with unit-normalized letters

$$B_r := \sqrt{2} \beta_r, \quad C^t := \gamma^t, \quad D_{\dot{\alpha}} := -i \partial_{\dot{\alpha}} c, \quad (4.185)$$

and the normalization of Definition 4.1 is restored at the end. In these letters the two ordered output carriers are

$$A_1 := \langle D^D, C^{tE} \rangle, \quad A_2 := \langle C^{tD}, D^E \rangle, \quad (4.186)$$

onto which every graph is projected without quotienting by total derivatives or exchanging the color indices D, E .

1. The insertion and the descendant. With the absorbed-frame scalings $\Phi_r = g\phi_r$, $\tilde{\Phi}^r = g\tilde{\phi}^r$ and the unit-normalized prepotential u , $\mathcal{V} = \sqrt{2}gu$, the gauge-covariant letter source expands as

$$B_{r,c} = \mathcal{D}_+ \phi_r + g\sqrt{2} [\mathcal{D}_+ u, \phi_r] + g^2 [(\mathcal{D}_+ u)u - u(\mathcal{D}_+ u), \phi_r] + O(g^3), \quad (4.187)$$

so the leading insertion of the channel is

$$\mathcal{O}_{B_r B_s}^{(0),AB} = g^2 (\mathcal{D}_+ \phi_r)^A (\mathcal{D}_+ \phi_s)^B. \quad (4.188)$$

The descendant of one letter follows from the tree system (4.6):

$$\nabla_- B_r = -2 \frac{\delta S}{\delta \tilde{\Phi}^r} - \sqrt{2} \varepsilon_{rst} (C^s \times C^t), \quad \frac{\delta S}{\delta \tilde{\Phi}^r} = -\frac{1}{4} \nabla^2 \Phi_r - \frac{1}{\sqrt{2}} \varepsilon_{rst} (C^s \times C^t), \quad (4.189)$$

with $(X \times Y)^A = f^{ABC} X^B Y^C$ as in Section 4.2.

Remark 4.32 (No pre-simplification before the occurrence census). The two potential structures in (4.189) — the Euler-potential piece $+\sqrt{2} \varepsilon_{rst} (C^s \times C^t)$ contained in $-2 \delta S / \delta \tilde{\Phi}^r$ and the explicit piece $-\sqrt{2} \varepsilon_{rst} (C^s \times C^t)$ — carry opposite signs and equal regulated kernels, and they cancel only *after* the regulated occurrence census of paragraph 7. It is not legitimate to pre-simplify (4.189) to $\frac{1}{2} \nabla^2 \Phi_r$ before that census: doing so erases the Euler-potential occurrence and corrupts the pole bookkeeping of Remark 4.6.

2. Graph inventory. The amputated ordered coefficient receives contributions from exactly three triangle parents. For $r \neq s$ with $\varepsilon_{rst} \neq 0$ they are: the two-matter-vertex triangle (TMM, ports (ϕ_r, ϕ_s)), and the two matter/superpotential triangles (TMH), route 002 with vertices (M_r, H_-) and external ports $(u^D, \tilde{\phi}^{tE})$, route 003 with vertices (H_-, M_s) and ports $(\tilde{\phi}^{tD}, u^E)$. Route 003 is computed as an independent orientation — reversed ports and graded source order — never inferred from route 002 by reflection. All three internal lines of every triangle are matter lines $\langle \phi \tilde{\phi} \rangle$; the external legs are one gauge leg u^D and one antichiral matter leg $\tilde{\phi}^{tE}$,

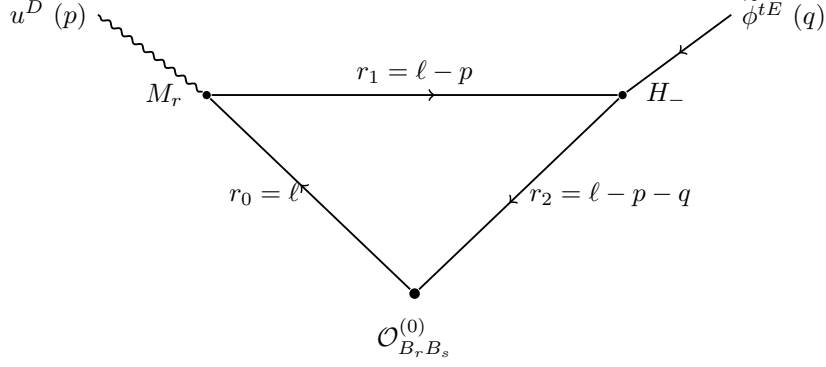


Figure 7: The ordered triangle of route 002 for the (β_r, β_s) channel, $r \neq s$, drawn in the unit normalization (4.185). The composite insertion $\mathcal{O}_{B_r B_s}^{(0)}$ (filled node) attaches to the matter vertex M_r and to the superpotential vertex H_- ; the three internal lines are chiral–antichiral matter propagators with the displayed routing; the external legs are the unit prepotential u^D (wavy) and the antichiral field $\tilde{\phi}^{tE}$, converted to the letters ∂c and γ^t by the external map (4.198). Route 003 is the independently oriented partner with reversed ports and the rebase (4.191).

the residual external operators that the external map of paragraph 4 converts to the letters ∂c and γ^t . The momentum routings are

$$r_0 = \ell, \quad r_1 = \ell - p, \quad r_2 = \ell - p - q; \quad \rho_0 = \ell, \quad \rho_1 = \ell - q, \quad \rho_2 = \ell - p - q, \quad (4.190)$$

for routes 002 and 003 respectively, with the canonical loop rebase

$$k_0 = -\rho_2, \quad k_1 = -\rho_1 = k_0 - p, \quad k_2 = -\rho_0 = k_0 - p - q, \quad (4.191)$$

and the kinematic bookkeeping

$$D_i = r_{i,d}^2, \quad \bar{r}_i^2 = r_{i,d}^2 + \mu_\ell^2, \quad \det r_i = -\bar{r}_i^2, \quad (4.192)$$

where p is the external momentum at the matter vertex and q the external momentum at H_- . The triangle of route 002 is displayed in Figure 7.

Beyond the triangles, the family contains: (a) two-point contact rows of bubble topology, the $\langle I_1 S_{H_-} \rangle$ rows T_2, T_4 of paragraph 7, where I_1 is the $O(g)$ commutator term of (4.187); (b) the Euler-kinetic cut rows of paragraph 7; (c) the Euler-potential and explicit-potential rows of Remark 4.32, which cancel each other before integration. Everything else vanishes exactly or is excluded: the twelve I_2 rows have no antichiral port; the $I_0 S_4$ matter rows cannot absorb both flavors r, s , and H_- has no quartic vertex; the gauge-fixing, ghost and auxiliary rows have no pair of matter-flavor ports; the coincident Jacobian contains $\delta_{rs} = 0$ for $r \neq s$; and the four spectator double-bridge parents (vertex pairs (H_-, H_+) or (M, M) with one letter as spectator) all have the source edge as an articulation edge, so they are one-particle-reducible external self-energy attachments and do not enter the amputated coefficient.

3. Wick contractions. With the Euclidean exponent $e^{-S/\hbar}$ the vertex factors are $+\sqrt{2}g/\hbar$ for M_r and $-\sqrt{2}g \varepsilon_{rst}/\hbar$ for H_- , and the propagators are

$$\langle u^A u^B \rangle = -\frac{\hbar \delta^{AB}}{p^2} \delta^4(\theta_{12}), \quad \langle \phi_r^A \tilde{\phi}^{sB} \rangle = \delta_r^s \frac{\hbar \delta^{AB}}{16 p^2} \bar{\mathcal{D}}_1^2 \mathcal{D}_1^2 \delta^4(\theta_{12}), \quad (4.193)$$

with the reversed chiral-line orientation derived explicitly rather than by reflection. The finite-Grassmann expansion of each triangle has two possible outer marks, according to which source

edge carries the descendant ∇_- -word of (4.189). The marks are sparse:

$$R_{001,L} = R_{001,R} = 0, \quad R_{002,L} = 0, \quad R_{002,R} = 128 v(r_0), \quad R_{003,R} = 0, \quad R_{003,L} = -128 v(\rho_2) = 128 v(k_0), \quad (4.194)$$

where $v(k) := (-k_{+\dot{2}}, k_{+\dot{1}})$ and the surviving mark of route 002 sits on the B_s source edge (selected edge r_2), that of route 003 on the B_r source edge (selected edge k_2 after the rebase (4.191)). Explicitly, the raw parent word before the selected inverse-kernel square is removed is $R^{\dot{1}} = -1024 r_{0,+\dot{2}} \det r_2$, $R^{\dot{2}} = +1024 r_{0,+\dot{1}} \det r_2$, and removing the selected square leaves the residual word $128 v(r_0) = (-128 r_{0,+\dot{2}}, 128 r_{0,+\dot{1}})$. The TMM route 001 has both marks zero and drops out. The contraction multiplicity is the Taylor/ordering factor

$$\frac{1}{2!} (1+1) = 1, \quad (4.195)$$

the second-order Taylor coefficient times the two vertex orderings. The color contraction of route 002 gives $+f^{ACD} f^{BCE}$, that of route 003 gives $-f^{ACD} f^{BCE}$; including the Koszul and source-attachment signs, the net route signs multiplying the common $+f^{ACD} f^{BCE}$ are

$$s_{002} = -i, \quad s_{003} = +i. \quad (4.196)$$

Remark 4.33 (Sign bookkeeping). The color and sign outcomes (4.196) are quoted at the granularity at which they were machine-verified (contraction value, ports, selected edge, net sign per route); we do not display an intermediate term-by-term sign chain. The quoted color outcomes and route signs are independent machine-checked rows, not a displayed factorization: the effective multiplier of the common $+f^{ACD} f^{BCE}$ is s_{00n} , with the color-contraction sign folded into it.

4. Prefactors and the external map. Assembling the vertex factors, propagator numerators and the multiplicity (4.195), the triangle and contact prefactors are

$$P_{\Delta} = g^2 \left(\frac{\sqrt{2}g}{\hbar} \right) \left(-\frac{\sqrt{2}g}{\hbar} \right) \left(\frac{\hbar}{16} \right)^3 \frac{1}{2!} (1+1) = -\frac{\hbar g^4}{2048}, \quad P_{I_1 H} = g^2 (\sqrt{2}g) \left(-\frac{\sqrt{2}g}{\hbar} \right) \left(\frac{\hbar}{16} \right)^2 = -\frac{\hbar g^4}{128}. \quad (4.197)$$

The factors read: g^2 from the two source legs; one M vertex and one H_- vertex; three chiral propagators $(\hbar/16)^3$ in the triangle; in $P_{I_1 H}$ the $O(g)$ commutator $\sqrt{2}[\mathcal{D}_+ u, \phi_r]$ of one insertion supplies $\sqrt{2}g$, one H_- vertex, and two chiral propagators. For the external legs, the source probe $u_{\dot{\alpha}} = \theta^+ \theta^- \bar{\theta}_{\dot{\alpha}}$ gives

$$\mathcal{D}^2 \bar{\mathcal{D}}_{\dot{\alpha}} u_{\dot{\alpha}} = -2, \quad \mathcal{D}^2 \bar{\mathcal{D}}_{\dot{\alpha}} u = -\frac{4\sqrt{2}}{g} D_{\dot{\alpha}}, \quad C^t = g \tilde{\phi}^t, \quad M_{\text{ext}} = \frac{2\sqrt{2}}{g^2}, \quad (4.198)$$

i.e. the probe ratio $(-4\sqrt{2}/g)/(-2) = 2\sqrt{2}/g$ expresses the gauge leg through the letter $D_{\dot{\alpha}}$, and $\tilde{\phi}^t = g^{-1} C^t$ expresses the antichiral leg through C^t . Each nonzero row then assembles as

$$\Gamma_{\text{row}} = P \times N^{\text{ev}} \times \int_{\ell}^{\text{DRED}} \frac{\mu_{\ell}^2}{D_0 D_1 D_2} \times M_{\text{ext}} \times (s_{00n} f^{ACD} f^{BCE}), \quad (4.199)$$

with $P \in \{P_{\Delta}, P_{I_1 H}\}$, the evanescent word N^{ev} of paragraph 6, and the simplex weights decomposing the result onto the carriers (4.186).

5. D -algebra chain. Let Π_b be the bridge projector (edge r_1) and Π_s the source-edge projector; both are Grassmann-even. Since $\mathcal{D}^2 = 2\mathcal{D}_- \mathcal{D}_+$, the Leibniz rule gives the complete identity

$$\mathcal{D}^2(\Pi_b \Pi_s) = (\mathcal{D}^2 \Pi_b) \Pi_s + \Pi_b (\mathcal{D}^2 \Pi_s) + 2(\mathcal{D}_- \Pi_b)(\mathcal{D}_+ \Pi_s) - 2(\mathcal{D}_+ \Pi_b)(\mathcal{D}_- \Pi_s). \quad (4.200)$$

In the ordered $(u, \tilde{\phi}^t)$ projection the first and third terms vanish,

$$(\mathcal{D}^2 \Pi_b) \Pi_s = 0, \quad 2(\mathcal{D}_- \Pi_b)(\mathcal{D}_+ \Pi_s) = 0, \quad (4.201)$$

leaving the two survivors $\Pi_b(\mathcal{D}^2 \Pi_s)$ and $-2(\mathcal{D}_+ \Pi_b)(\mathcal{D}_- \Pi_s)$. The pure-antichiral endpoint is

$$\theta^2 = -2\theta^+ \theta^-, \quad \boxed{\mathcal{D}^2 \theta^2 = -4}. \quad (4.202)$$

For route 002 the direct core before the endpoint is $N_{\text{core},002}^{\text{dir}} = 1024 v(r_0) \det r_2$, and the two survivors collapse exactly to

$$\Pi_b(\mathcal{D}^2 \Pi_s) \mapsto (-4) \cdot 1024 v(r_0) \det r_2 = 4096 v(r_0) \bar{r}_2^2, \quad -2(\mathcal{D}_+ \Pi_b)(\mathcal{D}_- \Pi_s) \mapsto -2048 v(r_0) \det r_1 = 2048 v(r_0) \bar{r}_1^2, \quad (4.203)$$

using $\det r_i = -\bar{r}_i^2$. Thus the channel has *two occurrences* per oriented route: the direct occurrence on edge $e = 2$ and the transported occurrence on the bridge edge $e = 1$, with ratio

$$\frac{O^{\text{tr}}}{O^{\text{dir}}} = \frac{-2}{-4} = \frac{1}{2}. \quad (4.204)$$

For route 003, computed in its independent orientation, the cores are

$$N_{\text{core},003}^{\text{dir}} = -1024 v(\rho_2) \det \rho_0 = 1024 v(k_0) \det k_2, \quad N_{\text{core},003}^{\text{tr}} = +2048 v(\rho_2) \det \rho_1, \quad (4.205)$$

and using $v(\rho_2) = -v(k_0)$ and $\det \rho_j = \det k_{2-j}$ the endpoint words are $4096 v(k_0) \bar{k}_2^2$ (direct, edge k_2) and $2048 v(k_0) \bar{k}_1^2$ (transported, edge k_1). The single-projector identity

$$\mathcal{D}^2 \bar{\mathcal{D}}^2 \mathcal{D}^2 \delta^4 = 16 \det(r) \mathcal{D}^2 \delta^4 \quad (4.206)$$

does not delete the neighboring occurrence, because

$$\mathcal{D}_+ \bar{\mathcal{D}}^2 \mathcal{D}^2 \delta^4 = 16 \det(r) \mathcal{D}_+ \delta^4 - \mathcal{D}^2 \bar{\mathcal{D}}^2 \mathcal{D}_+ \delta^4. \quad (4.207)$$

6. Evanescent extraction and tensor reduction. For every selected edge e , the full- d Schwinger–Dyson pair cancels identically,

$$\frac{r_{e,d}^2 R_e}{D_0 D_1 D_2} - \frac{R_e}{\prod_{j \neq e} D_j} = 0, \quad (4.208)$$

and in DRED the cutting-failure identity (4.15) leaves the universal remainder

$$\frac{\bar{r}_e^2 R_e}{D_0 D_1 D_2} - \frac{R_e}{\prod_{j \neq e} D_j} = \frac{\mu_\ell^2 R_e}{D_0 D_1 D_2}. \quad (4.209)$$

The direct and transported evanescent words are therefore

$$N_{\text{dir}}^{\text{ev}} = 4096 \mu_\ell^2 v(r_0), \quad N_{\text{tr}}^{\text{ev}} = 2048 \mu_\ell^2 v(r_0), \quad (4.210)$$

and identically on route 003 with $v(r_0) \rightarrow v(k_0)$ and edges k_2, k_1 . The ordered-simplex moments with the $\Gamma(3) = 2$ measure are

$$2 \int_{\Sigma_2} 1 = 1, \quad 2 \int_{\Sigma_2} y = \frac{1}{3}, \quad 2 \int_{\Sigma_2} z = \frac{1}{3}, \quad (4.211)$$

and with $r_0 = L + yp + z(p + q)$, respectively $k_0 = L + yp + z(p + q)$ after the rebase (4.191),

$$2 \int_{\Sigma_2} r_0 = \frac{2}{3} p + \frac{1}{3} q, \quad 2 \int_{\Sigma_2} k_0 = \frac{2}{3} p + \frac{1}{3} q. \quad (4.212)$$

The rank-one numerator $v(r_0)$ thereby reduces onto the ordered carriers (4.186) as

$$\text{route 002 : } \frac{2}{3} A_1 + \frac{1}{3} A_2, \quad \text{route 003 : } \frac{1}{3} A_1 + \frac{2}{3} A_2, \quad (4.213)$$

where the reversed ports of route 003 place $q/3$ on A_1 and $2p/3$ on A_2 . No factor $p^2, \bar{p}^2, q^2, \bar{q}^2$ multiplies these carriers, so no external-equation-of-motion admixture survives: the contact rows E_1, E_2 vanish by the external equations of motion ($\bar{q}^2 - q_d^2 = 0, \bar{p}^2 - p_d^2 = 0$), and T_1, T_3 vanish by spinor-derivative nilpotence ($\mathcal{D}_+^2 = 0$).

7. Pole cancellation. The transported occurrence cancels against the $\langle I_1 S_{H_-} \rangle$ contact row of its own route. The prefactor arithmetic is

$$P_\Delta(2048) + P_{I_1 H}(-128) = -\hbar g^4 + \hbar g^4 = 0, \quad (4.214)$$

so the full- d bubble pole cancels pointwise, before integration:

$$P_\Delta \frac{2048 r_{1,d}^2 v(r_0)}{D_0 D_1 D_2} + P_{I_1 H} \frac{-128 v(r_0)}{D_0 D_2} = 0, \quad (4.215)$$

leaving only the evanescent remainder

$$P_\Delta \frac{2048 \mu_\ell^2 v(r_0)}{D_0 D_1 D_2}. \quad (4.216)$$

Indeed, writing $\bar{r}_1^2 = r_{1,d}^2 + \mu_\ell^2$ in the transported occurrence, the $r_{1,d}^2$ part cancels D_1 and lands exactly on the contact row $T_2 = -128 v(r_0)/(D_0 D_2)$ with the matching prefactors (4.214); route 003 is identical with $T_4 = -128 v(k_0)/(K_0 K_2)$ on the edge k_1 . The two contact rows cannot be merged into one untagged polynomial: T_2 carries color $+i f^{ACD} f^{BCE}$, ports (u^D, C^{tE}) and edge r_1 , while T_4 carries $-i f^{ACD} f^{BCE}$, ports (C^{tD}, u^E) and edge k_1 ; each cancels its own route's transported occurrence. For the direct occurrence, the full- d pair (4.208) applies with $R_e = 4096 v(r_0)$: the parent term $4096 v(r_0) r_{2,d}^2/(D_0 D_1 D_2)$ is cancelled by the Euler-kinetic cut row, whose regulated word is

$$-4096 v(r_0)/(D_0 D_1) \text{ (route 002)}, \quad -4096 v(k_0)/(K_0 K_1) \text{ (route 003)}, \quad (4.217)$$

an Euler-operator insertion collapsing its propagator by the cutting rule $KK^{-1} = \delta$ of Section 4.3, leaving the DRED remainder $4096 \mu_\ell^2 v(r_0)/(D_0 D_1 D_2)$. Finally, the Euler-potential and explicit-potential rows of Remark 4.32 have identical regulated kernels and opposite signs,

$$(+\sqrt{2} K_{\text{potential}}) + (-\sqrt{2} K_{\text{potential}}) = 0, \quad (4.218)$$

and cancel before integration.

Remark 4.34 (Granularity of the cut row). The Euler-kinetic cut is presented through the universal Schwinger–Dyson identity (4.208) together with the recorded contact word (4.217); no standalone prefactor formula for this row exists in the underlying computation, and none is needed, since the cutting rule fixes the row up to the recorded word.

8. Loop integral and the assembled rows. With all bubble poles cancelled, the loop content is the finite master (4.16),

$$\int_\ell^{\text{DRED}} \frac{\mu_\ell^2}{D_0 D_1 D_2} = \frac{1}{32\pi^2}. \quad (4.219)$$

The assembly (4.199) gives the magnitude checks

$$\frac{\hbar g^4}{2048} \cdot 4096 \cdot \frac{1}{32\pi^2} \cdot \frac{2\sqrt{2}}{g^2} = 2\sqrt{2} \frac{\hbar g^2}{16\pi^2} \text{ (direct)}, \quad \frac{\hbar g^4}{2048} \cdot 2048 \cdot \frac{1}{32\pi^2} \cdot \frac{2\sqrt{2}}{g^2} = \sqrt{2} \frac{\hbar g^2}{16\pi^2} \text{ (transported)}. \quad (4.220)$$

Including the carrier reduction (4.213) and the signs (4.196), the direct rows are

$$\Gamma_{002}^{\text{dir}} = -2i\sqrt{2} \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \left(\frac{2}{3} A_1 + \frac{1}{3} A_2 \right), \quad \Gamma_{003}^{\text{dir}} = +2i\sqrt{2} \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \left(\frac{1}{3} A_1 + \frac{2}{3} A_2 \right), \quad (4.221)$$

hence

$$\Gamma_{\text{dir}} = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \left(-\frac{2i\sqrt{2}}{3} A_1 + \frac{2i\sqrt{2}}{3} A_2 \right). \quad (4.222)$$

The transported rows are half the direct ones by (4.204):

$$\Gamma_{002}^{\text{tr}} = -i\sqrt{2} \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \left(\frac{2}{3} A_1 + \frac{1}{3} A_2 \right), \quad \Gamma_{003}^{\text{tr}} = +i\sqrt{2} \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \left(\frac{1}{3} A_1 + \frac{2}{3} A_2 \right), \quad (4.223)$$

$$\Gamma_{\text{tr}} = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \left(-\frac{i\sqrt{2}}{3} A_1 + \frac{i\sqrt{2}}{3} A_2 \right). \quad (4.224)$$

Summing direct and transported, $-\frac{2i\sqrt{2}}{3} - \frac{i\sqrt{2}}{3} = -i\sqrt{2}$ and $\frac{2i\sqrt{2}}{3} + \frac{i\sqrt{2}}{3} = +i\sqrt{2}$, the 2 : 1 split reconstructs an integer vector. The $SU(3)$ lift attaches ε_{rst} : for every ordered $r \neq s$,

$$\Gamma_{B_r > B_s}^{(1)} = \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \varepsilon_{rst} \left[-i\sqrt{2} A_1 + i\sqrt{2} A_2 \right]. \quad (4.225)$$

9. Letter normalization: the final result. Restoring the letters of Definition 4.1, the two insertions contribute $(1/\sqrt{2})^2 = \frac{1}{2}$ by (4.185), while the carriers convert as

$$A_1 = \langle D^D, C^{tE} \rangle = -i \langle \partial c^D, \gamma^{tE} \rangle, \quad A_2 = \langle C^{tD}, D^E \rangle = -i \langle \gamma^{tD}, \partial c^E \rangle. \quad (4.226)$$

The net coefficient is therefore $\frac{1}{2} \cdot (-i\sqrt{2})(-i) \cdot \frac{\hbar g^2}{16\pi^2} = -\frac{1}{\sqrt{2}} \cdot \frac{\hbar g^2}{16\pi^2} = -\frac{\hbar g^2}{16\sqrt{2}\pi^2}$, and the unit-normalized result (4.225) becomes

$$\left. \nabla_-(\beta_r^A \beta_s^B) \right|_{1\text{-loop}} = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \varepsilon_{rst} \left\{ \langle \partial c, \gamma^t \rangle - \langle \gamma^t, \partial c \rangle \right\}^{DE}, \quad (4.227)$$

which is exactly the row (4.271) of the channel table.

Remark 4.35 (The diagonal (β_r, β_r) is an exact zero). For $r = s$ the only triangle topology is TMM, and its two marked words vanish by spinor-derivative nilpotence:

$$K_L = \int d^8 z_L d^8 z_R [\mathcal{D}_- \mathcal{D}_+ \bar{\mathcal{D}}^2 \mathcal{D}^2 \delta_{SL}] [\mathcal{D}_+ \bar{\mathcal{D}}^2 \mathcal{D}^2 \delta_{SR}] \delta_{LR} B_r(L) B_r(R) = 0, \quad K_R = 0, \quad (4.228)$$

with K_R the same expression with the two brackets exchanged, so $K_L - K_R = 0$. Every H_- -routed graph dies on the repeated-index obstruction

$$\varepsilon_{rtu} \delta_{rt} = \varepsilon_{rru} = 0, \quad \varepsilon_{rtu} \delta_{ru} = \varepsilon_{rtr} = 0, \quad (4.229)$$

a matter quartic cannot absorb two B_r source legs, and all spectator parents are one-particle-reducible. Hence $\nabla_-(\beta_r^A \beta_r^B)|_{1\text{-loop}} = 0$: class (iii) of Section 4.4.

Remark 4.36 (Why these are the accepted weights). An earlier independent evaluation of this family was rejected on three arithmetic grounds, and the displayed weights are the adjudicated ones: (i) its final-match claim summed the simplex weights as $-\sqrt{2} \frac{2}{3} + \sqrt{2} \frac{1}{3} = -\sqrt{2}$, which is false — the sum is $-\sqrt{2}/3$, and the correct integer vector arises only from the two-route sum (4.222)–(4.224); (ii) it multiplied the triangle prefactor by a spurious flavor-Wick factor 2, obtaining $+\hbar g^4/1024$ instead of the $-\hbar g^4/2048$ of (4.197); (iii) it dropped the $\langle I_1 S_{H_-} \rangle$ contact rows, thereby missing the transported occurrence (4.203) and the 2 : 1 split (4.204) whose full- d pole cancels by (4.214).

10. Comparison with the holomorphic twist. Budzik and Kulp compute the same channel in the holomorphic twist [1], their eq. (3.66):

$$Q_1((\beta_I)^A(\beta_J)^B) = \kappa^2 \varepsilon_{IJK} f_{ACD} f_{BCE} [\partial_{\dot{a}} c^D \partial^{\dot{a}} (\gamma^K)^E - \partial_{\dot{a}} (\gamma^K)^D \partial^{\dot{a}} c^E]. \quad (4.230)$$

With $\langle X, Y \rangle = (\partial_{\dot{a}} X)(\partial^{\dot{a}} Y)$ this is $\kappa^2 \varepsilon_{IJK} f_{ACD} f_{BCE} \{ \langle \partial c, \gamma^K \rangle - \langle \gamma^K, \partial c \rangle \}^{DE}$ — word by word the bracket of (4.227): the same antisymmetric ∂c – γ pair, the same ordering, the same ε and the same color tensor. On the prefactor, the kernel factor 2 of Section 4.13 (the $\Gamma(3)$ of the triangle Feynman parametrization, absent from the printed twist kernel) and the correspondence $Q_0 \leftrightarrow -\frac{1}{2} \nabla_-$ of Remark 4.2 map the coefficient κ^2 to

$$-2\kappa^2 \hbar_{\text{HT}} = -\frac{\hbar g^2}{16\sqrt{2}\pi^2}, \quad (4.231)$$

using the loop-scale identification $\hbar_{\text{HT}} \kappa^2 = \hbar g^2 / (32\sqrt{2}\pi^2)$. The diagonal vanishes on the twist side as well, since ε_{IJK} requires $I \neq J$, matching Remark 4.35.

Remark 4.37 (Status of the comparison). The agreement above is a comparison against an external target [1], not an internal completeness proof: the loop-scale identification entering (4.231) is a conditional dictionary value, and the seventeen-candidate adversarial census of Remark 4.8 remains open. The channel result (4.227) itself is established independently of the twist, by the derivation of paragraphs 1–9.

4.11 The channels (β_r, γ^s) and (γ^s, β_r)

The mixed matter family (4.270) is structurally different from every other nonzero family: its anomaly is *not* an ordinary parent graph. Every ordinary triangle parent of the ordered sources $(\nabla_+ \Phi_r, \tilde{\Phi}^s)$ can be enumerated, and each vanishes on the physical output word $\langle \partial c, \partial c \rangle$; the anomaly is instead the *regulated coincident Euler Jacobian* — the Schwinger density divergence $K_{-, \epsilon}^{AB}(z, z)$ of the kinetic Euler occurrence, evaluated through the cutting failure of Section 4.3 and the antichiral symbol trace. We present the ordinary inventory first and prove its vanishing, then the Euler-Jacobian mechanism; conflating the two was the historical error diagnosed in Remark 4.41. As in the seed channel, the orbit is computed with the unit-normalized insertion $\nabla_+ \Phi_r = \sqrt{2} \beta_r$; the letter normalization of Definition 4.1 is restored at the end.

1. Ordinary graph inventory: the route census. The ordinary one-matter-loop parents fall into two classes. The *TMM* triangles (two matter edges and one edge carrying the full $d^4\theta$ measure) give, for $r = s$, two routes: the vector-line bridge, whose output is a $\Phi\tilde{\Phi}$ pair and which cannot carry two external antichiral field strengths, and the matter bridge, whose output is a pair of vector lines — the only ordinary route that can carry the two ∂c letters. For $r \neq s$ there is a single TMM route, the vector-line bridge, with no two- ∂c output. The *THH* triangles (whose chiral and antichiral measure edges are evaluated directly rather than replaced by the full measure) give, for $r = s$, two routes, both with flavor sign $+1$, and, for $r \neq s$, a single route with flavor sign -1 ; none carries the two- ∂c output. Figure 8 shows the representative parent. The primitive coupling weights before Grassmann projection are

$$w_{\text{TMM, matter}} = \frac{\hbar g^2}{4096}, \quad w_{\text{TMM, vector}} = -\frac{\hbar g^2}{128}, \quad w_{\text{THH}} = \frac{\hbar g^2}{2048}, \quad w_{\text{Wick}} = \frac{1}{2!}(1+1) = 1, \quad (4.232)$$

from three matter propagators, $(1/16)^3$; from the vector line with its factor 2 and sign, $-2(1/16)^2$; from two matter propagators and the $\tilde{\mathcal{W}}$ projector, $(1/16)^2(1/8)$; and from the quadratic Dyson coefficient times the two equivalent contraction assignments. These weights belong to the ordinary parents, which vanish on the physical word; the physical coefficient is fixed by the Schwinger orbit below.

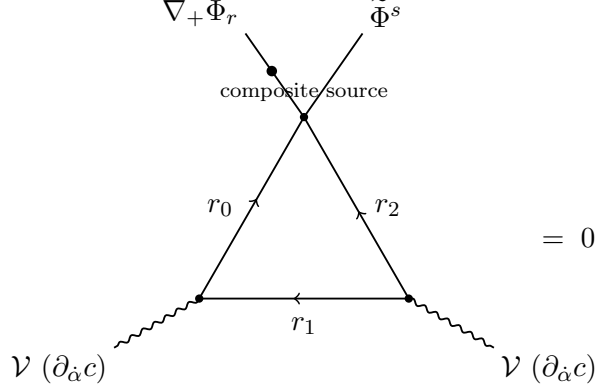


Figure 8: The representative ordinary matter-triangle parent of the (β_r, γ^s) family. The three edges are matter propagators with edge momenta r_0, r_1, r_2 , denominators $D_i = r_{i,d}^2$. The two ordered sources sit at one composite-source vertex: the marked insertion $\nabla_+ \Phi_r$ (filled node) and the antichiral leg $\tilde{\Phi}^s$. The two output letters enter as the field-strength preimages $\widetilde{\mathcal{W}}_{\dot{\alpha}}^{(1)} = -\frac{1}{8} \mathcal{D}^2 \bar{\mathcal{D}}_{\dot{\alpha}} \mathcal{V}$ on the two vector legs at the two distinct remaining vertices (two distinct superspace points). Every ordinary parent of the family vanishes on the physical output word $\langle \partial c, \partial c \rangle$, as displayed; the kernels themselves are nonzero (Remark 4.38). The anomaly is instead the regulated coincident Euler Jacobian of the kinetic occurrence (4.241).

Remark 4.38 (Machine-zero verdict with nonzero-kernel control). An external replay of the route census confirms the counts ($2 + 2$ routes for $r = s$, $1 + 1$ for $r \neq s$, exactly one ordinary route able to carry the two- ∂c word) and the physical-word projection: every ordinary parent — all four ∂c -preimage assignments of the TMM triangles, the evanescent residuals, the THH routes, the vector bridge, and the nonlinear contact rows, in both orderings — vanishes on $\langle \partial c, \partial c \rangle$. The kernels themselves are nonzero: a control evaluation of the same kernels on an off-physical test word returns a nonzero value for both orderings, so the zeros are projections, not zero kernels. Mechanically, two ∂c letters cannot be saturated at a single superspace point — all same-point products of their preimages vanish — and the anomaly needs the two independent antichiral Fourier variables of the coincident kernel of step 9.

2. The kinetic source expansion and the order- $\mathcal{V}^2$ orbit. The antichiral Euler operator was given in (4.5):

$$\frac{\delta S}{\delta \tilde{\Phi}^r} = -\frac{1}{4} \nabla^2 \Phi_r - \frac{1}{\sqrt{2}} \varepsilon_{rtu} (\gamma^t \times \gamma^u). \quad (4.233)$$

Its kinetic piece is expanded covariantly *before* any graph enumeration. With the undotted frame index $a = (+, -)$, the gauge connection and its expansion are

$$\Gamma_a := e^{-\mathcal{V}} \mathcal{D}_a e^{\mathcal{V}} = \mathcal{D}_a \mathcal{V} + \frac{1}{2} [\mathcal{D}_a \mathcal{V}, \mathcal{V}] + O(\mathcal{V}^3), \quad \nabla_a = \mathcal{D}_a + \Gamma_a, \quad (4.234)$$

and the graded Leibniz expansion on the even column Φ_r gives

$$\begin{aligned} \nabla^2 \Phi_r &= (\mathcal{D}^a + \Gamma^a)(\mathcal{D}_a \Phi_r + \Gamma_a \Phi_r) \\ &= \mathcal{D}^2 \Phi_r + 2\Gamma^a \mathcal{D}_a \Phi_r + (\mathcal{D}^a \Gamma_a) \Phi_r + \Gamma^a \Gamma_a \Phi_r, \end{aligned} \quad (4.235)$$

using $\Gamma_a \mathcal{D}^a = -\Gamma^a \mathcal{D}_a$. Therefore $\frac{1}{2}(\nabla^2 \Phi_r) \gamma^s = \mathcal{I}_0 + \mathcal{I}_1 + \mathcal{I}_2 + O(\mathcal{V}^3)$ with

$$\mathcal{I}_0 = \frac{1}{2}(\mathcal{D}^2 \Phi_r) \gamma^s, \quad (4.236)$$

$$\mathcal{I}_1 = \left[(\mathcal{D}^a \mathcal{V}) \mathcal{D}_a \Phi_r + \frac{1}{2}(\mathcal{D}^2 \mathcal{V}) \Phi_r \right] \gamma^s, \quad (4.237)$$

$$\mathcal{I}_2 = \left[\frac{1}{2}[\mathcal{D}^a \mathcal{V}, \mathcal{V}] \mathcal{D}_a \Phi_r + \frac{1}{4} \mathcal{D}^a [\mathcal{D}_a \mathcal{V}, \mathcal{V}] \Phi_r + \frac{1}{2}(\mathcal{D}^a \mathcal{V})(\mathcal{D}_a \mathcal{V}) \Phi_r \right] \gamma^s. \quad (4.238)$$

Writing M_1 and M_2 for the $\mathcal{V}$ and $\mathcal{V}^2/2$ terms of the matter-gauge chain $\tilde{\Phi} e^{\mathcal{V}} \Phi$, the complete order- $\mathcal{V}^2$ regulated orbit is the direct sum

$$\boxed{\mathcal{I}_0 \frac{1}{2!} M_1 M_1 \oplus \mathcal{I}_1 M_1 \oplus \mathcal{I}_0 M_2 \oplus \mathcal{I}_2} \quad (4.239)$$

whose four rows are respectively the triangle parent, the nonlinear-source collapsed contact, the matter seagull, and the source seagull. Their covariant sum is the coincident kernel K_- of step 9. The Schwinger source vector field of step 4 has no vector, ghost, or non-minimal component, so there is no gauge or ghost Jacobian orbit in this channel.

3. The three occurrence tags. The tree descendant of the unit-normalized insertion follows from (4.6):

$$\nabla_-(\nabla_+ \Phi_r) = -2 \frac{\delta S}{\delta \tilde{\Phi}^r} - \sqrt{2} \varepsilon_{rtu} (\gamma^t \times \gamma^u). \quad (4.240)$$

It must be expanded into three *separately tagged occurrences*:

$$\mathcal{I}_{E,\text{kin}}^{AB} := +\frac{1}{2}(\nabla^2 \Phi_r)^A \gamma^{sB}, \quad (4.241)$$

$$\mathcal{I}_{E,\text{pot}}^{AB} := +\sqrt{2} \varepsilon_{rtu} (\gamma^t \times \gamma^u)^A \gamma^{sB}, \quad (4.242)$$

$$\mathcal{I}_{X,\text{pot}}^{AB} := -\sqrt{2} \varepsilon_{rtu} (\gamma^t \times \gamma^u)^A \gamma^{sB}, \quad (4.243)$$

the kinetic Euler occurrence (containing the inverse kinetic edge), the Euler-potential occurrence, and the explicit-potential occurrence. The last two rows have identical regulated contact kernels and opposite source coefficients, and neither contains an inverse kinetic edge; they are retained until their regulated evaluation pairs them in step 7. Replacing the tagged descendant (4.240) by the single kinetic term $\frac{1}{2} \nabla^2 \Phi_r$ before regulation erases the occurrence tags and is not the computation performed here.

Remark 4.39 (No contribution from the gauge-current orbit). The vector equation of motion of (4.5) contains the current $-2i(\Phi_s \times \tilde{\Phi}^s)$. Acting on it with $-\nabla_+$ and using $\nabla_+ \tilde{\Phi}^s = 0$ gives $+2i(\nabla_+ \Phi_s \times \tilde{\Phi}^s)$, which cancels the explicit outer contact $-2i(\nabla_+ \Phi_s \times \tilde{\Phi}^s)$ pointwise, for arbitrary independent external momenta, before any propagator or loop integration; the gauge-fixing functional has only vector ports and contributes nothing on this source. Hence no gauge-current or gauge-fixing Jacobian feeds the (β_r, γ^s) family: the entire anomaly is the matter Euler Jacobian below.

4. The Schwinger vector field and the surviving contraction. For the ordered source (β_r^A, γ^{sB}) the Schwinger vector field is the antichiral field redefinition

$$X_{rs}^{AB}(z) := \tilde{\Phi}^{sB}(z) \frac{\vec{\delta}}{\delta \tilde{\Phi}^{rA}(z)}, \quad (4.244)$$

with support only on the integrated $\tilde{\Phi}$ block: its vector, Faddeev–Popov ghost, Nielsen–Kallosh and non-minimal components vanish identically. The regulated Schwinger–Dyson identity of Section 4.2 applied to X_{rs}^{AB} gives

$$0 = \left\langle \operatorname{div}_{E,\epsilon} X_{rs}^{AB} - \frac{1}{\hbar} X_{rs}^{AB}(S_E) \right\rangle, \quad \left\langle \gamma^{sB} \left(\frac{\delta S}{\delta \tilde{\Phi}^r} \right)^A \right\rangle = -\hbar g^2 \left\langle \operatorname{div}_{E,\epsilon} X_{rs}^{AB} \right\rangle, \quad (4.245)$$

the absorbed factor g^{-2} of the frame of Section 3 being restored in this step only. The differentiated insertion fixes the flavor tensor before any loop calculation:

$$\frac{\delta \tilde{\Phi}^{sB}(z)}{\delta \tilde{\Phi}^{rA}(z)} = \delta_{rs} \delta_A^B \delta_-, \quad \operatorname{div}_{E,\epsilon} X_{rs}^{AB} = \delta_{rs} K_{-,\epsilon}^{AB}(z, z), \quad (4.246)$$

with $\delta_- := -\frac{1}{4} \mathcal{D}^2 \delta_e$ the constrained coincident antichiral delta and $K_{-,\epsilon}^{AB}(z, z)$ the coincident Euler–Jacobian kernel. Thus $r \neq s$ vanishes independently of any external target. This coincident self-contraction of the source pair is the only Wick contraction surviving in the channel; at second order in the interaction the Dyson–Wick weight is $\frac{1}{2!}(1+1) = 1$, the two equivalent assignments of the two interaction copies cancelling the $1/2!$.

5. The marked-edge D -algebra word. The marked linear source carries the exact edge word

$$\mathcal{M}_B(r_e) \delta_e := \mathcal{D}_- \mathcal{D}_+ \bar{\mathcal{D}}^2 \mathcal{D}^2 \delta_e = \frac{1}{2} \mathcal{D}^2 \bar{\mathcal{D}}^2 \mathcal{D}^2 \delta_e = 8 \bar{r}_e^2 \mathcal{D}^2 \delta_e, \quad (4.247)$$

where the first equality uses $\mathcal{D}_- \mathcal{D}_+ = \frac{1}{2} \mathcal{D}^2$ and the second is the four-dimensional spinor D -algebra square of the edge momentum. Dividing by the constrained matter-propagator factor $16D_0$, $D_0 := r_{e,d}^2$,

$$\frac{\mathcal{M}_B(r_e) \delta_e}{16D_0} = \frac{1}{2} \frac{\bar{r}_e^2}{D_0} \mathcal{D}^2 \delta_e. \quad (4.248)$$

The full- d Schwinger contact on the same edge is $\frac{1}{2}(r_{e,d}^2/D_0) \mathcal{D}^2 \delta_e$, and their difference is

$$\frac{1}{2} \frac{\mu_\ell^2}{D_0} \mathcal{D}^2 \delta_e = -2 \frac{\mu_\ell^2}{D_0} \delta_-. \quad (4.249)$$

Remark 4.40 (Two recorded conventions for the spinor-square identity). The square $\mathcal{D}^2 \bar{\mathcal{D}}^2 \mathcal{D}^2 \delta^4 = 16 \det(r) \mathcal{D}^2 \delta^4$ of Section 4.10 (equation (4.206)) and the half square $8 \bar{r}_e^2 \mathcal{D}^2 \delta_e$ of (4.247) are recorded with opposite signs relative to the same determinant $\det r = -\bar{r}^2$. The two verification engines use different superspace-derivative conventions — one with real spinor derivatives, the other with i -loaded ones, under which the operator square scales by $i^6 = -1$ — and each machine-checks its own identity with its recorded sign ($+16 \det(r)$, respectively $-16 \det(r)$, for the full square). Each convention is used consistently within its own subsection, the physical coefficients of both channels were machine-verified independently, and no result is affected.

The factor 2 is the descendant factor of (4.240); it is not an additional graph multiplicity.

6. Cutting failure on the marked edge. On the selected Euler edge the four-dimensional square exceeds the d -dimensional propagator inverse by the evanescent square, $\bar{r}_e^2 = r_{e,d}^2 + \mu_\ell^2$. If the D -algebra square were the full loop square, the Schwinger cut would be exact and the row would vanish:

$$\frac{r_{e,d}^2}{D_0 D_1 D_2} - \frac{1}{D_1 D_2} = 0. \quad (4.250)$$

The spinor D -algebra supplies the four-dimensional square instead, so the universal cutting-failure identity (4.15) leaves only

$$\frac{\bar{r}_e^2}{D_0 D_1 D_2} - \frac{1}{D_1 D_2} = \frac{\bar{r}_e^2 - r_{e,d}^2}{D_0 D_1 D_2} = \frac{\mu_\ell^2}{D_0 D_1 D_2}. \quad (4.251)$$

Replacing $\bar{r}_e^2$ by $r_{e,d}^2$ before this subtraction makes the expression identically zero and removes the anomaly.

7. The potential-row pairing. The two potential occurrences (4.242) and (4.243) carry no inverse kinetic edge; their regulated evaluations $\mathcal{R}_\epsilon$ are ordinary contact rows built on one common kernel K_{pot} , and their opposite source coefficients pair them exactly:

$$\mathcal{R}_\epsilon \mathcal{I}_{E,\text{pot}}^{AB} + \mathcal{R}_\epsilon \mathcal{I}_{X,\text{pot}}^{AB} = \sqrt{2} K_{\text{pot}} - \sqrt{2} K_{\text{pot}} = 0. \quad (4.252)$$

The kinetic Euler occurrence (4.241) cannot be deleted with them: it contains the inverse edge, and its Schwinger cut fails by exactly the evanescent remainder (4.251).

8. The antichiral projector and the symbol trace. The external antichiral field strengths enter through the linearized preimage

$$\widetilde{\mathcal{W}}_\alpha^{(1)} = -\frac{1}{8} \mathcal{D}^2 \bar{\mathcal{D}}_\alpha \mathcal{V}, \quad (4.253)$$

and the locked antichiral projector of Section 3, normalized per closed projector,

$$\frac{1}{16} \nabla^2 \bar{\mathcal{D}}^2 U = \mathcal{D}_M \mathcal{D}^M U - (\bar{\mathcal{D}}_\beta U) \widetilde{\mathcal{W}}^{\dot{\beta}} - \frac{1}{2} U \nabla^a \mathcal{W}_a, \quad (4.254)$$

(M the summed superspace derivative index of the locked projector) contributes its $\widetilde{\mathcal{W}}^2$ coefficient through exactly two copies of the middle term, with squared coefficient $(-1)(-1) = +1$; the last term contains no odd antichiral Fourier variable and cannot saturate the trace below. Let $\bar{\pi}^{\dot{1}}, \bar{\pi}^{\dot{2}}$ be the odd Fourier variables of the antichiral delta and define the even symbol

$$\Xi_D := \widetilde{\mathcal{W}}_\alpha^D \bar{\pi}^{\dot{\alpha}} = \widetilde{\mathcal{W}}_1^D \bar{\pi}^{\dot{1}} + \widetilde{\mathcal{W}}_2^D \bar{\pi}^{\dot{2}}. \quad (4.255)$$

All four $\widetilde{\mathcal{W}}$ and $\bar{\pi}$ generators are odd, so anticommuting them gives

$$\Xi_D \Xi_E = (\widetilde{\mathcal{W}}_2^D \widetilde{\mathcal{W}}_1^E - \widetilde{\mathcal{W}}_1^D \widetilde{\mathcal{W}}_2^E) \bar{\pi}^{\dot{1}} \bar{\pi}^{\dot{2}} = \langle \widetilde{\mathcal{W}}^D, \widetilde{\mathcal{W}}^E \rangle \bar{\pi}^{\dot{1}} \bar{\pi}^{\dot{2}}, \quad (4.256)$$

the dotted pairing of (4.3). With the Berezin rule $\int d\bar{\pi}^{\dot{2}} d\bar{\pi}^{\dot{1}} \bar{\pi}^{\dot{1}} \bar{\pi}^{\dot{2}} = 1$, the exact dotted trace is

$$\boxed{\int d^2 \bar{\pi} \Xi_D \Xi_E = \langle \widetilde{\mathcal{W}}^D, \widetilde{\mathcal{W}}^E \rangle.} \quad (4.257)$$

For one color label, $\Xi_D^2 = \widetilde{\mathcal{W}}^{D\dot{\alpha}} \widetilde{\mathcal{W}}_\alpha^D \bar{\pi}^{\dot{1}} \bar{\pi}^{\dot{2}}$. The quadratic Dyson coefficient $1/2!$ and the four-dimensional Gaussian $\int \frac{d^4 L}{(2\pi)^4} e^{-L^2/M^2} = \frac{M^4}{16\pi^2}$ fix the coincident quadratic coefficient at

$$\frac{1}{2!} \cdot \frac{1}{M^4} \cdot \frac{M^4}{16\pi^2} = \frac{1}{32\pi^2}, \quad (4.258)$$

the same normalization as the evanescent master integral (4.16).

9. Color chains and the coincident kernel. For the ordered source (β_r^A, γ^{sB}) the open adjoint chain of the orbit is

$$(T_D)^A{}_Z (T_E)^Z{}_C \kappa^{CB} = -f_{DZA} f_{EBZ} = f_{AZD} f_{BZE} = f^{ACD} f^{BCE}, \quad (4.259)$$

with the summed index relabeled in the last step. The independently routed reverse source (γ^{sA}, β_r^B) gives

$$(T_E)^B{}_Z (T_D)^Z{}_C \kappa^{CA} = -f_{EZB} f_{DAZ} = f_{BZE} f_{AZD} = f^{ACD} f^{BCE}, \quad (4.260)$$

the same ordered tensor. Since the dotted pairing of two odd letters is symmetric (Section 4.1), only the $(D \leftrightarrow E)$ -symmetric part of the color tensor contributes. Combining the symbol trace (4.257), the color chain (4.259), the cutting failure (4.251), and the master value (4.16), the coincident kernel at quadratic order in the antichiral field strength is

$$K_{-, \epsilon}^{AB}(z, z) \Big|_{\widetilde{\mathcal{W}}^2} = \frac{1}{32\pi^2} f^{ACD} f^{BCE} \langle \widetilde{\mathcal{W}}^D, \widetilde{\mathcal{W}}^E \rangle. \quad (4.261)$$

10. The evanescent master value and the tensor-integral pair. The triangle master integral of the channel is the universal one (4.16),

$$\lim_{\epsilon \rightarrow 0} \mu^{2\epsilon} \int \frac{d^{4-2\epsilon} L}{(2\pi)^{4-2\epsilon}} \frac{\mu_\ell^2}{(L^2 + \Delta)^3} = \frac{1}{32\pi^2}, \quad (4.262)$$

independent of the Feynman-parameter combination Δ , so the anomaly is a polynomial in external momenta: a local operator. There is no graph-specific factor $4 - d$; the finite number is entirely the evanescent square multiplying the logarithmic ultraviolet residue. Two further integrals belong to the component cross-check of Section 5: with $I_n(\Delta) := \int \frac{d^d \ell}{(2\pi)^d} \mu_\ell^2 / (\ell^2 + \Delta)^n$ one has $\lim_{\epsilon \rightarrow 0} I_3(\Delta) = 1/(32\pi^2)$ and $\lim_{\epsilon \rightarrow 0} \Delta I_4(\Delta) = 0$, and the rank-two box-type pair

$$J := \frac{1}{d} \int \frac{d^d \ell}{(2\pi)^d} \frac{\mu_\ell^2 \ell^2}{(\ell^2 + \Delta)^4} = \frac{1}{d} [I_3(\Delta) - \Delta I_4(\Delta)], \quad \lim_{\epsilon \rightarrow 0} J = \frac{1}{128\pi^2}, \quad \lim_{\epsilon \rightarrow 0} 2J = \frac{1}{64\pi^2}. \quad (4.263)$$

The first number is the tensor integral; the second includes the explicit factor 2 of the four-dimensional sigma-matrix numerator. They cannot be identified with each other, and neither is the triangle master (4.262): their role is to fix the free sign and the loop-counting power (exactly one power of $\hbar$) of the component projection quoted in step 13.

11. Final result. Since $\nabla_- \gamma^s = 0$, only the descendant of the first letter acts. Assembling the three occurrences, the Schwinger identity (4.245), the pairing (4.252), and the kernel (4.261),

$$\begin{aligned} \nabla_- ((\nabla_+ \Phi_r)^A \gamma^{sB}) \Big|_{1\text{-loop}} &= \left\langle \left[-2 \left(\frac{\delta S}{\delta \widetilde{\Phi}^r} \right)^A - \sqrt{2} \varepsilon_{rtu} (\gamma^t \times \gamma^u)^A \right] \gamma^{sB} \right\rangle_{\text{ev}} \\ &= \left\langle -2 \left(\frac{\delta S}{\delta \widetilde{\Phi}^r} \right)^A \Big|_{\text{kin}} \gamma^{sB} \right\rangle_{\text{ev}} + (\sqrt{2} K_{\text{pot}} - \sqrt{2} K_{\text{pot}}) \\ &= +2 \delta_{rs} \hbar g^2 K_{-, \epsilon}^{AB}(z, z) \Big|_{\widetilde{\mathcal{W}}^2} \\ &= 2 \delta_{rs} \hbar g^2 \cdot \frac{1}{32\pi^2} f^{ACD} f^{BCE} \langle \widetilde{\mathcal{W}}^D, \widetilde{\mathcal{W}}^E \rangle \\ &= \delta_{rs} \frac{\hbar g^2}{16\pi^2} f^{ACD} f^{BCE} \langle \widetilde{\mathcal{W}}^D, \widetilde{\mathcal{W}}^E \rangle. \end{aligned} \quad (4.264)$$

The normalization trace is: Schwinger descendant $(-2)(-\hbar g^2) = +2\hbar g^2$; coincident kernel $\frac{1}{2!} \cdot \frac{1}{16\pi^2} = \frac{1}{32\pi^2}$; product $2\hbar g^2/(32\pi^2) = \hbar g^2/(16\pi^2)$ — exactly one power of $\hbar$, the universal one-loop coefficient. Restoring the letter normalization, $\beta_r = \frac{1}{\sqrt{2}} \nabla_+ \Phi_r$ absorbs one factor $\sqrt{2}$, and $\partial_{\dot{\alpha}} c = i \widetilde{\mathcal{W}}_{\dot{\alpha}}$ contributes $i^2 = -1$ to the dotted pairing, so

$$\nabla_- (\beta_r^A \gamma^{sB}), \quad \nabla_- (\gamma^{sA} \beta_r^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \delta_{rs} \langle \partial c, \partial c \rangle^{DE}, \quad (4.265)$$

exactly the row (4.270) of the channel table. The two orderings are equal and independent: $|\gamma^s| = 0$ and $\nabla_- \gamma^s = 0$ give $\nabla_- (\gamma^{sA} \beta_r^B) = \gamma^{sA} \nabla_- \beta_r^B$, and the independent reverse Schwinger vector field with the reverse color chain (4.260) produces the same ordered tensor. The twelve flavor-off-diagonal pairs, $r \neq s$, are exact zeros by the δ_{rs} of (4.246) — the singlet contraction $\delta_{rs} \langle \partial c, \partial c \rangle$ has no off-diagonal support, census class (ii) of Section 4.4.

Remark 4.41 (Where the ordinary-parent obstruction applies, and where it does not). The ordinary bottom-component census remains correct on its own terrain: the mixed quadratic blocks vanish, $G_{\phi\tilde{\psi}} = G_{\psi\tilde{\phi}} = 0$, so no ordinary component triangle with two external- ∂c Yukawa vertices and the bottom sources exists. That statement does not evaluate the Schwinger occurrence. The occurrence first differentiates the coincident insertion, $\tilde{\Phi}^{sB}\tilde{\delta}/\delta\tilde{\Phi}^{rA} \rightarrow \delta_{rs}K_{-,e}^{AB}(z,z)$, and only then admits a bottom projection; promoting the ordinary-parent obstruction to a statement about the regulated density divergence is rejected. The order of operations in this subsection — occurrence tags, Schwinger–Dyson identity, regulated pairing, coincident kernel — is the accepted one; an external audit of the full family passes all 84 of its checks, and the physical verifier passes all 33 of its sector checks.

12. Holomorphic-twist comparison and the generalized Konishi anomaly. The holomorphic-twist computation of [1] prints, for this family (their eq. (3.65)),

$$Q_1((\beta_I)^A(\gamma^J)^B) = \kappa_H^2 \delta_I^J f_{ACD} f_{BCE} \partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} c^E. \quad (4.266)$$

With the loop-scale identification $\hbar_{\text{HT}}\kappa_H^2 = \hbar g^2/(32\sqrt{2}\pi^2)$ of Section 7, the twist prefactor $-2\kappa_H^2\hbar_{\text{HT}}$ is exactly the uniform prefactor $-\hbar g^2/(16\sqrt{2}\pi^2)$ of (4.265), and the output words agree for both orderings — only (β, γ) is printed in [1]; (γ, β) follows from supercommutative normal ordering with no sign — and for all off-diagonal pairs, which vanish on both sides. Physically, the (β_r, γ^s) channel is the regulated generalized Konishi anomaly: the density divergence of the functional measure under the antichiral source redefinition (4.244). The full eighty-one-channel table is its completion over the BPS letter alphabet, obtained here by a direct superspace supergraph computation with no twist-formalism input.

Remark 4.42 (Status of the comparison). The loop-scale identification is a conditional dictionary entry, not a theorem of this paper; the comparison is performed on output words, and it is a cross-check, not an internal completeness proof of the graph census. The Konishi identification is stated at the level of the mechanism — the regulated density divergence of the measure; an independent re-derivation of the classic generalized-Konishi formula in the present conventions is not claimed: the channel computation itself is the derivation offered. The full dictionary, the derivative-tower kernel, and the factor-two adjudication are in Section 7.

13. Component bottom-projection cross-check. The bottom components of the letters are $\beta_r = \psi_{r+}, \gamma^s = \tilde{\phi}^s$, $\partial_{\dot{\alpha}} c = -\tilde{\lambda}_{\dot{\alpha}}$ (Section 5), so at zero extra holomorphic derivative $\langle \partial c, \partial c \rangle^{DE} = \tilde{\lambda}_{\dot{\alpha}}^D \tilde{\lambda}^{E\dot{\alpha}}$, and the left-hand side of (4.265) projects to $Q_-(\psi_{r+}^A \tilde{\phi}_s^B)$. The bottom projection of the superfield result is therefore

$$Q_-(\psi_{r+}^A \tilde{\phi}_s^B) \Big|_{1\text{-loop}} = -\frac{\sqrt{2}\hbar g^2}{32\pi^2} \delta_{rs} f^{ACD} f^{BCE} \tilde{\lambda}_{\dot{\alpha}}^D \tilde{\lambda}^{E\dot{\alpha}}, \quad (4.267)$$

using $1/(16\sqrt{2}) = \sqrt{2}/32$. This is an exact projection of (4.265), not an independent component-box derivation: the mixed propagators vanish, so no ordinary two-Yukawa parent exists at the component level, and the dotted gaugino bilinear is color-symmetric, $\tilde{\lambda}_{\dot{\alpha}}^D \tilde{\lambda}^{E\dot{\alpha}} = \tilde{\lambda}_{\dot{\alpha}}^E \tilde{\lambda}^{D\dot{\alpha}}$, so any antisymmetric color word would annihilate it. The full component treatment, including the role of the tensor pair (4.263), is in Section 5.

4.12 The complete channel table

We now state all seven nonzero families of the master identity (4.8). The letter definitions (4.1) induce a uniform rescaling between the unit-normalized field-strength composites used in the graph computations and the letters of Definition 4.1. The bookkeeping is immediate: in the seed channel the two insertions contribute $(-i/\sqrt{2})^2 = -\frac{1}{2}$ and the output bilinears contribute

$\sqrt{2}$, giving the net prefactor $-\frac{1}{2} \cdot \sqrt{2} \cdot \frac{\hbar g^2}{16\pi^2} = -\frac{\hbar g^2}{16\sqrt{2}\pi^2}$ of equation (4.96). The same arithmetic, applied family by family, yields the common prefactor $-\hbar g^2/(16\sqrt{2}\pi^2)$ for every family, including the $(b, \beta)/(\beta, b)$ family, whose output word is fixed by the residual-supersymmetry closure discussed below. With $\langle X, Y \rangle$ as in (4.3) and $\partial_{\dot{\alpha}}(\partial c)$ the holomorphic derivative of the letter ∂c , the complete set of nonzero channels is

$$\nabla_{-}(b^A b^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \left\{ \langle \partial c, b \rangle - \langle b, \partial c \rangle + \sum_r (\langle \beta_r, \gamma^r \rangle - \langle \gamma^r, \beta_r \rangle) \right\}^{DE}, \quad (4.268)$$

$$\nabla_{-}(b^A \gamma^{rB}), \quad \nabla_{-}(\gamma^{rA} b^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \left\{ \langle \partial c, \gamma^r \rangle - \langle \gamma^r, \partial c \rangle \right\}^{DE}, \quad (4.269)$$

$$\nabla_{-}(\beta_r^A \gamma^{sB}), \quad \nabla_{-}(\gamma^{sA} \beta_r^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \delta_{rs} \langle \partial c, \partial c \rangle^{DE}, \quad (4.270)$$

$$\nabla_{-}(\beta_r^A \beta_s^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \varepsilon_{rst} \left\{ \langle \partial c, \gamma^t \rangle - \langle \gamma^t, \partial c \rangle \right\}^{DE}, \quad (4.271)$$

$$\nabla_{-}(b^A \beta_r^B), \quad \nabla_{-}(\beta_r^A b^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \left\{ \langle \beta_r, \partial c \rangle + \langle \partial c, \beta_r \rangle + \varepsilon_{rst} \langle \gamma^s, \gamma^t \rangle \right\}^{DE}, \quad (4.272)$$

$$\nabla_{-}(b^A \partial_{\dot{\alpha}} c^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \left\{ \frac{1}{3} \langle \partial_{\dot{\alpha}}(\partial c), \partial c \rangle + \frac{2}{3} \langle \partial c, \partial_{\dot{\alpha}}(\partial c) \rangle \right\}^{DE}, \quad (4.273)$$

$$\nabla_{-}(\partial_{\dot{\alpha}} c^A b^B) = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} f^{ACD} f^{BCE} \left\{ \frac{2}{3} \langle \partial_{\dot{\alpha}}(\partial c), \partial c \rangle + \frac{1}{3} \langle \partial c, \partial_{\dot{\alpha}}(\partial c) \rangle \right\}^{DE}. \quad (4.274)$$

The free dotted index in (4.273) and (4.274) is the letter index of $\partial_{\dot{\alpha}} c$ on the left and the extra holomorphic derivative on the right; the weights $(\frac{1}{3}, \frac{2}{3})$ are the $(m, n) = (1, 0)$ instance of the tower kernel of Section 4.13. Together with the fifty-two zeros of Section 4.4, the rows (4.268)–(4.274) exhaust the eighty-one ordered channels: $1 + 6 + 6 + 4 + 6 + 6 = 29$ nonzero, as in (4.21).

Residual-supersymmetry closure of the $(b, \beta)/(\beta, b)$ channels

The family (4.272) is the only one whose output word is not read off a cut parent graph directly. The raw graph carrier of this family contains an equation-of-motion (total-derivative) layer; the physical answer is the unique residual-supersymmetry-closed vector obtained after removing that layer and adding a finite normal-product (scheme) term, with the remaining one-parameter freedom anchored to the (b, b) scale. The closure works as follows. The residual supercharges of the bottom-letter sector, $q_r := \frac{1}{\sqrt{2}} Q_{E,+}^r$, act on the letters by

$$q_r b = 0, \quad q_r \beta_s = \delta_{rs} b, \quad q_r \gamma^s = -\varepsilon_{rst} \beta_t, \quad q_r (\partial_{\dot{\alpha}} c) = \partial_{\dot{\alpha}} \gamma^r. \quad (4.275)$$

At twisted dimension $\frac{9}{2}$, the space of odd $SU(3)$ -singlet letter bilinears annihilated by all q_r is one-dimensional, spanned by the seed output word

$$\left\{ \langle \partial c, b \rangle - \langle b, \partial c \rangle + \sum_r (\langle \beta_r, \gamma^r \rangle - \langle \gamma^r, \beta_r \rangle) \right\}^{DE} \quad (4.276)$$

of (4.268). The (b, β_r) output word is its unique triplet q -primitive:

$$q_s \left\{ \langle \beta_r, \partial c \rangle + \langle \partial c, \beta_r \rangle + \varepsilon_{rtu} \langle \gamma^t, \gamma^u \rangle \right\} = -\delta_{sr} \left\{ \langle \partial c, b \rangle - \langle b, \partial c \rangle + \sum_{r'} (\langle \beta_{r'}, \gamma^{r'} \rangle - \langle \gamma^{r'}, \beta_{r'} \rangle) \right\}. \quad (4.277)$$

Equivalently, in the ordered basis $(\langle \partial c, \beta_r \rangle, \langle \beta_r, \partial c \rangle, \langle \gamma^s, \gamma^t \rangle, \langle \gamma^t, \gamma^s \rangle)$ with $\varepsilon_{rst} = +1$, the three Ward equations $q_s[\cdot] = 0$ ($s = 1, 2, 3$) cut out the one-dimensional ray spanned by $(1, 1, 1, -1)$, and anchoring this ray to the (b, b) scale through (4.277) fixes the unit normalization displayed in (4.272). Since $q_r(b^A b^B) = 0$, the seed channel's own forward orbit is only itself: the physical families were each computed directly, and the closure is a cross-check that fixes the (b, β) normalization.

Remark 4.43 (The finite shift is a scheme term, not an anomaly graph). The passage from the raw graph carrier to the renormalized coefficient vector of the $(b, \beta)/(\beta, b)$ family removes an equation-of-motion (total-derivative) layer and adds a finite normal-product term. This is a finite scheme adjustment, not an additional anomaly graph: all bare-graph anomaly in every channel comes only from the μ_ℓ^2 cutting failure (4.15). The closure was verified by thirty-one independent finite Ward-identity checks. Through this finite shift and the anchoring to the (b, b) scale, the $(b, \beta)/(\beta, b)$ family lands on the same universal prefactor $-\hbar g^2/(16\sqrt{2}\pi^2)$ as every other nonzero family, with the output word displayed in (4.272).

4.13 The holomorphic-derivative tower

Shifting the second letter of a channel bilinear by a formal vector w — equivalently, resumming all holomorphic derivatives — inserts the leg phase $e^{iw \cdot (aq+bp)}$ into the ordered Feynman parametrization of the triangle. The ordered-simplex moment is

$$\int_{0 < a < b < 1} a^p b^q da db = \frac{1}{(p+1)(p+q+2)}, \quad (4.278)$$

so that

$$2 \int_0^1 db \int_0^b da a^{k+\ell} b^{(m-k)+(n-\ell)} = \frac{2}{(k+\ell+1)(m+n+2)}. \quad (4.279)$$

Restoring the binomial weights of the derivative expansion gives the one-loop holomorphic-derivative kernel

$$\boxed{K_{m,n;k,\ell} = \frac{2 \binom{m}{k} \binom{n}{\ell}}{(m+n+2)(k+\ell+1)}} \quad (m, n \in \mathbb{Z}_{\geq 0}, 0 \leq k \leq m, 0 \leq \ell \leq n). \quad (4.280)$$

Remark 4.44 (The overall factor 2 is $\Gamma(3)$). The numerator factor 2 in (4.280) is $\Gamma(3)$ from combining the three denominators, $\frac{1}{D_0 D_1 D_2} = 2 \int_{\Delta_2} \frac{1}{(r^2 + \Delta)^3}$; it is not an orientation multiplicity and not a second loop orientation.

Because the $\partial_{\dot{\alpha}}$ are gauge-covariant and do not commute, derivative monomials are understood as shuffle-symmetrized averages over the (m, n) orderings of $\partial_1^m \partial_2^n$. The full kernel operator on letter pairs is

$$\mathcal{K}_{m,n}(f^D, g^E) = \sum_{k=0}^m \sum_{\ell=0}^n \frac{2 \binom{m}{k} \binom{n}{\ell}}{(m+n+2)(k+\ell+1)} (\partial_1^k \partial_2^\ell \partial_{\dot{\alpha}} f)^D (\partial_1^{m-k} \partial_2^{n-\ell} \partial_{\dot{\alpha}} g)^E. \quad (4.281)$$

Its first instances are

$$\mathcal{K}_{0,0} = \langle f, g \rangle, \quad \mathcal{K}_{1,0} = \frac{1}{3} \langle \partial_1 f, g \rangle + \frac{2}{3} \langle f, \partial_1 g \rangle, \quad (4.282)$$

$$\mathcal{K}_{2,0} = \frac{1}{6} \langle \partial_1^2 f, g \rangle + \frac{1}{2} \langle \partial_1 f, \partial_1 g \rangle + \frac{1}{2} \langle f, \partial_1^2 g \rangle, \quad (4.283)$$

$$\mathcal{K}_{1,1} = \frac{1}{6} \langle \partial_1 \partial_2 f, g \rangle + \frac{1}{4} \langle \partial_1 f, \partial_2 g \rangle + \frac{1}{4} \langle \partial_2 f, \partial_1 g \rangle + \frac{1}{2} \langle f, \partial_1 \partial_2 g \rangle. \quad (4.284)$$

In particular the $(b, \partial c)$ weights $(\frac{1}{3}, \frac{2}{3})$ of (4.273)–(4.274) are the $(m, n) = (1, 0)$ instance: $K_{1,0;1,0} = \frac{2}{3 \cdot 2} = \frac{1}{3}$ for the derivative on the first letter and $K_{1,0;0,0} = \frac{2}{3 \cdot 1} = \frac{2}{3}$ for the derivative on the second. The comparison of (4.280) with the holomorphic-twist kernel of [1] is the subject of Section 7.

5 The component Feynman-diagram approach

5.1 The program: component quantization of the same action

The one-loop anomalous supersymmetry Ward identity of this paper was derived in Euclidean $\mathcal{N} = 1$ superspace (Section 4). The present section develops the parallel component-field approach: the same Euclidean action of Sections 2 and 3, expanded in component fields and quantized by ordinary Feynman diagrams. An independent component check is desirable for three reasons: the component expansion makes the anomaly directly readable as a bottom-component transformation law; the off-shell formulation carries an off-shell Noether current for the supersymmetry, in whose divergence the anomaly would appear as an Euler-operator insertion, checkable term by term [2, 3]; and ordinary Wick diagrams, regulable by dimensional reduction [5], give a graph-by-graph cross-check of the superspace Schwinger–Dyson derivation, with no superspace D -algebra anywhere in the loop evaluation.

The component content of the theory is the $\mathcal{N} = 1$ vector multiplet (A_m, λ, D) together with three adjoint chiral multiplets (ϕ_r, ψ_r, F_r) , $r = 1, 2, 3$, and their independent Euclidean antichiral partners $(\tilde{\phi}_r, \tilde{\psi}_r, \tilde{F}_r)$ and $\tilde{\lambda}$: tilded and untilded fields are independent integration variables in Euclidean signature and become Lorentzian conjugates only after a real integration cycle is selected. With the color commutator $[X, Y]^A := f^{BCA} X^B Y^C$ and $\kappa_{AB} = \delta_{AB}$ (Section 2), the exact Euclidean off-shell $\mathcal{N} = 4$ component Lagrangian is

$$\mathcal{L}_{E,\text{off}}^{(4)} = \frac{1}{g^2} \delta_{AB} \left[\frac{1}{4} F_{mn}^A F_{mn}^B + \tilde{\lambda}^A \bar{\sigma}_E^m \mathcal{D}_m \lambda^B - \frac{1}{2} D^A D^B + (\mathcal{D}_m \tilde{\phi}_r)^A (\mathcal{D}_m \phi_r)^B + \tilde{\psi}_r^A \bar{\sigma}_E^m \mathcal{D}_m \psi_r^B - \tilde{F}_r^A F_r^B - i D^A [\phi_r, \tilde{\phi}_r] \right. \\ \left. - \frac{\sqrt{2}}{g^2} f^{ABC} (\tilde{\phi}_r^A \psi_r^B \lambda^C + \phi_r^A \tilde{\psi}_r^B \tilde{\lambda}^C) + \frac{1}{\sqrt{2} g^2} \varepsilon_{rst} f^{ABC} (F_r^A \phi_s^B \phi_t^C + \tilde{F}_r^A \tilde{\phi}_s^B \tilde{\phi}_t^C - \phi_t^A \psi_r^B \psi_s^C - \tilde{\phi}_t^A \tilde{\psi}_r^B \tilde{\psi}_s^C) \right], \quad (5.1)$$

plus a topological term that plays no role below and is omitted from the display. Here D^A is the vector-multiplet auxiliary scalar (not a derivative), $\bar{\sigma}_E^m$ are the Euclidean sigma matrices, and repeated flavor indices are summed. Eliminating the auxiliaries through $D = -i[\phi_r, \tilde{\phi}_r]$, $\tilde{F}_r = \frac{1}{\sqrt{2}} \varepsilon_{rst} [\phi_s, \phi_t]$, $F_r = \frac{1}{\sqrt{2}} \varepsilon_{rst} [\tilde{\phi}_s, \tilde{\phi}_t]$ gives the on-shell component action. The two Yukawa families $-\frac{\sqrt{2}}{g^2} f^{ABC} \tilde{\phi}_r^A \psi_r^B \lambda^C$ and $-\frac{\sqrt{2}}{g^2} f^{ABC} \phi_r^A \tilde{\psi}_r^B \tilde{\lambda}^C$ in (5.1) are the only vertices carrying the fields of the anomaly; they are the vertices from which a naive ordinary component derivation would have to be built (Section 5.2).

The component framework also locks the BV–BRST structure exactly. With chiral and antichiral Faddeev–Popov ghost superfields c and $\tilde{c}$, both of ghost number $+1$, the BRST differential acts on the unrestricted vector prepotential $\mathcal{V}$ and the matter superfields as

$$s e^{\mathcal{V}} = i \tilde{c} e^{\mathcal{V}} - i e^{\mathcal{V}} c, \quad s c = i c^2, \quad s \tilde{c} = i \tilde{c}^2, \quad s \Phi_r = i c \Phi_r, \quad s \tilde{\Phi}_r = -i \tilde{\Phi}_r \tilde{c}. \quad (5.2)$$

This differential is nilpotent, $s^2 = 0$. The minimal Batalin–Vilkovisky action obeys the classical master equation and, after the finite-cutoff regulator insertion, the order-by-order quantum master equation [4]. Gauge fixing uses a nonminimal sector with the antighost $\bar{c}$ and the gauge functions $\mathcal{F}_{\pm}(\mathcal{V}) = -\frac{1}{4} \bar{\mathcal{D}}^2 \mathcal{V}$, $-\frac{1}{4} \mathcal{D}^2 \mathcal{V}$; the Wess–Zumino surface is not imposed before gauge fixing.

Remark 5.1 (Partial scope of the component Feynman rules). Locked exactly are the component actions, the BV–BRST structure, and the all-order primitive graph grammar with labeled-graph weights; this grammar becomes a numerical Feynman rule only after every edge inverse and integration cycle is locked. The Fermi–Feynman gauge-averaged propagators exist only conditionally: the corresponding gauge-fixing kernel contains $\square^{-1}$ and is nonlocal, so an admissible completed perturbative slice is not claimed (status `PASS_EXACT_PARTIAL_SCOPE`). The accepted superspace computation of Section 4 therefore adopts the Fermi–Feynman Gaussian

average as a recorded scheme decision and keeps the Nielsen–Kallosh factor as an external, field-independent measure. Former claims of complete component momentum-space Feynman rules are NOT_ACCEPTED_COMPLETE_FEYNMAN_RULES.

This component approach is one of four mutually independent computations into which the proof program is organized: signature (Euclidean or Lorentzian) times formulation (superspace or component). Each works from the shared frozen conventions and its own derivations only, and results are compared only at the final canonical level. Gating rules order them: the Euclidean component computation begins once the conventions are frozen, the Euclidean superspace one starts after the Euclidean component canonical result is locked, and the two Lorentzian ones start only after a Euclidean bridge is frozen. The accepted component statement below is in fact obtained as the bottom projection of the Euclidean superspace result—explicitly not as an independent component-box derivation.

5.2 The accepted result: bottom projection of the (β, γ) anomaly

Section 4 establishes, for the ordered letter pair (β_r, γ^s) and equally for the reversed ordering (γ^s, β_r) , the exact one-loop identity

$$\nabla_- (\beta_r^A \gamma^s B) \Big|_{1\text{-loop}} = -\frac{\hbar g^2}{16\sqrt{2}\pi^2} \delta_{rs} f^{ACD} f^{BCE} \langle \partial c^D, \partial c^E \rangle, \quad (5.3)$$

where $\langle X, Y \rangle := (\partial_{\dot{\alpha}} X)(\partial^{\dot{\alpha}} Y)$ is the dotted pairing and ∇_- is the anomalous frame component of the gauge-covariant spinor derivative. Its component content is read off by bottom projection. The bottom components of the letters are

$$\beta_r^A \Big| = \psi_{r+}^A, \quad \gamma^s B \Big| = \tilde{\phi}_s^B, \quad \partial_{\dot{\alpha}} c^D \Big| = -\tilde{\lambda}_{\dot{\alpha}}^D, \quad (5.4)$$

where ψ_{r+} is the $+$ -frame bottom component of the r -th matter Weyl fermion, $\tilde{\phi}_s$ the bottom scalar of the antichiral multiplet, and $\tilde{\lambda}$ the dotted gaugino. The component variation Q_- is the bottom action induced by the same locked ∇_- acting on the letters; no independent rephasing is introduced. The dotted pairing projects as

$$\langle \partial c^D, \partial c^E \rangle \Big| = (\partial_{\dot{\alpha}} c^D)(\partial^{\dot{\alpha}} c^E) \Big| = \tilde{\lambda}_{\dot{\alpha}}^D \tilde{\lambda}^{E\dot{\alpha}}. \quad (5.5)$$

Combining (5.3)–(5.5) gives the accepted component formula

$$\boxed{Q_- \left(\psi_{r+}^A \tilde{\phi}_s^B \right) \Big|_{1\text{-loop}} = -\frac{\sqrt{2}\hbar g^2}{32\pi^2} \delta_{rs} f^{ACD} f^{BCE} \tilde{\lambda}_{\dot{\alpha}}^D \tilde{\lambda}^{E\dot{\alpha}}.} \quad (5.6)$$

Remark 5.2 (Status of the boxed formula). Equation (5.6) is an exact projection of the accepted superfield result of Section 4, not an independent component-box derivation. It fixes the former free sign of the component-box proposal to -1 , and the loop order contains exactly one power of $\hbar$.

The dotted gaugino bilinear on the right of (5.6) has a color symmetry that constrains every component realization. With $\epsilon^{12} = +1$, define

$$S^{DE} := \tilde{\lambda}_{\dot{\alpha}}^D \tilde{\lambda}^{E\dot{\alpha}} = \tilde{\lambda}_1^D \tilde{\lambda}_2^E - \tilde{\lambda}_2^D \tilde{\lambda}_1^E, \quad S^{ED} = S^{DE}, \quad (5.7)$$

where the second relation is Grassmann reordering. Consequently, any antisymmetric color coefficient $A_{DE} = -A_{ED}$ obeys

$$A_{DE} S^{DE} = \frac{1}{2} (A_{DE} + A_{ED}) S^{DE} = 0 : \quad (5.8)$$

antisymmetric color words vanish, the former crossed word $W_{DE} - W_{ED}$ gives zero, and a nonzero result for the ordered pair (β, γ) must carry the symmetric color projection $f^{ACD} f^{BCE} S^{DE}$, which is indeed symmetric in $D \leftrightarrow E$.

The evanescent tensor integrals behind the loop coefficient are standard dimensional-reduction integrals [5]: with $d = 4 - 2\epsilon$, $\Delta > 0$, μ_ℓ^2 the evanescent part of the loop-momentum square, and $I_n(\Delta) := \int \frac{d^d \ell}{(2\pi)^d} \mu_\ell^2 (\ell^2 + \Delta)^{-n}$,

$$\lim_{\epsilon \rightarrow 0} I_3(\Delta) = \frac{1}{32\pi^2}, \quad \lim_{\epsilon \rightarrow 0} \Delta I_4(\Delta) = 0. \quad (5.9)$$

The tensor integral used in the former component-box evaluation is

$$J := \frac{1}{d} \int \frac{d^d \ell}{(2\pi)^d} \frac{\mu_\ell^2 \ell^2}{(\ell^2 + \Delta)^4} = \frac{1}{d} [I_3(\Delta) - \Delta I_4(\Delta)], \quad \lim_{\epsilon \rightarrow 0} J = \frac{1}{128\pi^2}, \quad \lim_{\epsilon \rightarrow 0} 2J = \frac{1}{64\pi^2}. \quad (5.10)$$

The first value is the tensor integral itself; the second includes the explicit factor 2 from the four-dimensional sigma/Fierz numerator. The two numbers cannot be identified with each other, and their separation rejects the former component-box evaluation and fixes its free sign and loop-counting power.

Why can the boxed formula (5.6) not be reproduced by an ordinary component graph? The locked component quadratic action pairs scalars with scalars (the $(\phi, \tilde{\phi})$ kinetic term) and fermions with fermions (the $(\psi, \tilde{\psi})$ kinetic term), so the mixed propagators vanish exactly,

$$G_{\phi_r \tilde{\psi}_s} = 0, \quad G_{\psi_r \tilde{\phi}_s} = 0. \quad (5.11)$$

A naive ordinary Wick box or triangle built from the two Yukawa families of (5.1), with external bottoms ψ_{r+} , $\tilde{\phi}_s$ and an output pair $\tilde{\lambda}\tilde{\lambda}$, would need precisely these scalar–fermion converting propagators; by (5.11) no such graph exists, and the ordinary two-Yukawa box is rejected as the parent of the (β, γ) anomaly. (An earlier argument had incorrectly promoted this ordinary-Wick obstruction to a statement about the regulated density divergence.)

What the nonzero (β, γ) anomaly actually is, at the component level, is the regulated coincident Euler Jacobian, differentiated before the bottom projection:

$$\tilde{\Phi}_s^B \frac{\vec{\delta}}{\delta \tilde{\Phi}_r^A} \longrightarrow \delta_{rs} K_{-, \epsilon}^{AB}(z, z) \longrightarrow \delta_{rs} f^{ACD} f^{BCE} \tilde{\mathcal{W}}_\alpha^D \tilde{\mathcal{W}}^{E\alpha}, \quad (5.12)$$

with the regulated coincident kernel

$$K_{-, \epsilon}^{AB}(z, z)|_{\tilde{\mathcal{W}}^2} = \frac{1}{32\pi^2} f^{ACD} f^{BCE} \tilde{\mathcal{W}}_\alpha^D \tilde{\mathcal{W}}^{E\alpha}. \quad (5.13)$$

Any future component realization must prove equality with (5.12); a nonzero ordinary box cannot be substituted for it. The supporting orbit evaluation fixes every factor: the regulated Schwinger–Dyson equation for the source vector field $X_{rs}^{AB} = \tilde{\Phi}_s^B \vec{\delta} / \delta \tilde{\Phi}_r^A$ shows that $r \neq s$ vanishes independently of any external target; the two opposite-sign superpotential occurrences have identical regulated contact kernels and cancel; the kinetic Euler occurrence is evaluated by (5.9) with no graph-specific factor $4 - d$; both color routings give the same tensor $f^{ACD} f^{BCE}$; and the interaction-copy factor is 1. Since $\partial_\alpha c = i\tilde{\mathcal{W}}_\alpha$, the dotted bilinear in (5.3) is minus the $\tilde{\mathcal{W}}$ bilinear in (5.13), which accounts for the overall sign of (5.6).

5.3 Open items

Remark 5.3 (Open and blocked items in the component route). Three component-route items remain outside the accepted result; none of them affects the boxed formula (5.6).

- (i) **The off-shell Noether identity is not accepted** (NOT_ACCEPTED_NOETHER_B1_B4_MISSING): the complete Euler operators and the announced termwise checks B1–B4 are absent; only the sign pattern of the proposed right-hand side is on record, unverified, so a component-level off-shell supersymmetry current with a fully verified divergence identity does not exist.
- (ii) **The Lorentzian vector cycle is blocked** (BLOCKED_STEP3D_LC_VECTOR_CYCLE): the Lorentzian vector integration cycle remains unfixed, so no Lorentzian component graph calculation can start.
- (iii) **The target-seeded component sweep of the 81 ordered letter pairs is not accepted** (NOT_ACCEPTED): the former sweep inherited the nonexistent ordinary two-Yukawa parent and classified pairs by comparison with a target-shaped table. The authoritative all-81 result is the target-blind superspace Schwinger-cut table of Section 4, with 29 nonzero and 52 zero ordered pairs.

The physical one-loop anomaly sector has no unresolved coefficient row; the open items above concern independent component re-derivations and the Lorentzian continuation only.

6 The heat-kernel approach

6.1 Motivation: a semigroup-based second derivation

The one-loop anomalous Ward identity of Section 4, with its universal coefficient $\frac{\hbar g^2}{16\pi^2}$, is derived perturbatively: a Schwinger–Dyson identity plus the cutting-failure mechanism of dimensional reduction, graph by graph. A second, independent and non-perturbative derivation of the same coefficient is highly desirable. The heat-kernel strategy would regulate the one-loop fluctuation determinant of the full $\mathcal{N} = 4$ Hessian complex by the semigroup e^{-sH} of a single elliptic operator H on the graded fluctuation bundle, and would extract the anomaly from a short-time Seeley–DeWitt coefficient of that semigroup [6]. Such a derivation would (i) cross-check the perturbative Schwinger-cut result without enumerating graphs; (ii) make the finiteness and locality of the anomaly manifest from short-time heat-kernel asymptotics; and (iii) connect directly to the evanescent structure of dimensional reduction through the same master integral $\frac{1}{32\pi^2}$ [5].

This section reports the exact status of that programme. The scalar auxiliary identities on which any such derivation must rest are established and machine-verified (Subsection 6.2). The full derivation, however, is blocked at the regulator level: no single elliptic generator on the full $\mathcal{N} = 4$ fluctuation complex has been constructed, and the anomaly coefficient used elsewhere in this paper is *not* obtained from this route. Subsection 6.3 states the obstruction precisely, and Subsection 6.4 lists what would close the route.

6.2 Established scalar and semigroup lemmas

Every identity in this subsection is exact; each is proved by direct scalar computation and is machine-verified in the source repository. We write $H(g) = H_0 + gH_1 + g^2H_2$ for the one-loop fluctuation operator expanded in the coupling, and $s > 0$ for the proper-time parameter.

Lemma 6.1 (Second-order Duhamel coefficient). *Let $E \rightarrow M$ be a fixed graded bundle, let $H_0 : E \rightarrow E$ be an even operator, and let H_1, H_2 be even endomorphism-valued differential operators. Set*

$$H(g) = H_0 + gH_1 + g^2H_2. \tag{6.1}$$

Then the coefficient of g^2 in the semigroup $e^{-sH(g)}$ is the two-term ordered integral

$$\begin{aligned} [g^2] e^{-sH(g)} = & - \int_0^s dt e^{-(s-t)H_0} H_2 e^{-tH_0} \\ & + \int_{0 < t_1 < t_2 < s} dt_1 dt_2 e^{-(s-t_2)H_0} H_1 e^{-(t_2-t_1)H_0} H_1 e^{-t_1H_0}. \end{aligned} \quad (6.2)$$

Remark 6.2 (The quadratic background block cannot be omitted). The first line of (6.2) is at the same background order as the second line. Hence a heat-kernel derivation of the physical coefficient must derive the full H_2 seagull, connection-square, gauge and ghost blocks; the quadratic background block cannot be omitted in favor of two linear H_1 insertions.

Lemma 6.3 (Ordered-simplex moment). *For integers $p, q \geq 0$,*

$$M_{p,q} := \int_{0 < a < b < 1} a^p b^q da db = \frac{1}{(p+1)(p+q+2)}. \quad (6.3)$$

Consequently, for $0 \leq k \leq m$ and $0 \leq \ell \leq n$, the tower coefficient is

$$T_{m,n;k,\ell} := \binom{m}{k} \binom{n}{\ell} M_{k+\ell, m+n-k-\ell} = \frac{\binom{m}{k} \binom{n}{\ell}}{(m+n+2)(k+\ell+1)}. \quad (6.4)$$

Proof. Integrating a at fixed b gives $M_{p,q} = \int_0^1 db b^q \frac{b^{p+1}}{p+1} = \frac{1}{(p+1)(p+q+2)}$, and (6.4) follows by substitution of (6.3). $\square$

Remark 6.4 (A second ordering supplies no automatic factor two). The moment (6.3) sums a single ordering of the simplex; no factor from the reversed ordering is included. A factor of two may be added only after an independent ordered-vertex calculation proves equality of the two orderings, including color order and Koszul sign.

Lemma 6.5 (Gaussian convolution and bridge means). *For $s > 0$ and $x \in \mathbb{R}^4$ let*

$$K_s(x) = \frac{1}{(4\pi s)^2} e^{-x^2/4s}. \quad (6.5)$$

Then K_s is normalized, solves the heat equation, and has diagonal value

$$\int_{\mathbb{R}^4} d^4x K_s(x) = 1, \quad \partial_s K_s = \Delta_{\mathbb{R}^4} K_s, \quad K_s(0) = \frac{1}{16\pi^2 s^2}, \quad (6.6)$$

and it obeys the convolution (Chapman–Kolmogorov) identity

$$\int_{\mathbb{R}^4} d^4y K_s(x-y) K_t(y) = K_{s+t}(x). \quad (6.7)$$

Moreover, for the three-segment bridge with $t_1, t_2, t_3 > 0$, $T = t_1 + t_2 + t_3$, and Gaussian weight $Q(x, y) = \frac{(x-w)^2}{4t_1} + \frac{(y-x)^2}{4t_2} + \frac{y^2}{4t_3}$ in one Cartesian component, the stationary means are

$$\langle x \rangle = \frac{t_2 + t_3}{T} w, \quad \langle y \rangle = \frac{t_3}{T} w, \quad (6.8)$$

and the inverse Hessian of Q is homogeneous of degree one under $(t_1, t_2, t_3) \mapsto (\rho t_1, \rho t_2, \rho t_3)$, $\rho > 0$.

Remark 6.6. Equation (6.8) proves the exact bridge mean and covariance scaling; it does not remove the endpoint factor $K_T(w)$.

Lemma 6.7 (Master-integral consistency with dimensional reduction). *Let $d = 4 - 2\varepsilon$, and let μ_ℓ^2 be the evanescent component of the loop momentum squared (Section 3). Then*

$$\lim_{\varepsilon \rightarrow 0} \mu^{2\varepsilon} \int \frac{d^d \ell}{(2\pi)^d} \frac{\mu_\ell^2}{(\ell^2 + \Delta)^3} = \frac{1}{32\pi^2}. \quad (6.9)$$

Remark 6.8. The value (6.9) agrees with the master integral of the accepted derivation (Sections 4 and 5), so the scalar heat-kernel layer is consistent with the evanescent mechanism of dimensional reduction [5]. Equality of this scalar integral does not, however, determine the $\mathcal{N} = 4$ operator pairing, the color tensor, the orientation sign, or the anomaly multiplicity.

Lemma 6.9 (Vanishing short-time limit at fixed nonzero shift). *For fixed $w \neq 0$,*

$$\lim_{s \downarrow 0} \frac{1}{(4\pi s)^2} e^{-w^2/4s} = 0. \quad (6.10)$$

Consequently a local derivative tower requires distributional coefficient extraction at $w = 0$; it cannot be obtained by deleting $K_s(w)$ before taking the limit.

6.3 The obstruction: the regulator proposal fails

The proposed continuation defines its would-be generator blockwise from a Hessian two-form $\delta^2 S$ and a field-space identification $\mathcal{G}$. Its own displayed free matter blocks give

$$(\mathcal{G}^{-1} \delta^2 S)^2 \Big|_{E_+} = \bar{\nabla}^2 \nabla^2 = 16 \square \mathcal{P}_+, \quad (6.11)$$

whereas the declared generator is $-\square \mathcal{P}_+$; the accompanying verifier inserts a factor $-1/16$ that appears nowhere in the proposal. Both the normalization (16) and the sign are wrong relative to the declared generator. Likewise, the proposal's own vector-block weights give

$$\mathcal{G}_V^{-1} \delta^2 S_{VV} = (2g^2) \frac{1}{2g^2} \square = \square, \quad (6.12)$$

whereas the declared generator is $-\square$; the verifier adds an undeclared Euclidean sign. Finally, the prescription applies the squared operator on the matter and ghost rows but the unsquared operator on the vector row. This is not a single operator functional calculus when nonzero vector–matter mixing blocks are retained: different powers on different blocks define different operators, so the off-diagonal mixing blocks have no defined heat evolution under this prescription.

Remark 6.10 (Status of the route: blocked). For the three independent reasons above, the full heat-kernel derivation of the anomaly coefficient is currently blocked; the source repository records the status `BLOCKED_HEAT_KERNEL_TYPED_REGULATOR_AND_COEFFICIENT_DERIVATION`. The identities of Subsection 6.2 are admitted as verified auxiliary lemmas only: they do not define a regulator on the full $\mathcal{N} = 4$ fluctuation complex, and they do not independently derive the coefficient $\frac{hg^2}{16\pi^2}$, which remains the result of the Schwinger-cut derivation of Section 4.

6.4 What would close the route

The route would produce a regulator only when the following are derived together, not declared:

- a *single elliptic generator* on the full graded $\mathcal{N} = 4$ fluctuation bundle—one operator, one functional calculus, with the correct sign and normalization on every block (matter, vector, ghost), covering also the off-diagonal vector–matter mixing blocks;

- its *dual Euler action* on the letter bilinears, consistent with the Schwinger–Dyson/Euler reduction of Section 4;
- the *full quadratic coefficient* H_2 —the seagull, connection-square, gauge and ghost blocks at order g^2 —which by Lemma 6.1 sits at the same background order as two linear insertions and cannot be dropped;
- the *distributional short-time limit*: coefficient extraction at $w = 0$ in the distributional sense, as required by Lemma 6.9.

Reproducing the accepted coefficient $\frac{\hbar g^2}{16\pi^2}$ with the correct color tensor and the correct census of nonzero channels would in addition require the $\mathcal{N} = 4$ operator pairing, the orientation sign and the anomaly multiplicity to be derived from that generator, since none of them follow from the verified scalar identities alone. Until then, the heat-kernel approach stands as an auxiliary consistency layer, not as a second derivation.

7 Comparison with the holomorphic twist

The one-loop coefficients of Section 4 were derived, validated, and sealed before the holomorphic-twist literature for this problem was consulted. The present section compares that sealed result with the independent prediction of Budzik and Kulp [1], who compute the one-loop correction to the supersymmetry algebra of the holomorphic twist of $\mathcal{N} = 4$ super-Yang–Mills theory. The comparison is an external cross-check: no coefficient of [1] enters the derivation of Section 4, and everything concluded here is collected in the admissibility statement of Subsection 7.5.

7.1 The external target

In the holomorphic twist [1, 7] the $\mathcal{N} = 1$ vector multiplet becomes a holomorphic bc system and each chiral multiplet a holomorphic $\beta\gamma$ system on $\mathbb{C}^2$. The Batalin–Vilkovisky fields are

$$\begin{aligned} b &= b^{(0)} + b^{(1)} + b^{(2)} \in \Omega^{0,\bullet}(\mathbb{C}^2, \mathfrak{g}^\vee), & c &= c^{(0)} + c^{(1)} + c^{(2)} \in \Omega^{0,\bullet}(\mathbb{C}^2, \mathfrak{g})[1], \\ \beta_I &= \beta_I^{(0)} + \beta_I^{(1)} + \beta_I^{(2)} \in \Omega^{0,\bullet}(\mathbb{C}^2, R^\vee)[1], & \gamma^I &= \gamma^{I(0)} + \gamma^{I(1)} + \gamma^{I(2)} \in \Omega^{0,\bullet}(\mathbb{C}^2, R), \end{aligned} \quad (7.1)$$

with action (their eq. (3.1))

$$S_{\text{BV}} = \int_{\mathbb{C}^2} d^2z \left\{ \text{Tr} \left[b \left(\bar{\partial}c + \frac{1}{2}[c, c] \right) \right] + \beta_I (\bar{\partial}\gamma^I + c\gamma^I) + W[\gamma] \right\}. \quad (7.2)$$

The tree differential acts as

$$\begin{aligned} Q_0 c &= \frac{1}{2}[c, c], & Q_0 b &= [c, b] - \mu(\beta, \gamma), \\ Q_0 \gamma^I &= c\gamma^I, & Q_0 \beta_I &= \beta_I c + \partial_{\gamma^I} W, \end{aligned} \quad (7.3)$$

with $\mu(\beta, \gamma)$ the moment map. The letter gradings are $|b| = |\gamma| = 0$ and $|c| = |\beta| = 1 \bmod 2$, and normal ordering is supercommutative on letters. The untwisted identifications are $\gamma \sim \bar{\phi}$, $\beta \sim \psi_+$, $\partial_{\dot{\alpha}} c \sim \bar{\lambda}_{\dot{\alpha}}$, $b \sim F_{++}$, and $\partial_{\dot{\alpha}} \sim D_{+\dot{\alpha}}$, consistent with the component table of Section 5. The Lie algebra is normalized as (their eq. (3.11))

$$[t_A, t_B] = f_{AB}^C t_C, \quad \text{Tr}(t_A t_B) = \kappa_H \delta_{AB}, \quad \text{Tr}(t_A [t_B, t_C]) = \kappa_H f_{ABC}, \quad (7.4)$$

with κ_H the dual Coxeter number; the missing factor of i relative to our convention $[T_A, T_B] = i f_{AB}^C T_C$ is supplied by the frame map (7.11) below. For the $\mathcal{N} = 4$ theory the matter representation is $R = \mathfrak{g} \otimes \mathbb{C}^3$, the superpotential is $W[\gamma] = \text{Tr} \gamma^1 [\gamma^2, \gamma^3] = \frac{1}{3!} \varepsilon_{IJK} \kappa_H f_{ABC} \gamma^{IA} \gamma^{JB} \gamma^{KC}$,

the twisting supercharge is Q_-^4 , and $\gamma^I \sim \Phi^{4I}$, $\beta_I \sim \Psi_{I+}$, $\partial_{\dot{\alpha}} c \sim \bar{\Psi}_{\dot{\alpha}}^4$, $b \sim F_{++}$. The 1/16-BPS fields assemble into a single superfield on $\mathbb{C}^{2|3}$ (their eq. (3.56)),

$$C = c + \theta_I \gamma^I + \frac{1}{2} \varepsilon^{IJK} \theta_I \theta_J \beta_K + \theta_1 \theta_2 \theta_3 b, \quad (7.5)$$

in terms of which the action is a holomorphic Chern–Simons theory,

$$S_{\text{BV}} = \frac{1}{2} \int_{\mathbb{C}^{2|3}} d^2 z d^3 \theta \operatorname{Tr} \left[C \left(\bar{\partial} C + \frac{1}{3} [C, C] \right) \right]. \quad (7.6)$$

The classical supercharge obeys the tree identity (their eq. (1.9))

$$Q_0 C(\theta) = \frac{1}{2} [C(\theta), C(\theta)]. \quad (7.7)$$

Their one-loop result is a compact prediction for the loop-corrected supercharge Q_1 (their eqs. (1.10) and (3.69)),

$$Q_1(C^A(\theta) C^B(\theta')) = -\kappa_H^2 f_{ACD} f_{BCE} (\theta_1 - \theta'_1)(\theta_2 - \theta'_2)(\theta_3 - \theta'_3) \partial_{\dot{\alpha}} C^D(\theta) \partial^{\dot{\alpha}} C^E(\theta'). \quad (7.8)$$

(The introduction of [1] prints the coefficient $-1/4$ where its main text prints $-\kappa_H^2$; this is the second source inconsistency, see Subsection 7.4.) Expanded in the components of (7.5), the prediction reads (their eqs. (3.63)–(3.68))

$$\begin{aligned} Q_1(b^A c^B) &= \kappa_H^2 f_{ACD} f_{BCE} \partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} c^E, \\ Q_1(b^A b^B) &= \kappa_H^2 f_{ACD} f_{BCE} [\partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} b^E - \partial_{\dot{\alpha}} b^D \partial^{\dot{\alpha}} c^E + \partial_{\dot{\alpha}} \beta_I^D \partial^{\dot{\alpha}} \gamma^{IE} - \partial_{\dot{\alpha}} \gamma^{ID} \partial^{\dot{\alpha}} \beta_I^E], \\ Q_1(\beta_I^A \gamma^{JB}) &= \kappa_H^2 \delta_I^J f_{ACD} f_{BCE} \partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} c^E, \\ Q_1(\beta_I^A \beta_J^B) &= \kappa_H^2 \varepsilon_{IJK} f_{ACD} f_{BCE} [\partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} \gamma^{KE} - \partial_{\dot{\alpha}} \gamma^{KD} \partial^{\dot{\alpha}} c^E], \\ Q_1(b^A \gamma^{IB}) &= \kappa_H^2 f_{ACD} f_{BCE} [\partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} \gamma^{IE} - \partial_{\dot{\alpha}} \gamma^{ID} \partial^{\dot{\alpha}} c^E], \\ Q_1(\beta_I^A b^B) &= \kappa_H^2 f_{ACD} f_{BCE} [\partial_{\dot{\alpha}} \beta_I^D \partial^{\dot{\alpha}} c^E + \partial_{\dot{\alpha}} c^D \partial^{\dot{\alpha}} \beta_I^E + \varepsilon_{IJK} \partial_{\dot{\alpha}} \gamma^{JD} \partial^{\dot{\alpha}} \gamma^{KE}], \end{aligned} \quad (7.9)$$

while $Q_1(cc) = Q_1(\gamma\gamma) = Q_1(\gamma c) = Q_1(\beta c) = 0$. In the L_∞ language of [1] the $(\ell + 2)$ -ary bracket is the ℓ -loop correction, with $\{\mathcal{I}, \cdot\}_0 = Q_0$ and $\frac{1}{2!} \{\mathcal{I}, \mathcal{I}, \cdot\}_0 = Q_1$; the Maurer–Cartan identities give $\{Q_0, Q_1\} = 0$ and $Q_1^2 + \{Q_0, Q_2\} = 0$. Hence Q_1 is nilpotent only on Q_0 -cohomology and higher corrections Q_n are necessarily present: the construction of [1] is not one-loop exact, and the comparison below concerns its one-loop operation Q_1 .

7.2 The dictionary

Let R_A^P be a color frame, U_I^r a flavor frame, $S_{\dot{a}}^{\dot{b}}$ a dotted-spin frame, and $\rho \neq 0$ an overall scale. The exact map between the letters $b, \beta_r, \gamma^r, \partial_{\dot{\alpha}} c$ of Section 4 and the alphabet of [1] is

$$\begin{aligned} b^A &\mapsto \rho R_A^P b^P, \\ \beta_I^A &\mapsto \rho R_A^P U_I^r \beta_r^P, \\ \gamma^{IA} &\mapsto \rho R_A^P (U^{-1})_r^I \gamma^{rP}, \\ \partial_{\dot{a}} c^A &\mapsto \rho R_A^P S_{\dot{a}}^{\dot{b}} \partial_{\dot{b}} c^P, \end{aligned} \quad (7.10)$$

where the left-hand sides are the fields of (7.1) and the right-hand sides use the letters of Section 4. Every row carries the common scale ρ : the factors $-i/\sqrt{2}$ and $1/\sqrt{2}$ familiar from the raw superfield composites are precisely those absorbed into the letter definitions of Definition 4.1, so that in the letters of this paper the map is uniform. The γ row carries the inverse flavor frame and the ∂c row the spin frame. At $\rho = 1$ with identity frames the map reduces to

the identity, $b \mapsto b$, $\beta_I \mapsto \beta_r \delta_I^r$, $\gamma^I \mapsto \gamma^r \delta^I_r$, and $\partial_{\dot{a}} c \mapsto \partial_{\dot{a}} c$. The frames enter the color and trace relations through

$$R_A^P R_B^Q \delta_{PQ} = \delta_{AB}, \quad t_A = -i R_A^P T_P, \quad f_{AB}^C = R_A^P R_B^Q f_{PQ}^R (R^{-1})_R^C, \quad \text{Tr}_H = -\kappa_H \text{Tr}, \quad (7.11)$$

where Tr_H is the trace of (7.4) and Tr ours, with $\kappa_{AB} = \delta_{AB}$. Applied to (7.5), the map sends

$$C(\theta) \mapsto \rho \left[c + \theta_r \gamma^r + \frac{1}{2} \varepsilon_{rst} \theta_r \theta_s \beta_t + \theta_1 \theta_2 \theta_3 b \right]; \quad (7.12)$$

the first entry is the ghost (the formal ghost placeholder of the letter algebra), which takes part in the twisted theory but not in the physical letter pairs compared below. Matching the one-loop coefficients of the two formalisms on the gaugino ($\mathcal{W}_+$) component gives the loop-scale relation

$$\hbar_{\text{HT}} \kappa_H^2 = \frac{\hbar g^2}{32\sqrt{2}\pi^2} = \frac{1}{2\sqrt{2}} \frac{\hbar g^2}{16\pi^2}. \quad (7.13)$$

Remark 7.1 (conditional status of (7.13)). The relation (7.13) is obtained under a conditional total-differential scale in the dictionary analysis; it is a conditional dictionary entry, not a theorem proved in this paper. The four physical letter rows of (7.10) are independent of this conditionality, and the comparison of Subsection 7.3 is performed on output words only. A single κ_H presupposes a simple gauge algebra; for a general reductive gauge algebra the trace relations hold factorwise.

7.3 Comparison results

The comparison proceeds word by word. Each ordered pair of the nine letters $(b, \beta_r, \gamma^r, \partial_{\dot{a}} c)$ is mapped by (7.10); the predicted output monomials are translated into the letters of Section 4; and the result is compared with the sealed coefficient table over $\mathbb{Q}(i, \sqrt{2})$. With the identification $Q_0 \leftrightarrow -\frac{1}{2}\nabla_-$ of Section 4, the quantity compared with $Q_1(XY)$ is the one-loop anomalous variation $\nabla_-(XY)|_{1\text{-loop}}$, normalized by (7.13).

Proposition 7.2 (ordered-word comparison). *For every ordered pair of letters, the sealed coefficient word of Section 4, its independent expansion, and the translated holomorphic-twist word agree exactly:*

$$N_{\text{pairs}} = 81, \quad N_{\text{agree}} = 81, \quad N_{\text{disagree}} = 0, \quad 81 = 29_{\text{nonzero}} + 52_{\text{zero}}. \quad (7.14)$$

The nonzero census decomposes as

$$29 = \frac{1}{(b,b)} + \frac{6}{(b,\beta_r),(\beta_r,b)} + \frac{6}{(b,\gamma^r),(\gamma^r,b)} + \frac{6}{(\beta_r,\beta_s),r \neq s} + \frac{6}{(\beta_r,\gamma^r),(\gamma^r,\beta_r)} + \frac{4}{(b,\partial_{\dot{a}}c),(\partial_{\dot{a}}c,b)}, \quad (7.15)$$

comprising $8 + 6 \cdot 4 + 6 \cdot 2 + 6 \cdot 2 + 6 \cdot 1 + 4 \cdot 2 = 70$ nonzero output words; the remaining 52 ordered pairs vanish on both sides.

The mixed rows $(b, \partial_{\dot{a}} c)$ and $(\partial_{\dot{a}} c, b)$ display the intrinsic split $K_{(e_{\dot{a}});0} = \frac{2}{3}$ and $K_{(e_{\dot{a}});(e_{\dot{a}})} = \frac{1}{3}$ for a unit dotted derivative $e_{\dot{a}}$, the two orderings being exchanged by the Leibniz rule; this is the $\frac{1}{3} - \frac{2}{3}$ distribution already recorded in the channel table of Section 4. At arbitrary derivative order the comparison uses the Appendix-B tower of [1] (their eq. (B.5)),

$$m! n! C_{mn} = \frac{1}{m+n+2} \sum_{k=0}^m \sum_{\ell=0}^n \frac{1}{k+\ell+1} \binom{m}{k} \binom{n}{\ell} (p_1)_1^k (p_1)_2^\ell (p_2)_1^{m-k} (p_2)_2^{n-\ell}, \quad (7.16)$$

whose component formulas (their eqs. (B.6)–(B.11)) multiply this tower by $\kappa_H^2 f_{ACD} f_{BCE}$ with the same flavor structures as the zero-derivative pairs (7.9); we write their holomorphic form variables as p_i , $p_1 + p_2 + p_3 = 0$, to avoid a clash of notation.

Proposition 7.3 (all-orders kernel identity). *For all $m, n \in \mathbb{Z}_{\geq 0}$ and $0 \leq k \leq m$, $0 \leq \ell \leq n$,*

$$K_{m,n;k,\ell} = \frac{2 \binom{m}{k} \binom{n}{\ell}}{(m+n+2)(k+\ell+1)} = 2 T_{m,n;k,\ell}^{\text{HT,printed}} = T_{m,n;k,\ell}^{\text{HT,corrected}}, \quad (7.17)$$

where $K_{m,n;k,\ell}$ is the derivative kernel of the one-loop variation of Section 4,

$$T_{m,n;k,\ell}^{\text{HT,printed}} = \frac{\binom{m}{k} \binom{n}{\ell}}{(m+n+2)(k+\ell+1)} \quad (7.18)$$

is the coefficient tower of (7.16), and $T^{\text{HT,corrected}} := 2 T^{\text{HT,printed}}$. At the origin $T_{0,0;0,0}^{\text{HT,printed}} = \frac{1}{2}$ while $K_{0,0;0,0} = 1$.

The factor 2 in (7.17) is the $\Gamma(3) = 2$ of the triangle Feynman parametrization,

$$\frac{1}{D_0 D_1 D_2} = 2 \int_{\Delta_2} \frac{1}{(r^2 + \Delta)^3}, \quad 2 \int_{\Delta_2} 1 = 1, \quad (7.19)$$

not an orientation multiplicity. The triangle operator of [1] (their eq. (3.8)), with $\partial = (\partial_1, \partial_2)$ the holomorphic derivative on $\mathbb{C}^2$,

$$\mathcal{D}_{w,z}^\Delta(f, g) := \sum_{m,n=0}^{\infty} \frac{1}{(n+m+2)!} e^{z \cdot \partial} ((w-z) \cdot \partial)^n \partial_{\dot{\alpha}} f(y) e^{w \cdot \partial} ((z-w) \cdot \partial)^m \partial^{\dot{\alpha}} g(y) \Big|_{y=0}, \quad (7.20)$$

generates the generic shifted pairs, each Q_1 pair action being $\kappa_H^2 f_{ACD} f_{BCE}$ times the corresponding $\mathcal{D}^\Delta$ bilinear; its printed zero-shift value (their eq. (3.9)),

$$\mathcal{D}_{0,0}^\Delta(f, g) = \frac{1}{2} \partial_{\dot{\alpha}} f \partial^{\dot{\alpha}} g, \quad (7.21)$$

misses precisely the prefactor 2 of (7.19), while the printed component formulas (7.9) carry the complete-amplitude normalization (see Subsection 7.4). The identity (7.17) is proved symbolically for all $m, n \geq 0$ and checked coefficient by coefficient on the finite rectangle $0 \leq m, n \leq 8$:

$$\sum_{m=0}^8 \sum_{n=0}^8 (m+1)(n+1) = \left(\sum_{j=1}^9 j \right)^2 = 45^2 = 2025 \quad \text{kernel checks, all passing (2025/2025)}. \quad (7.22)$$

Multiplication by the 70 nonzero base output words lifts this to the full derivative tower,

$$2025 \times 70 = 141750 \quad \text{lifted coefficient checks, all passing (141750/141750)}, \quad (7.23)$$

while the 52 vanishing rows remain zero at every derivative order. The independent physical verification suite accepts the physical one-loop anomaly sector with 33/33 checks passed and 0 failed.

7.4 Two source-internal inconsistencies and their adjudication

The source [1] contains two internal normalization conflicts. Both are quarantined as matters of source provenance: the coefficients of this paper are derived independently in Section 4, and the comparison above uses only output words, so no contested normalization of [1] is imported.

The factor two. Printed inside [1] are: the zero-shift kernel (7.21), carrying $1/2 = 1/\Gamma(3)$; the shifted-pair formulas multiplying $\mathcal{D}^\Delta$ by κ_H^2 , hence $\kappa_H^2/2$ at $z = w = 0$; the zero-shift component formulas (7.9), printing κ_H^2 times the derivative bilinears; and the tower (7.16), printing $\kappa_H^2/(m + n + 2) = \kappa_H^2/2$ at $m = n = 0$. The evaluated branches give

$$\mathcal{I}_\Delta[p; 0] = \mathcal{D}_{0,0}^\Delta = C_{00} = \frac{1}{2} , \quad \frac{c_{\text{component}}}{c_{\text{triangle}}} = \frac{c_{\text{compact}}}{c_{\text{triangle}}} = 2 , \quad (7.24)$$

with $\mathcal{I}_\Delta[p; 0] = \frac{1}{2} p_1 \wedge p_2$ the zero-shift master triangle (their eqs. (2.35) and (2.39)). The adjudication selects the complete-amplitude branch: the zero-shift component block and the compact block of [1] carry the complete normalization, while its master-integral, generic shifted-pair, and Appendix-B branches are uniformly smaller by the factor 2 and must be multiplied by 2 before comparison with the complete amplitude, consistently with (7.17) holding at every derivative degree. The conflict remains unresolved inside the source: the classification recorded here repairs nothing in [1] and changes no relative coefficient of Section 4.

The compact coefficient. The introduction of [1] prints the compact prediction (7.8) with coefficient $-1/4$; its main text prints $-\kappa_H^2$. The two displays agree only if

$$\kappa_H^2 = \frac{1}{4} . \quad (7.25)$$

The independently fixed coefficient of Section 4 selects the main-text branch $-\kappa_H^2$ and rejects the introductory display unless $\kappa_H^2 = 1/4$; again this classifies the conflict without repairing the source.

7.5 The admissibility verdict

Theorem 7.4 (admissibility of the holomorphic-twist comparison). *Under the dictionary (7.10), the identification $Q_0 \leftrightarrow -\frac{1}{2}\nabla_-$ of Section 4, and the loop-scale relation (7.13), the direct superspace calculation of Section 4 exactly reproduces the holomorphic-twist prediction for the one-loop semi-chiral anomaly: the 81 ordered output words agree as in (7.14) (29 nonzero, 52 zero), and the derivative kernels agree to all orders,*

$$K_{m,n;k,\ell} = T_{m,n;k,\ell}^{\text{HT,corrected}} \quad \text{for all } m, n \geq 0 , \quad (7.26)$$

i.e. the sealed one-loop coefficient table coincides with the factor-two-corrected one-loop supercharge Q_1 of [1] on every output word, with no mismatch.

Remark 7.5 (agreement of totals is not an absence proof). Theorem 7.4 compares total coefficients. Writing $C_{\text{complete}} = C_{\text{tabulated}} + \sum_{j \in \mathcal{O}} c_j$ for the contributions c_j of any set $\mathcal{O}$ of omitted Wick routings,

$$C_{\text{tabulated}} = C_{\text{HT}} \quad \text{does not imply} \quad \mathcal{O} = \emptyset \text{ or } c_j = 0 \text{ for every } j : \quad (7.27)$$

cancellations such as $c_1 = t, c_2 = -t$ with $t \neq 0$ leave the total unchanged. Equality of the total coefficient with the holomorphic-twist prediction therefore does not prove termwise absence of omitted Wick routings; the agreement is a final cross-check, not a target-blind absence proof.

Remark 7.6 (the adversarial census remains open). The completeness question for the graph census is not settled by Theorem 7.4 and carries the status `OPEN_FINAL_CENSUS_17_CANDIDATES`: the ghost–Nielsen–Kallosh sector has not run, the zero/cut-orbit sector has not run, the automated three-lens refutation has not run, and four physics-relevant Wick routings remain deferred to a future per-routing derivation from the locked action, without reading the target.

Remark 7.7 (the accepted coefficients are unchanged). The accepted coefficient table of Section 4 rests on the target-blind Schwinger-cut derivation presented there; it was read, validated, and sealed (SHA-256 `ec77327d...b3016d`) before the source of [1] was consulted, and it is unaffected by the open census: within the physical one-loop anomaly sector no coefficient row is unresolved.

8 Conclusions and open problems

We have determined the complete one-loop anomalous supersymmetry transformation law of the BPS-letter composites in Euclidean $\mathcal{N} = 4$ super-Yang–Mills theory quantized in $\mathcal{N} = 1$ superspace. For each of the 81 ordered letter pairs, the semi-chiral derivative of the composite equals the universal coefficient $\hbar g^2/(16\pi^2)$ times the ordered color tensor $f^{ACD}f^{BCE}$ times a fixed, explicitly known letter bilinear: 29 channels are nonzero and 52 vanish exactly, with the zeros accounted for by four independent structural mechanisms (pairs with no tree descendant, flavor-off-diagonal mixed pairs, repeated-index diagonal pairs, and pairs killed by the charge and vertex analysis). The sole microscopic origin of the anomaly is the evanescent failure of the DRED cutting rule, (1.1): one universal μ_ℓ^2 insertion per marked line, whose master integral is the finite constant $1/(32\pi^2)$. The anomaly is therefore ultraviolet-finite, local in the external momenta, and — after a finite normal-product term fixed by the residual supersymmetries, under which the nonzero channels close — scheme-independent. Resummed over the holomorphic derivatives, the anomaly is governed by the closed-form kernel (1.3), whose prefactor is the $\Gamma(3) = 2$ of the triangle Feynman parametrization.

Three independent bodies of evidence support the result. First, the superspace derivation is target-blind: every channel was computed from the locked action by the Schwinger–Dyson reduction and the cutting failure, with exact arithmetic, before any external target was consulted. Second, the holomorphic-twist comparison of Sec. 7 is exact: all 81 ordered output words and all checked tower coefficients agree with the one-loop prediction of [1] once its printed tower coefficient is corrected by the factor 2, and the same comparison selects the main-text branch of the source’s compact coefficient. Third, the component bottom projection (1.4) provides an exact component-level corollary, with the color-symmetric dotted-gaugino bilinear. The heat-kernel route contributes exact auxiliary lemmas but not a second derivation.

Remark 8.1. The scope of the accepted result is the physical one-loop anomaly sector in Euclidean signature. The holomorphic-twist agreement is an external cross-check, not an internal completeness proof, and no Lorentzian-signature statement is made.

Open problems

- **Adversarial census completeness.** Seventeen candidate graph sectors or routings raised by the adversarial census review remain adjudicated: the ghost–Nielsen–Kallosh sector, the zero/cut-orbit sector, the automated refutation pass, and four physics-relevant Wick routings have not been derived from the locked action without reference to the target. Status: `OPEN_FINAL_CENSUS_17_CANDIDATES`. This does not alter any accepted coefficient; it is the remaining completeness-type question, since equality with the external target cannot by itself exclude omitted contributions that cancel in the total.
- **Heat-kernel second derivation.** A non-perturbative derivation requires, derived together rather than declared: a single elliptic generator with the correct sign and normalization on every block (matter, vector, ghost, and the off-diagonal mixing blocks), a field-space super-metric, the full second-order coefficient of the semigroup expansion, and distributional short-time extraction at coincident points. Status: `BLOCKED_HEAT_KERNEL_TYPED_REGULATOR_AND_COEFFICIENT_1`.
- **Component off-shell Noether identity.** The component-level off-shell supersymmetry current whose divergence reproduces the anomalous transformation law term by term has not been established: the complete component Euler operators and the announced termwise checks are absent. Status: `NOT_ACCEPTED_NOETHER_B1_B4_MISSING`. The complete component Feynman rules and an independent all-pairs component sweep are likewise not accepted; the only accepted component statement is the exact bottom projection (1.4).

- **Lorentzian vector integration cycle.** The Lorentzian vector integration cycle of the regulated path-integral measure remains unfixed. Status: `BLOCKED_STEP3D_LC_VECTOR_CYCLE`.
- **Lorentzian sector.** All results of this paper are Euclidean. The Lorentzian component and Lorentzian superspace calculations are gated on a frozen Euclidean bridge and on the vector cycle above; the continuation of the anomalous Ward identity to Lorentzian signature is deferred.

Remark 8.2. The holomorphic-twist algebra of [1] requires higher-loop corrections (the one-loop operator squares to zero only on the tree cohomology). Extending the present exact one-loop census of the anomalous supersymmetry transformation law beyond one loop is an open problem on both sides of the comparison.

A Verification status

All coefficient-level statements of this paper are backed by machine-locked exact-arithmetic checks, performed over the rational field extended by i and $\sqrt{2}$ (no floating-point approximation at any stage). This appendix records what was checked, what the independent reviews concluded, and what remains open. Tokens in typewriter font are verbatim status strings of the underlying audits.

Machine-locked exact-arithmetic checks

- **81-channel census.** The target-blind 81-row coefficient table of Sec. 4: all 81 ordered channels complete and exact, 29 nonzero and 52 exactly zero, 0 open and 0 unresolved rows; the table was read, validated, canonicalized and sealed (SHA-256) before the external holomorphic-twist target was read.
- **Seed-channel replays.** The field-strength seed channel: 269 external-slot replay checks, 0 failures. Color-mask replay: 9216 full color-mask checks plus a 2048-entry sparse replay, 0 equality failures.
- **DRED audits.** The cutting-failure identity (1.1) for every marked line, and the evanescent μ_ℓ^2 -moment master integrals — the triangle value $1/(32\pi^2)$ and the rank-two values $J = 1/(128\pi^2)$, $2J = 1/(64\pi^2)$ — are locked in a machine-regenerated master-integral table, including the identities $\Delta J_3 = (\epsilon/2)J_2$ and $\text{Res}_{\epsilon=0} J_2 = 1/(16\pi^2)$.
- **Sector checks.** Finite residual-Ward checks of the mixed-chirality channels: 31/31. Regulated Schwinger–Dyson audit of the channel underlying (1.4): 84/84. Epsilon-canonicalization audit of the mixed flavor sector: 42/42. Heat-kernel auxiliary lemmas (Gaussian semigroup, bridge moments, short-time limit, simplex moment, Duhamel coefficient, DRED master integral): 8/8.
- **Holomorphic-twist round trip.** Direct ordered output-word matches: 81/81, 0 mismatches. Finite-rectangle kernel checks on $0 \leq m, n \leq 8$: 2025/2025 ($= 45^2$), 0 mismatches. Lifted to all 70 nonzero base output words: $2025 \times 70 = 141750/141750$ lifted coefficient checks, 0 mismatches. The external source import itself passed 267 consistency checks.
- **Physical verifier.** 33/33 checks passed, 0 failed; status `ACCEPTED`, accepted scope `PHYSICAL_ONE_LOOP_ANOMALIES`.

Independent review trail

An independent physics re-derivation review of the one-loop result was conducted on 2026-07-16 (non-authority review evidence). Its spot-check verdict, `SPOT_CHECKS_PASSED_WITH_BOUNDED_RESIDUAL_SURFA` confirms the seed-channel derivation chain, the 81-channel census, and the factor-two adjudication, leaving four named bounded residual surface items, none of which is a defect in any accepted coefficient. An external review (GPT Pro, Gate-15) independently returned `FINAL_ONE_LOOP_HT_SETTLEMENT_ACCEPTED` for the physical one-loop anomaly sector, covering the universal cutting-failure identity, the exact census $29 = 1 + 6 + 6 + 6 + 6 + 4$ with $81 - 29 = 52$, the residual-Ward-compatible finite renormalization of the mixed-chirality channels, the anchor-free scale fixing, and the holomorphic-twist kernel identity.

Remark A.1. The adversarial-census review remains open: `OPEN_FINAL_CENSUS_17_CANDIDATES`. Its proposed completeness certificate — exact agreement with the holomorphic-twist target — was rejected, because equality of totals, $C_{\text{enum}} = C_{\text{HT}}$, does not imply an empty omission set: additional routings with contributions c_j satisfying $\sum_j c_j = 0$ would leave the total unchanged, e.g. $c_1 = t, c_2 = -t$ with $t \neq 0$. The holomorphic-twist agreement is therefore a final cross-check, not a target-blind absence proof. The accepted coefficient table is unaffected: its acceptance rests on the independent target-blind superspace derivation, not on either census proposal.

Remark A.2. The machine checks verify exact-arithmetic equalities, replays of locked derivation chains, and target comparisons; they are not by themselves an independent re-derivation. The independent reviews cited above are advisory evidence; the authority for the accepted result is the derivation of Sec. 4 itself.

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